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HP407 – EVIDENCE REVIEW AND SYNTHESIS FOR DECISION MAKING

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Effectiveness of training and support for caregivers of people with dementia in low- and middle-income countries: a systematic review and meta-analysis

ABSTRACT

Background: Caregivers of people with dementia (PwDs) in low- and middle-income countries (LMICs) needed training and support to protect themselves from overburdening and to protect their care recipient from harm, which called for systematic review and meta-analysis focusing on caregiver interventions in LMICs to inform evidence-based intervention.

Objective: To evaluate the effectiveness of caregiver interventions on improving caregiver burden in LMICs.

Design: Five reviewers screened titles and abstracts. The risk of bias of individual studies was assessed using RoB 2 by at least two reviewers independently, with disagreements resolved through group discussion. Two reviewers independently extracted data, with disagreements resolved through group discussion. Pairwise meta-analyses were performed for caregiver burden and caregiver depression, with the robustness of effect estimates tested by leave-one-out analysis and the heterogeneity explored through subgroup analysis and meta-regression.

Data sources: Medline, Embase, Global Health and PsycINFO (via Ovid) and CINAHL (via EBSCO) databases were used to identify studies. Records from an in-house dataset were also included if meeting the eligibility criteria.

Eligibility criteria: We included randomised control trials (RCTs) comparing caregiver intervention to no intervention in the caregivers of PwD in LMICs. The trials must be published in English during 2019-20 and must report outcomes measuring or affecting caregiver burden.

Results: Using random-effects models, we estimated the effect of caregiver interventions on caregiver burden and depression to be standardised mean differences (SMD) of -1.13 (95% CI: [-1.76, -0.51]) and -1.71 (95% CI: [-2.73, -0.69]), which were robust in the leave-one-out analysis. We also observed substantial and significant heterogeneity across studies ($I^2 = 87\%$, $p < 0.01$ and $I^2 = 89\%$, $p < 0.01$ for caregiver burden and depression, respectively). Most prediction intervals (PIs) overlapped with zero, indicating the treatment effects may be insignificant for future studies due to the heterogeneity. Supplementary analysis for three caregiver outcomes and four patient outcomes showed significant improvements, yet one should be careful when generalising the results to all interventions, as meta-analyses for different outcomes included different studies, of which the scalability was yet to be tested.

Conclusions: We estimated a medium-to-large effects of caregiver interventions on caregiver burden and depression respectively. The effect sizes were larger than similar trials in high-income countries (HICs), potentially due to lower care quality in LMICs and/or the bias in the included study due to study design. While this study provided evidence for the effectiveness of caregiver interventions, future studies should examine the scalability of the interventions. Prospective and cumulative meta-analyses to update the latest evidence and network meta-analysis to identify the most effective interventions should be performed.

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ABBREVIATIONS

- ADAS-cog	Alzheimer's Disease Assessment Scale-Cognitive Subscale
- ADL	Activities of daily
- BADL	Barthel's Activities of Daily Living
- BDI	Beck Depression Inventory; BAI; Beck Anxiety Inventory
- BEHAVE-AD	Behaviour Pathology in Alzheimer's Disease Rating Scale
- CBI	Caregiver Burden Inventory
- CBT	Cognitive-behavioural therapy
- CCI	Comprehensive care intervention
- CCM	Counselling/case management
- CES-D	Center for Epidemiologic Studies- Depression Scale
- CFT	Cognitive and functional training
- CI	Confidence interval
- Ctrl.	Control group
- CSI	Caregiver Strain Index
- DEMQOL	Dementia Quality of Life measure
- EASI	Everyday Abilities Scale for India
- Expt.	Experimental group
- GDS	Global Deterioration Scale; MMSE: Mini-Mental State Examination
- GHQ-12	12-Item General Health Questionnaire
- GHQ-28	28-item General Health Questionnaire
- GQOLI-74	74-item Generic quality of life inventory
- GS	Group support
- Ham-D	Hamilton Depression Rating Scale
- HIC	High-income country

- ITT	Intention to treat
- LMIC	Low- and middle-income country
- MCI	Multi-component intervention
- MD	Mean difference
- Misc.	Miscellaneous intervention
- MoCA	Montreal Cognitive Assessment
- NPI-S	Neuropsychiatric Symptoms Inventory (distress subscale)
- NPI-S	Neuropsychiatric Symptoms Inventory (severity subscale)
- NULL	No intervention
- PACQ	Positive Aspects of Caregiving Scale
- PEI	Psychoeducational interventions
- PHQ-4	4-item Patient Health Questionnaire
- PP	Per protocol
- PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
- PwD	People with dementia
- QoL-AD	Quality of Life in Alzheimer's disease
- QoL	Quality of life
- RCT	Randomised controlled trial
- SD	Standard deviation
- SF36	36-Item Short Form Survey
- SMD	Standardised mean difference
- SQ	Spielberger's questionnaires
- SRQ-20	20-item self-report screening tool
- TMAS	Taylor Manifest Anxiety Scale
- TSWBS	Thai Subjective Well-being scale
- WHO-Bref-QoL	World Health Organization Quality-of-Life Scale
- ZBI	Zarit Burden Interview

1 INTRODUCTION

Caregivers are the invisible second patients of dementia, as they often struggle to balance caregiving with other demands, such as work and relationships, which increases their stress, depression and anxiety and causes other health complications.¹ Time and effort spent on caregiving entails role strain, termed caregiver burden. The role strain leads to stress in caregiving and negatively affects caregiver well-being.²⁻⁴ Thus, caregivers need and deserve knowledge and training to protect themselves from overburdening. Caregiver support also contributes to more competent caregiving that protects care recipients from harm.⁵

In 1995, the National Institutes of Health acknowledged that despite increasing knowledge about caregiver burden, evidence-based approaches to alleviating caregiver burden were lacking.⁶ Since then, numerous trials have been conducted to evaluate the effectiveness of existing and new caregiver interventions. To estimate the intervention effect with greater power and precision,⁷ meta-analyses, which calculate a pooled estimate of intervention effects from individual studies, have been performed.⁸⁻¹² A recent meta-analysis by Walter and Pinquart (2020) showed caregiver interventions produced significant, small-to-moderate improvements in various health outcomes in PwDs and their caregivers.¹²

However, these studies regarding dementia care may not represent LMICs, as only 10% of dementia research is conducted in LMICs.¹³ This has triggered the 10/66 Dementia Research Group to design a caregiver intervention suitable for LMICs, with 3 RCTs.¹⁴⁻¹⁶ Although a meta-analysis for current evidence in LMICs has since been called for,¹⁵ as we examined the meta-analyses included in a 2020 meta-review,¹⁷ none focused on LMICs.¹⁸ A recent scoping review of caregiver interventions in Asia suggests that only including English-language literature may underestimate evidence that could be synthesised in LMICs.¹⁹ A systematic review of caregiver interventions in China showed different effects on PwDs and caregivers, but it did not quantify the effects or the heterogeneity across the effects.

Therefore, a systematic review was warranted to synthesise evidence from more LMICs and more languages. Meta-analyses were especially needed to quantify the evidence from all eligible studies and evaluate the generalisability and consistency of intervention effects across studies.²⁰ Thus, the review aimed to evaluate the effectiveness of caregiver interventions for caregiver and care recipient health, thus providing evidence for designing and implementing effective caregiver interventions in LMICs.

2 METHODS

2.1 Eligibility criteria

We included studies that met the following inclusion criteria:

- **Population:** PwD and their caregivers living in LMICs
- **Interventions:** Interventions with a significant proportion of caregiver training and support
- **Comparator:** No intervention
- **Outcomes:** We considered caregiver burden and depression as our primary outcomes because they were the most measured outcomes in caregiver interventions and often the reasons why we designed caregiver interventions.^{21–23} The secondary outcomes are any outcome mentioned in Van Houtven et al. (2021).²¹ As Zarit (2018)²² suggested, alleviating depression and burden was not necessarily what caregivers wanted from an intervention. Thus, considering other outcomes as secondary outcomes provided a fuller understanding about the effectiveness of caregiver interventions.
- **Study design:** We only included RCTs because RCTs are the gold standard for evaluating the effectiveness of an intervention,²⁴ where their rigorous study designs minimised threats to internal validity.²⁵

We excluded studies not in English or Chinese because we were unable to read and extract data from them.

2.2 Search methods

A database search of Medline, Embase, Global Health and PsycINFO via Ovid, plus CINHL via EBSCO, was performed to identify relevant studies on 21 September 2022, with the search strategy described in Appendix 2. To update an in-house dataset including dementia intervention studies published from 2008 to 2018, we restricted the years of publication between 2019 and 2020.

2.3 Study selection

Four reviewers first performed title and abstract screening independently, with disagreement settled through group consensus. Then, four reviewers divided the studies evenly and retrieved the full text of the eligible studies, after which four independently selected included studies. Finally, any disagreement on included studies was settled through group discussion.

2.4 Data extraction

Two reviewers independently extracted and recorded the data, with disagreement settled through group discussions. We performed data extraction as guided in the Cochrane Handbook.²⁶ We included the mean value, standard deviation (SD) of the outcome measurements, plus the number of participants measured, in each arm. Change-from-baseline data were collected if

For each study, we also recorded its study settings, randomisation methods, measurement time points and funding sources. When multiple measurement points were available, we selected the one closest to the end of intervention. We considered this appropriate because, based on the existing literature,^{12,27,28} the intervention effects usually remained stable at follow-up.

For each intervention, we recorded its description, duration, frequency, and delivery modes (e.g. home visits and phone interviews). For the study samples, we recorded their demographic and clinical characteristics, including the mean age, the female percentage of both PwDs and caregivers, and dementia stage of PwD.

2.5 Intervention classification

For most interventions before 2019, their classification was given in Walter and Pinquart (2020) and their previous works.^{9,11,12} Those not classified in previous meta-analyses were classified by two reviewers in accordance with the definitions adapted from previous meta-analyses through discussion (Table 1).¹²

Table 1 Definitions of intervention types defined by Walter and Pinquart (2020)

Type (Abbreviation)	Definition
Psychoeducational interventions (PEI)	Structured educational programmes in which the interventionists provide information about dementia and AD and train caregivers to deal with the disease-related problems of care recipients effectively. Additional support may be provided, but it needs to be secondary to the educational content.
Cognitive-behavioural therapy (CBT)	Structured or unstructured programmes involving identification and modification of the beliefs underlying caregivers or other techniques from CBT to develop a new behavioural patterns for caregiving
Counselling/case management (CCM)	Individualised advisory services and support aiming to help caregivers solve their specific personal needs not necessarily related to caregiving

General support (GS)	Peer- or professional-led groups to enable discussion and information sharing about dementia- and caregiving-related issues and to allow people to share their personal feelings
Respite	Provision of substitute care upon caregivers' request or when caregivers are in need, thus allowing caregivers to be temporarily exempted from care responsibility
Cognitive and functional training (CFT)	Memory clinics or training programmes aimed at the improvement of caregivers' cognitive and functional impairments rather than care recipients'
Multi-component intervention (MCI)	A combination of the above-mentioned types of interventions. Further information must be provided about the intervention types of each component of the intervention.
Miscellaneous (Misc.)	Caregiver interventions that do not belong to any above-mentioned type
Comprehensive care intervention (CCI)	Interventions for PwD with caregiver involvement

Note: As proposed by Cheng and Zhang (2020),¹⁷ the interventions for PwD with caregiver involvement were classified into an additional intervention type, termed "comprehensive care intervention (CCI)", in this review.

2.6 Risk of bias assessment

The risk of bias within each study was assessed as guided by the RoB2 tool.²⁹ Each of the newly included studies was evaluated by > 2 reviewers, with the disagreement settled through group consensus. Publication bias was assessed through funnel plots and Egger's test only if the outcome was measured in > 10 studies, or else the test would be underpowered.^{30,31} In each meta-analysis, we also compared fixed- and random-effects estimates to evaluate small-study effects.^{32,33} By checking the sources of funding, we also evaluated industry bias.

2.7 Synthesis of results

All the collected data were combined and saved in a pre-defined data extraction table (Appendix 4). We conducted a qualitative analysis as guided by Eden et al. (2011),³⁴ in which we evaluated the comparability of study designs, participants, intervention contents, and outcome measures and assessed the feasibility and scalability of each intervention type.

For comparable studies, we performed a meta-analysis with R (see Appendix 3 for details). To compare the outcomes measured with different instruments, we converted the outcome data into standardised mean difference (SMD).³⁵ To be consistent with previous meta-analyses,¹¹ we estimated SMD with Hedge's *g*, which estimates SMD more accurately than Cohen's *d* when the sample size is small.^{36,37}

The heterogeneity across studies was evaluated with I^2 statistics and Cochran's Q test.³⁸ We used random-effects models when $I^2 > 0$, or else we used a fixed-effects model. We performed subgroup analysis for intervention types per previous meta-analyses.^{9,11,12} To explore other potential sources of heterogeneity, we performed subgroup analysis and meta-regression for exploratory variables. As guided by the Cochrane Handbook, we only considered an outcome eligible for meta-regression when measured in > 10 studies, or else our method would be underpowered.

3 RESULTS

3.1 Study selection

After reviewing 2653 titles and abstracts and assessing 255 full texts, we included 6 studies^{39–44} from the database search (Figure 1). After screening 733 records and 24 full texts, we further included 20 studies from the in-house dataset (Figure 1).^{14,15,27,28,45–59} We did not find any eligible non-English study in the database search but included nine Chinese-language studies from the in-house dataset.^{45,46,53,55,57–60}

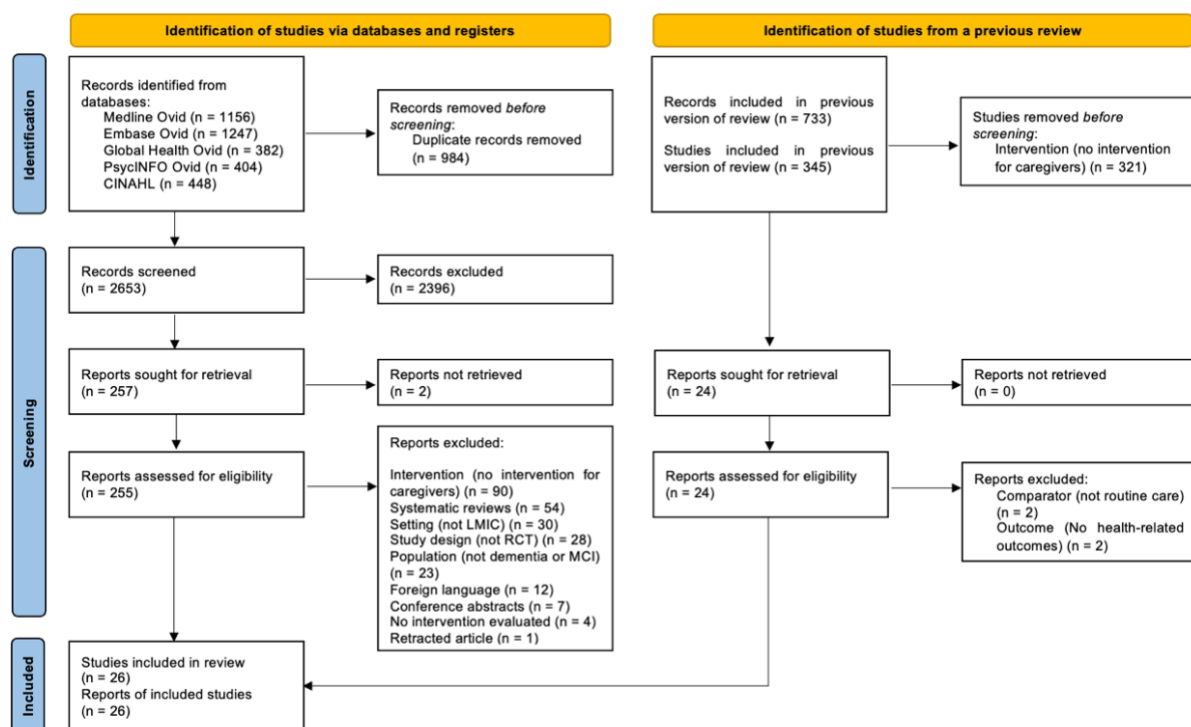


Figure 1. PRISMA flow chart of study selection

3.2 Study and participant characteristics

Study characteristics are summarised in Table 2. All but one study were two-arm RCTs comparing the intervention with routine care. Thus, we treated the only three-arm study⁴⁸ as two studies sharing the same control group for meta-analysis. Only two studies performed cluster randomisation, which blinded the participants from the intervention not assigned to them while blinding was not assured in the rest.⁶¹ Noticeably, two trials,^{46,59} published in 2010 with shared authors in the same Chinese hospital, described similar interventions, which need further clarification about their impendence.

The studies reported 12 caregiver outcomes and four patient outcomes, which were measured with different instruments (Table 3 and Table 4). However, only six studies measured both caregiver outcomes and patient outcomes. Consistent with previous studies,^{21,22} burden and depression were the most and second most measured caregiver outcomes, measured in 12 and 6 studies, respectively. Measurement scales for caregiver burden included ZBI, CBI and CSI. Measurement scales for caregiver depression included the GHQ-28, CES-D, Ham-D and BDI.

Different outcome measurements affected whether the study randomised PwD or caregivers, thus leading to baseline differences for the mean age and female percentage of either caregivers or PwD, although they were similar in the control and intervention arms but different across studies (Table 5). Most studies were conveniently sampled from hospitals or patient registration, which may introduce bias.^{62,63} Only six studies recruited the participants from the communities, which may better represent the general population (Table 2).

Table 2 Key characteristics of included studies

Table 2.1 Key characteristics of 6 new studies

Study identifier	Participants					Intervention				Method			
	Country	Source	Sample size for analysis			Descriptive name	Content	Frequency	Modes	Study design	Duration from baseline		Type of funding
			Ctrl.	Expt.	Attrition%						Intervention	Measurement	
Chen 2020 ⁴⁰	China	Hospital admissions	51	51	N/A	Extended Nursing	GS	Multiple	WeChat, home visits, group activities	Parallel group RCT, randomised by patient	8 wk	1 yr	National/federal government
Ghaffari 2019 ⁴¹	Iran	Hospital admissions	27	27	7.4	Resilience Education	CBT	Twice a week	Hospital visits	Parallel group RCT, block-randomised by caregiver	1 wk	1 mo	University
Hinton 2020 ⁴²	Vietnam	Community sampling	26	25	15.0	Multi-component interventions based on REACH	MCI	Weekly	Home visits	Parallel group RCT, randomised by community	3 wk	3 mo	National/federal government
Pan 2019 ⁴³	China	Hospital admissions	45	54	11.6	Nurse-led cognitive behavioural sessions and subsequent consultation calls	CBT	Monthly	Home visits, telephone interviews	Parallel group RCT, block-randomised by caregiver	5 mo	7 mo	Local government
Uyar 2019 ⁴⁴	Turkey	Hospital admissions	26	28	11.5	Multi-component psychosocial intervention	MCI	Multiple	Telephone interviews, WhatsApp & Home visits	Parallel group RCT, randomised by patient-caregiver pair	16 wk	16 wk	N/A
Zarepour 2020 ³⁹	Iran	Hospital admissions	35	35	0	On-site educational intervention + informational support on the anxiety of family	PEI	7 sessions in 4 weeks	Group sessions, plus virtual messaging programs or email	Parallel group RCT, block-randomised by caregiver	4 wk	4 wk + 1 mo 4 wk + 3 mo	University

Table 2.2 Key characteristics of 20 previously included studies

Study identifier	Participants					Intervention				Method			
	Country	Source	Sample size for analysis			Descriptive name	Content	Frequency	Delivery mode	Study design	Duration from baseline		Type of funding
			Ctrl.	Expt.	Attrition%						Intervention	Measurement	
Amit 2008 ¹⁴	India	Community	26	33	27.2	Home Care Program for supporting caregivers of PwD	MCI	Multiple	Home visits, hospital visits	Parallel group RCT, randomised by patient-caregiver pair	6 mo	3 mo 6 mo	International organisation
Guerra 2011 ¹⁵	Peru	Community	29	29	3.4	'Helping Carer to Care' training	PEI	Weekly	Home visits	Parallel group RCT, stratified permuted block randomisation by caregiver	5 wk	6 mo	International organisation
JIANG 2012 ⁴⁵	China	Hospital admissions	48	46	2.1	Whole course rehabilitation nursing intervention	CCI	Multiple	Home visits, telephone interviews	Parallel group RCT, randomised by patient	12 mo	12 mo	N/A
Lin 2017 ⁶⁴	China	Community	39	39	0	Family visits of community nurses	CCI	Multiple	Home visits, telephone interviews, community events	Parallel group RCT, randomised by caregiver	3 mo	3 mo	University
Lök 2017 ⁴⁷	Turkey	Hospital admissions	20	20	N/A	"First You Should Get Stronger" Caregiving Program	PEI	Weekly	Home visits	Parallel group RCT, randomised by caregiver	7 wk	7 wk	N/A
Mahdavi 2017 ⁴⁸	Iran	Patient registration	31	28/30	11.0	Spiritual Group Therapy	Misc.	Weekly	Group sessions	Parallel group RCT, block-randomised by caregiver	5 wk	5 wk	University

Mercedeh 2014 ⁴⁹	Iran	Patient registration	10	10	N/A	Mindfulness based cognitive therapy	CBT	Weekly	Group sessions	Parallel group RCT, randomised by caregiver	8 wk	8 wk 8 wk + 2 mo	University
Orawan 2018 ²⁸	Thailand	Hospital admissions	36	36	10.0	Enhancing Positive Aspects of Caregiving Program	MCI	Weekly	Hospital visits or home visits	Parallel group RCT, randomised by caregiver	7 wk	8 wk 12 wk 20 wk	N/A
Pahlavanzadeh 2010 ⁵⁰	Iran	Hospital admissions	25	25	16.7	Family education program	PEI	Weekly	Group sessions	Parallel group RCT, randomised by caregiver	5 wk	5 wk 5 wk + 1 mo	University
Salamizadeh 2017 ⁵¹	Iran	Hospital admissions	28	26	10.0	Spiritual Care Education	Misc.	Weekly	Group sessions	Parallel group RCT, randomised by caregiver	5 wk	5 wk	N/A
Shata 2017 ²⁷	Egypt	Hospital admissions	59	55	5.0	A multi-component psychosocial intervention program	MCI	Weekly	Group sessions	Parallel group RCT, randomised by patient-caregiver pair	8 wk	8 wk 8 wk + 3 mo	Self-fund
Söylemez 2016 ⁵²	Turkey	Hospital admissions	34	35	1.4	Progressively Lowered Stress Threshold-based intervention	PEI	Bi-weekly	Group sessions, phone interviews, home visits	Parallel group RCT, randomised by patient-caregiver pair	3 mo	5 mo + 2 wk	University
SU 2012 ⁵³	China	Hospital admissions	20	20	0	Comprehensive nursing intervention	CCI	Multiple	Home visits, phone interviews, hospital visits	Parallel group RCT, randomised by patient	14 days	14 days + 6 mo	N/A
SUN 2010 ⁵⁴	China	Hospital admissions	19	19	N/A	Nursing intervention	CCI	Multiple	Home visits, hospital visits, phone interviews, community events	Parallel group RCT, randomised by patient	12 mo	12 mo	N/A
TAN 2010 ⁴⁶	China	Hospital admissions	26	26	N/A	Early family intervention	CCI	Multiple	Home visits, phone interviews, community events	Parallel group RCT, randomised by patient	12 mo	12 mo	N/A
Wang 2010b ⁵⁵	China	Community	40	40	N/A	Community nursing intervention	CCI	Multiple	Home visits, phone interviews, group sessions	Parallel group RCT, randomised by patient	12 mo	12 mo	Local government
Wang 2012e ⁵⁶	China	Hospital admissions	39	39	7.7	Mutual support group	GS	Bi-weekly	Group sessions	Parallel group RCT, randomised by caregiver	24 wk	24 wk 24 wk + 1 mo	N/A
Wang 2014a ⁵⁷	China	Community	61	62	3.9	Home care intervention	CCI	Multiple	Home visits	Parallel group RCT, randomised by patient	12 mo	6 mo 12 mo	N/A
Yang 2017a ⁵⁸	China	Community	70	71	N/A	Family medical intervention model	CCI	Multiple	Group sessions, home visits, telephone interviews	Parallel group RCT, randomised by community	12 mo	6 mo 12 mo	Local government
ZHAO 2010 ⁵⁹	China	Unclear	25	25	N/A	Family nursing intervention	CCI	Multiple	Home visits, telephone interviews, community events	Parallel group RCT, randomised by patient	12 mo	12 mo	N/A

Note: Mahdavi 2017 had three arms. In this study, we consider Control group 1 (Group sessions without spiritual therapy; N = 28) and the Intervention group (Spiritual group therapy; N = 30) as the intervention arms and compared the two to the Control group 2 (No intervention) separately.

Table 3 Instruments used to measure caregiver and care recipient outcomes

Study	Caregiver						Care recipient			
	Burden	QoL	Depression	Anxiety	Distress	Mental health	Cognitive function	QoL	ADL	Neuro-psychiatric symptoms
Amit 2008	ZBI					GHQ-12			EASI	
Chen 2020							MoCA		BADL	BEHAVE-AD
Ghaffari 2019			GHQ-28	GHQ-28		GHQ-28				
Guerra 2011	ZBI	WHO-QoL			NPI-D			DEMQL		NPI-S
Hinton 2020	ZBI		PHQ-4							
JIANG 2012		WHO-QoL					MMSE ADAS-cog			GDS NPI-S
Lin 2017	CBI									
Lök 2017	ZBI									
Mahdavi 2017	CSI									
Mercedeh 2014	CBI		Ham-D							
Orawan 2018										
Pahlavanzadeh 2010	ZBI									
Pan 2019			CES-D							
Salamizadeh 2017										
Shata 2017	ZBI		Ham-D	TMAS						
Söylemez 2016	CBI	WHO-QoL						QoL-AD		
SU 2012							MMSE		BADL	
SUN 2010										
TAN 2010							MMSE		ADL	
Uyar 2019	ZBI	SF36	BDI	BAI	NPI-D			QoL-AD		NPI-S
Wang 2010b							MMSE		ADL	
Wang 2012e		WHO-QoL			NPI-D					
Wang 2014a							MMSE		ADL	
Yang 2017a		GQOLI-74					MMSE	QoL-AD	ADL	
Zarepour 2020				SQ						
ZHAO 2010										

Note: We highlighted the studies measuring both caregiver and care recipient outcomes in sky blue. Most Chinese-language studies claimed to use the “ADL scale”. We consulted the Chinese database CNKI using the keyword “ADL scale” in Chinese and found that the ADL scale often exclusively referred to an adapted version of the ADL scale by Lawton and Brody (1969)⁶⁵ in the context of Chinese-language literature.⁶⁶⁻⁶⁸ Uyar (2019) measured two QoL scores, one for mental health and the other for physical health. PHQ-4 measures anxiety or depression with a single score.

Table 4 Number of included studies and sample sizes for each outcome

Outcome measures	# of studies	Sample size		
		Intervention	Control	Total
Caregiver outcomes				
▪ Caregiver burden	12	353	327	680
▪ Caregiver depression	6	205	201	406
▪ Caregiver distress	5	152	151	303
▪ Caregiver anxiety	4	141	147	288
▪ Caregiver QoL				
○ Overall	4	140	141	281
○ Physical	2	53	57	110
○ Psychological	2	53	57	110
○ Environmental	1	27	29	56
▪ Caregiver mental health	3	83	77	160
▪ Caregiver coping	1	54	45	99
▪ Caregiver mutuality	1	54	45	99
▪ Caregiver positive experience	1	36	36	72

▪ Caregiver subjective well-being	1	36	36	72
▪ Caregiver self-efficacy	1	26	28	54
▪ Caregiver healthy behaviour	1	20	20	40
Patient outcomes				
▪ Patient ADL	8	349	342	691
▪ Patient cognitive function	7	316	316	632
▪ Patient neuropsychiatric symptoms	6	254	252	506
▪ Patient QoL	4	150	150	300

Note: Primary outcomes are highlighted in sky blue. Caregiver healthy behaviour was measured in six domains: feeding, moral development, healthy responsibility, interpersonal communication, physical activity, and stress management. Three studies^{49,54,59} calculated overall SF-36 scores to measure QoL, which was considered to be a measurement error, as SF-36 measures distinct concepts, and it is meaningless to calculate overall scores. We noted that one study⁴⁶ used data that contradicted the claimed treatment effect on patient ADL, which was likely a mistake or an error. Thus, we excluded these data from the meta-analysis to reduce bias.

Table 5 Baseline characteristics of included studies

Study Identifier	Experimental group				Control group			
	Patient		Caregiver		Patient		Caregiver	
	Female %	Mean age	Female %	Mean age	Female %	Mean age	Female %	Mean age
Chen 2020	52.8	72.4			52.9	71.4		
Lök 2017	45.0	70.7	70.0	53.0	35.0	74.2	65.0	51.8
Mahdavi 2017		75.4		53.0		75.4		53.0
Mercedeh 2014			100.0				100.0	
Orawan 2018			83.3	55.1			72.2	55.7
Pahlavanzadeh 2010	44.0	72.0	84.0	42.9	56.0	67.7	68.0	46.6
Ghaffari 2019			88.0	42.6			72.0	43.4
Hinton 2020	76.0	83.7	92.0	59.0	76.9	82.8	80.8	58.7
Salamizadeh 2017			80.8	47.4			89.3	50.6
Söylemez 2016	63.0	78.6	74.0	61.2	54.0	76.3	71.0	62.3
Pan 2019	71.4	79.0	55.4	63.3	63.4	80.0	69.6	62.1
Wang 2012e	51.3		59.0		53.8		64.1	
JIANG 2012	58.3	71.9			54.2	71.3		
Liu 2017	64.1		66.7	66.5	64.1		56.4	62.8
SU 2012	50.0	68.0			55.0	68.0		
SUN 2010	47.4	74.1			52.6	73.2		
TAN 2010	61.5	68.4			61.5	68.4		
Wang 2010b								
Wang 2014a								
Yang 2017a	56.3	76.1			54.3	74.7		
ZHAO 2010	60.0	68.4			60.0	68.4		
Uyar 2019	76.7	76.7	83.3	53.6	67.7	76.7	74.2	53.6
Zarepour 2020		73.6	68.6	46.8		73.6	71.4	46.8
Guerra 2011	79.3	81.7	85.7	53.3	69.0	82.0	89.7	47.6
Amit 2008	36.6	79.4	90.2	53.2	32.5	77.3	85.0	53.8
Shata 2017	67.5	69.3	61.8	49.4	67.5	69.3	69.5	48.0
Median	59.2	74.1	82.1	59.2	55.5	74.2	71.7	53.0
Max	79.3	83.7	79.3	66.5	76.9	82.8	100.0	62.8
Min	36.6	68.0	55.4	42.6	32.5	67.7	56.4	43.4

Note: We considered the values in each group the same when the study only provided an overall value of each variable. Subgroup summary statistics thus imputed from group summary statistics were highlighted.

3.3 Intervention characteristics

The included interventions were highly heterogeneous (Table 2). Besides ten studies using CCI, three used CBT, two used GS, six used MCI, and six used PEI, while the remaining two used spiritual care interventions were classified as miscellaneous interventions. Typical delivery modes included one or a combination of home visits, hospital visits, phone interviews and group sessions on a regular basis, or most often a weekly basis, while two studies published during 2019-2020 used online chats, suggesting a new trend for intervention delivery. CCI was the main caregiver support in China. All studies using CCI were conducted in China, while only two studies in China did not use CCI. Most interventions were delivered on a regular basis, e.g. weekly, while CCI typically used multiple delivery modes with different frequencies. Therefore, it may be useful to analyse CCI in comparison to other included studies.

Most studies had low total attrition rates, with a median attrition rate of 6.5% for all studies reporting attrition. The included studies covered all types of interventions except for respite care, potentially due to a lack of such services in LMICs.⁶⁹ The scalability of the interventions was questionable, as most interventions were trial-specific and were not tested beyond the original trial settings. Only three interventions were validated across settings, including the 10/66 intervention,¹⁴⁻¹⁶ the REACH intervention,^{42,61,70} and the mutual support group.^{56,71,72}

3.4 Risk of bias

The risk of bias was summarised in Figure 2, 3 and 4, with the rationale stated in Appendix 5. Among the six newly included studies, only one study (Hinton 2020) was judged to have a low risk of bias. Two studies were found to have some concerns with randomisation due to poor descriptions of randomisation or allocation sequence concealment. Two were at high risk due to significant baseline differences. Most studies poorly described or did not describe deviations from the intended interventions and did not blind participants and interventionists from the intervention allocation, leading to a high risk of some concerns with the bias due to deviations from the intended intervention.⁷³ Self-reported psychological scales and the nonblinding of outcome assessors both contributed to bias in outcome measurements.⁷⁴ As most studies did not publish their protocols or statistical analysis plans, there was insufficient information to identify potential selective reporting, resulting in some concerns.

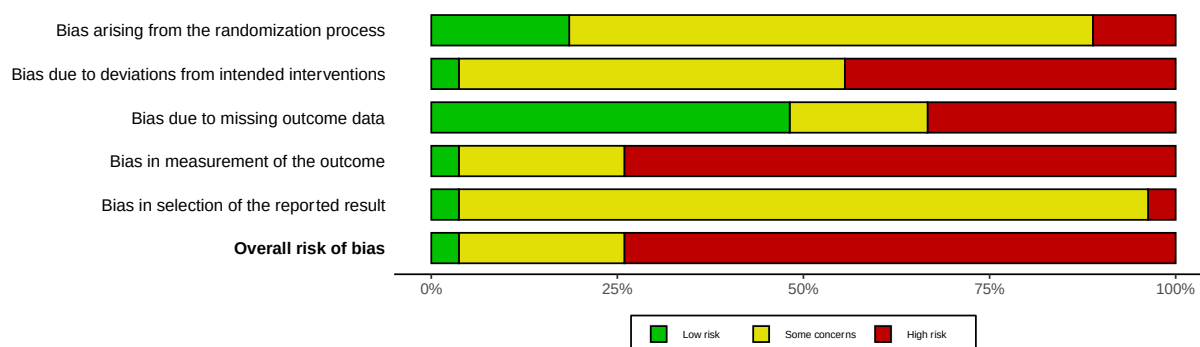


Figure 2 Bar plot of risk of bias information for all included studies

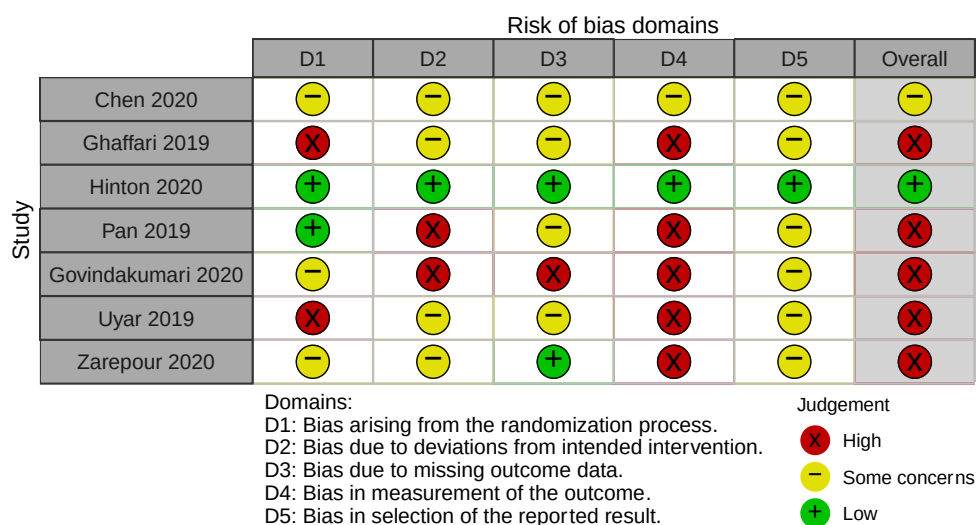


Figure 3 Traffic light plot of the risk of bias information for six new studies

Note: Govindakumari 2020 was assigned to our group, but it is ineligible as it did not contain any component of caregiver intervention. However, we still performed a risk of bias assessment for the study. The risk of bias information of the 20 previously included studies was obtained from the in-house dataset and is displayed in Appendix 5.

The funnel plot and Egger's test ($p = 0.8080$) did not show significant publication bias in the data regarding caregiver burden (Figure 4). However, other outcome data were not powered for Egger's test. Thus, we were unable to determine whether there was a significant publication bias in these outcome data. Among 15 studies reporting the sources of funding, none was funded by a commercial company. Most studies were funded by governments and universities. Thus, we believed that the risk of industry bias was low.

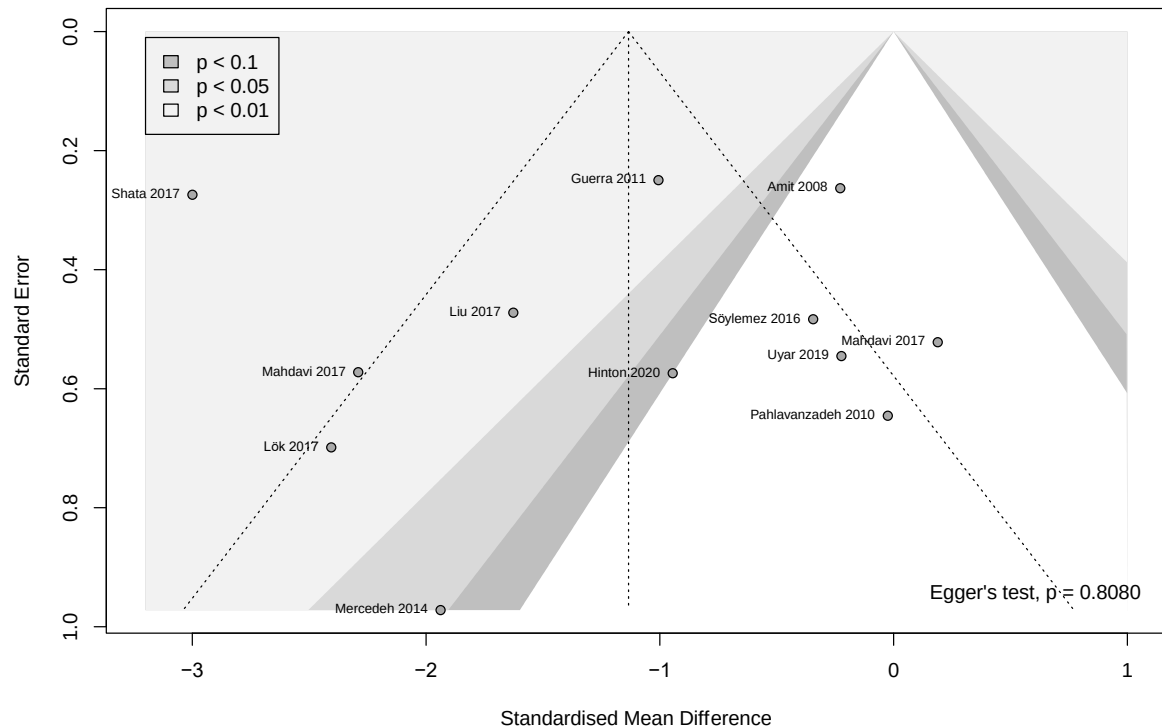


Figure 4 Funnel plot of effects on caregiver burden

3.5 Intervention effects

3.5.1 Caregiver burden

Twelve studies, with a total of 680 caregivers, were included in the meta-analysis of the intervention effects on caregiver burden (Figure 5). The overall heterogeneity is significant ($I^2 = 87\%$, $p < 0.01$). Using a random-effects model, the pooled estimate of all RCTs was calculated as -1.13 (95% CI: $[-1.76, -0.51]$), showing that the interventions included yielded a significant, medium-to-large treatment effect on caregiver burden. However, PIs overlapping zero indicated potential non-treatment effects for future studies due to heterogeneity.

Despite significant subgroup differences for intervention types ($p = 0.03$), our meta-analyses for MCI and PEI both showed significant, high heterogeneity ($I^2 = 95\%$, $p < 0.01$ and $I^2 = 62\%$, $p = 0.05$, respectively). MCI only showed an insignificant improvement in caregiver burden (SMD: -1.13 ; 95% CI: $[-2.47, 0.21]$), while PEI showed a significant improvement (SMD: -0.90 ; 95% CI $[-1.75, -0.05]$). The only study using GS did not improve caregiver burden.

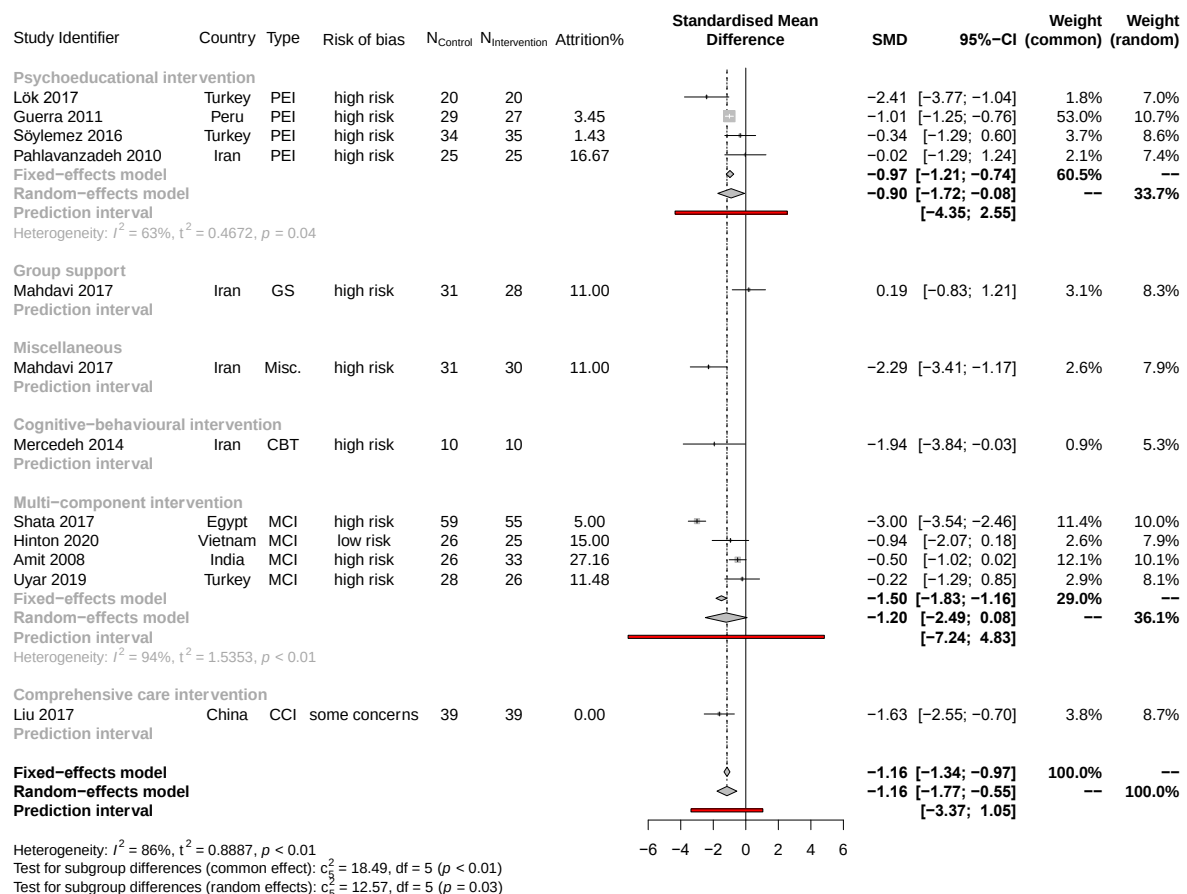


Figure 5 Forest plot of meta-analysis of effects on caregiver burden

Note: The fixed effect(s) is also occasionally described as the common effect in the figure. Though we performed meta-analysis using a random-effects model, the fix-effects estimates are also given in the figure to detect potential small-study effects, which are not obvious by visually inspecting the differences in the figure.

Using the leave-one-out analysis, we found that the treatment effect remained significant when omitting any single study (Figure 6). However, Shata 2017²⁷ was identified as an influential case, which had the largest effect size and contributed to the most heterogeneity. After excluding the case (Figure 7), the I^2 dropped from 86.6% to 65.2%, and the summative effect size was reduced to -0.90 (95% CI: [-1.41, -0.39]). After examining the study, we found that our analysis results were consistent with the study, which reported a very large effect size (partial $\eta^2 = 0.77$),²⁷ much larger than Cohen's 0.14 cut-off for large effect sizes.⁷⁵

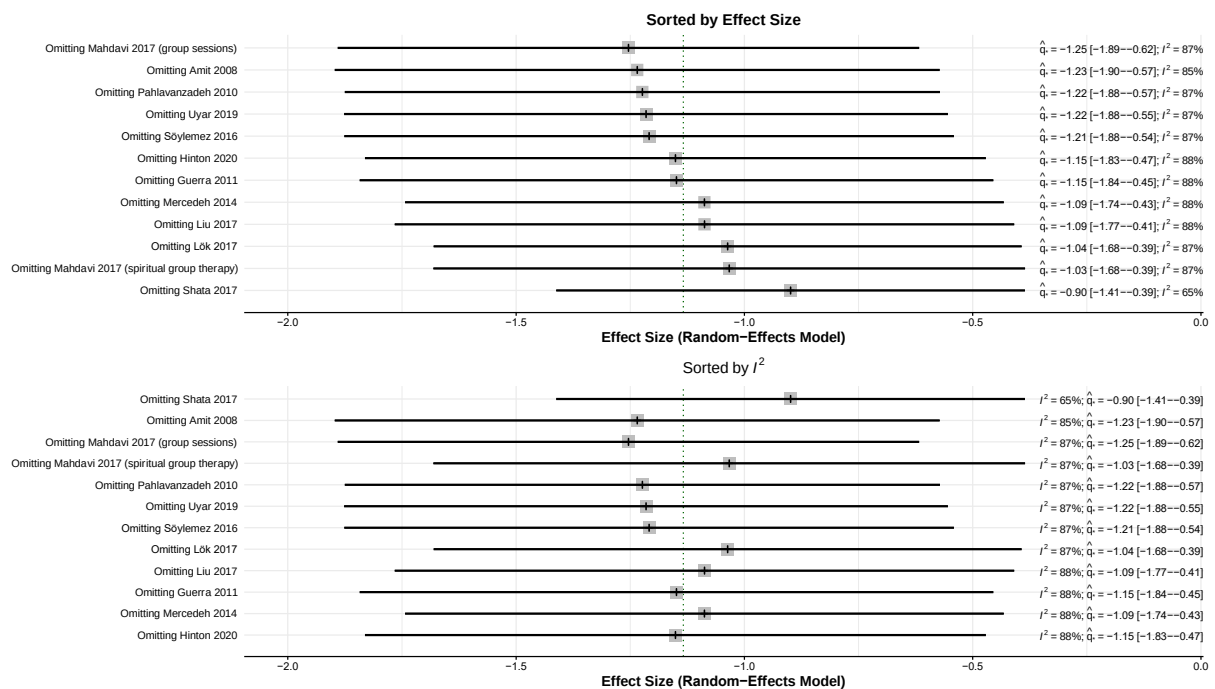


Figure 6 Leave-one-out analysis of effects on caregiver burden

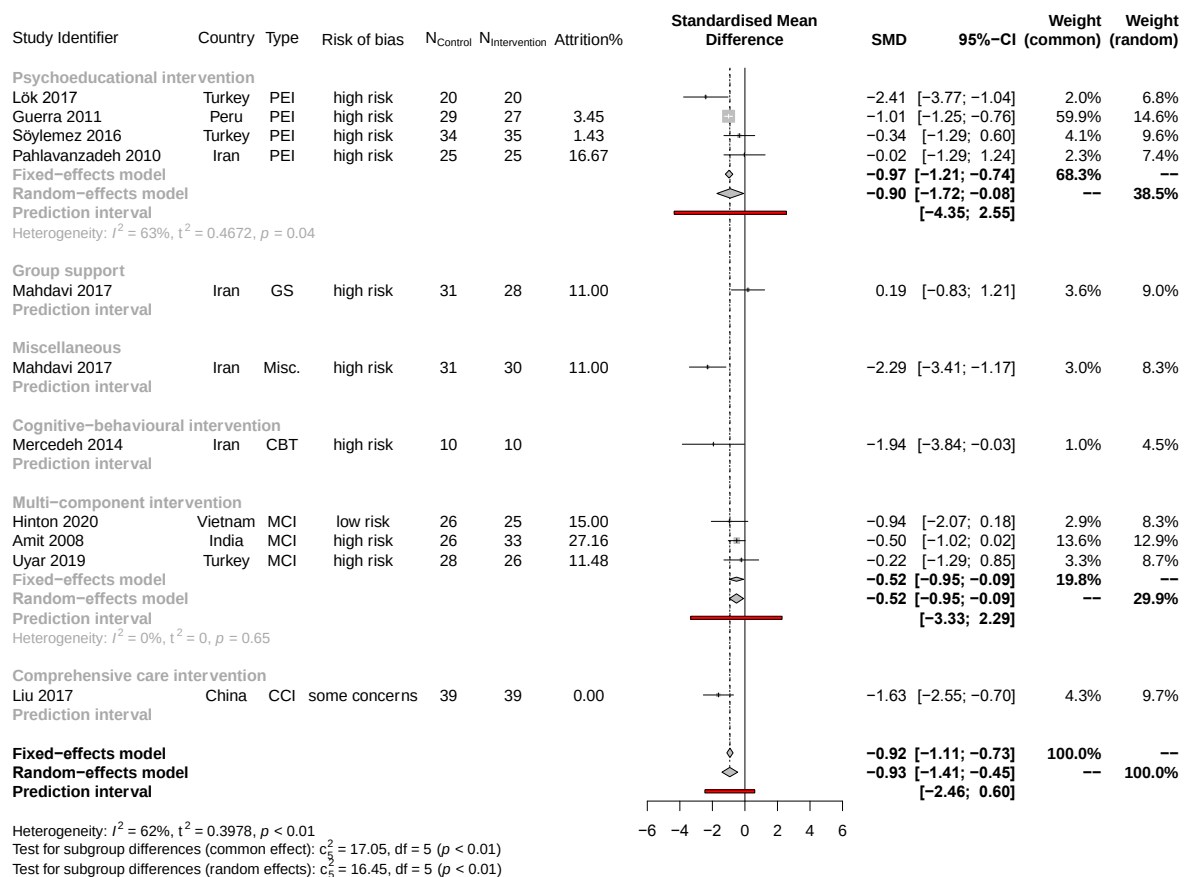


Figure 7 Forest plot of meta-analysis of effects on caregiver burden excluding the outlier

To further explore the heterogeneity, we performed meta-regression with years of publication, total attrition rates, measurement durations, intervention durations, caregiver mean age, caregiver female percentage, patient mean age and female percentage of patients. None showed a significant correlation (Appendix 6). We also performed subgroup analysis for exploratory variables, including country, funding types and intervention frequency, which did not show significant subgroup differences (Appendix 7).

3.5.2 Caregiver depression

Six studies, with a total of 406 caregivers, were included in the meta-analysis (Figure 8). The overall heterogeneity is substantial and significant ($I^2 = 89\%$, $p < 0.01$). Using a random-effects model, the pooled estimate of all RCTs was calculated as -1.71 (95% CI: $[-2.73, -0.69]$), showing that the interventions yielded a significant, medium-to-large treatment effect on caregiver burden. However, PIs overlapping zero indicated potential non-treatment effects for future studies due to heterogeneity. Subgroup differences between intervention types were not significant, probably due to too few studies included for each type.

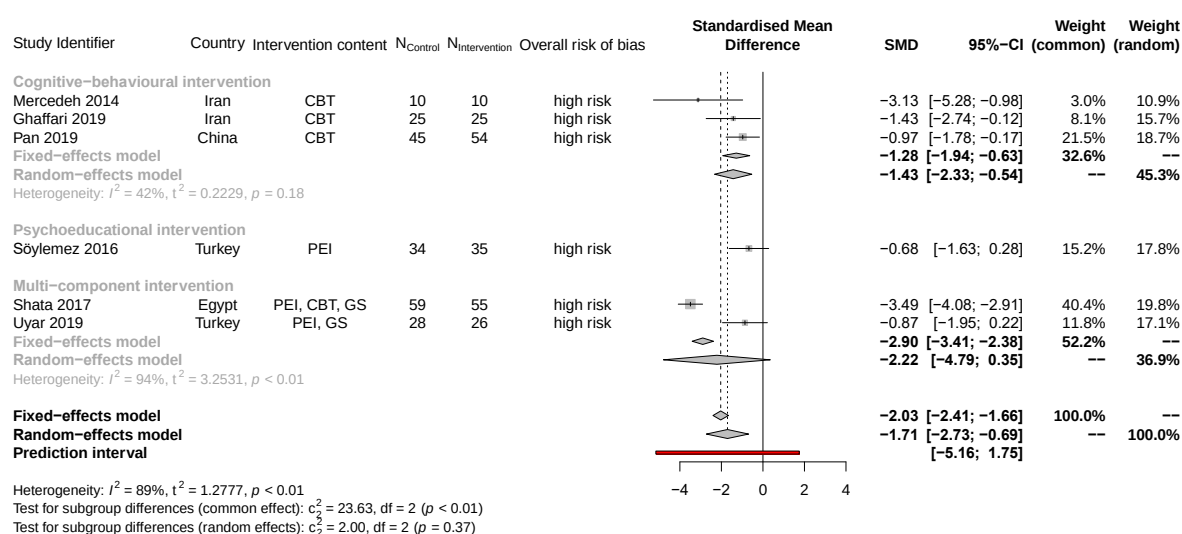


Figure 8 Forest plot of meta-analysis of effects on caregiver depression

Note: The fixed effect(s) is also occasionally described as the common effect in the figure. Though we performed meta-analysis using a random-effects model, the fix-effects estimates were also given in the figure to detect potential small-study effects, which is obvious by visually inspecting the differences in the figure. As shown in the figure, the fixed-effects estimate is larger than the random-effects estimate because the random-effects model weighs more on smaller studies with a smaller effect size.

The leave-one-out analysis showed that this result remained significant when omitting any single study (Figure 9). After excluding the most influential case of Shata 2017 (Figure 10),

the heterogeneity dropped to 14.0%, with the summative effect size reduced to -1.05 (95% CI: [-1.53, -0.56]). This was consistent with the excluded study, which reported large effects for all outcomes but a larger effect on caregiver depression than on caregiver burden. This also contributes to the discrepancy between fixed- and random-effects estimates suggested a small-study effect, where larger studies tend to have larger effect sizes (Figure 8).

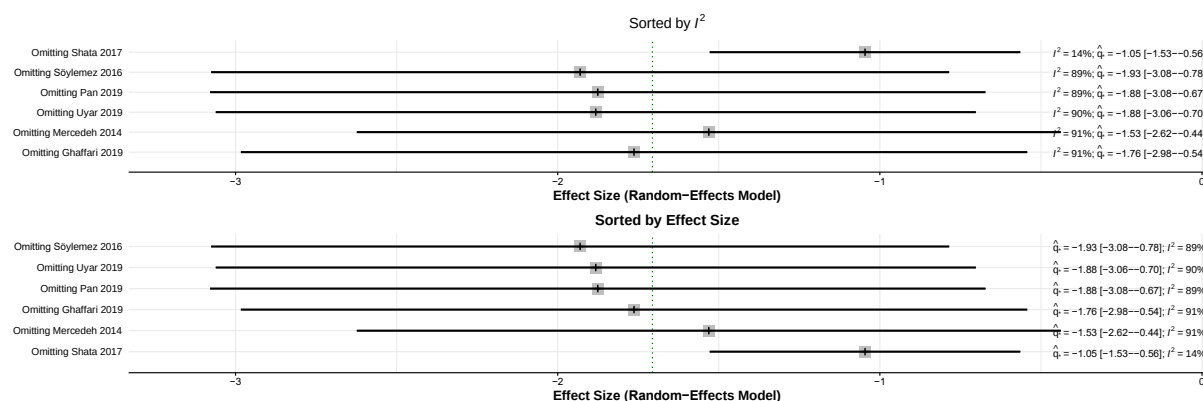


Figure 9 Leave-one-out analysis of effects on caregiver depression

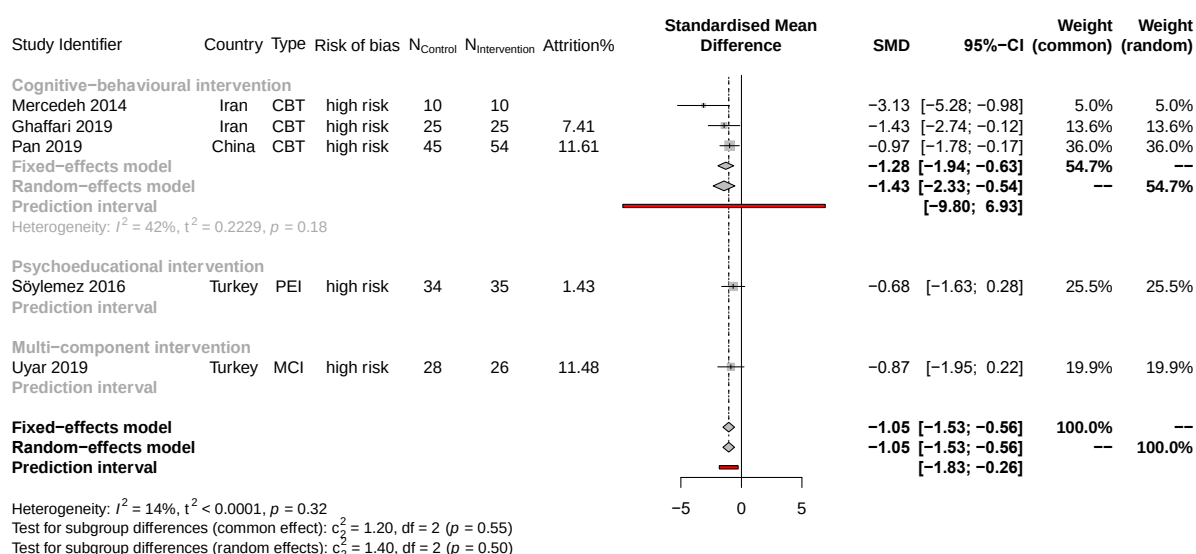


Figure 10 Forest plot of meta-analysis of effects on caregiver depression excluding the outlier

3.5.3 Other outcomes

Meta-analyses for other outcomes were unintended but included in Appendix 6 and Figure 11, which all showed significant improvements. However, caregiver distress was not robust as shown in the leave-one-out analysis. Although the effects on patient outcomes were highly homogeneous, they were often measured in studies only measuring patient outcomes, thus not generalisable to studies measuring other outcomes.

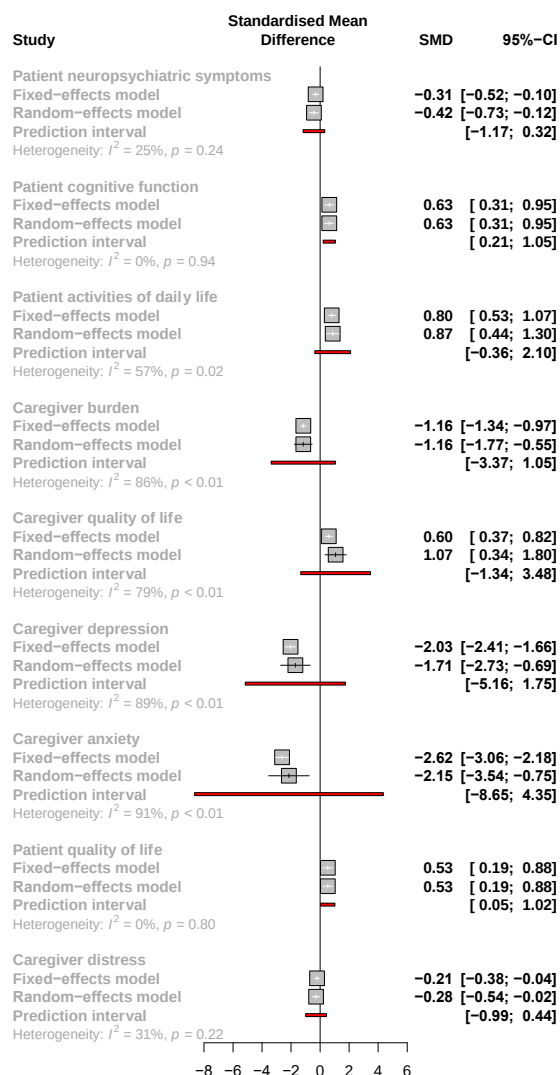


Figure 11 Summary of meta-analysis of effects on all outcomes (see also Appendix 6)

4 DISCUSSION

4.1 Summary of findings

Our systematic review shows that caregiver interventions significantly improved caregiver burden and depression in LMICs, with medium-to-large effect sizes, which contrasted previous meta-analyses showing only small-to-medium effects. However, among the five included studies reporting effect sizes,^{14,15,27,42,56} three reported larger effect sizes than the similar trials in HICs,^{15,42,56} while three studies attributed larger effect sizes to poor quality of care in LMICs.^{15,27,42} As our supplementary analysis further showed the significant treatment effects on all three caregiver outcomes and all four care recipient outcomes, our review provided robust evidence to improve care quality with training and support for caregivers in LMICs.

4.2 Strengths and weaknesses

This study is among the first meta-analyses for caregiver interventions in all LMICs, providing evidence for policymakers and health providers in LMICs to improve care quality and caregiver health with caregiver interventions. With 9 Chinese-language studies included and the outcome measurements beyond caregiver burden and depression explored, we expanded the evidence to broader literature, thus providing more comprehensive insights into the effects of caregiver interventions. Also, we used the intervention types defined in previous meta-analyses. Thus, our results were comparable with the results of these studies.

However, as most included studies in this review were considered to have a high risk of bias, the effects in this study may be exaggerated.^{76–78} We also recognised that our methods might be underpowered, especially for subgroup analysis with exploratory variables.⁷⁹

As each meta-analysis contained different interventions, the results may not be generalised to all interventions. Cheng and Zhang (2020)¹⁷ suggested that different interventions had different effects on different outcomes. An intervention that can improve caregiver burden may not be beneficial to dementia patients and *vice versa*. Thus, outcome measurements in future studies should not be limited to caregiver burden and depression to enable more comprehensive evaluation of the effectiveness of caregiver interventions.

Our review may not represent wider populations in LMICs, as we did not include data from low-income countries, although we searched the literature for low-income countries. 2/3 of the included studies were conducted in China and Iran. This is consistent with the lack of research from low-income countries, which published only 0.6% of the world's health research.⁸⁰ Also, as most participants were conveniently sampled from hospital admissions, our results may have low generalisability to the wider population in the same country due to selection bias,⁸¹ especially when considering those living in rural and remote areas where medical resources were often inaccessible.⁸²

Compared with systematic reviews by Hinton et al. (2019)¹⁹ and Wu et al. (2020),¹⁸ we quantitatively synthesised the evidence from LMICs with meta-analysis. Compared with previous comprehensive meta-analyses by Walter and Pinquart (2020)¹² and Cheng and Zhang (2020),¹⁷ and a recent network meta-analysis by Sun et al. (2022),⁸³ our review exclusively

focused on LMICs and included literature from more LMICs. However, we did not evaluate the comparative effectiveness of each intervention type with network meta-analysis.⁸⁴

4.3 Future directions

Despite overall effectiveness of caregiver interventions, their scalability remains unclear. Most meta-analysed outcomes had PIs overlapping zero, suggesting potential non-treatment effect due to the heterogeneity, of which the sources require future research. Considering cultural differences in caregiving affected the scalability,^{42,56} further trials need to be done to test their effectiveness across settings. Also, cultural tailoring will be needed in some interventions,^{28,85} e.g. those designed for Buddhists and Muslims,^{28,48,51} when tested in different settings. Besides, more holistic approaches including both caregiver and patient outcomes should be used in future trials, to make outcome measurements more comparable.²²

James et al. (2021)⁸⁶ systematically reviewed culturally tailored caregiver interventions for LMICs and ethnic minorities, suggesting that the importance of keeping the therapeutic parts of these interventions during culture tailoring. However, our current evidence for what works in an intervention may be inconclusive. For example, Pan et al. (2019) found that the improvements in caregiver outcomes were not mediated by mutuality and coping behaviour as expected, while other components in the CBT may contribute to the effectiveness. Thus, clinical trials and network meta-analyses comparing multiple interventions are still needed to identify the most effective interventions and therapeutic components.⁸⁷

Different intervention classification may lead to different results and interpretations.^{12,17,83} Thus, it is meaningful to develop a cumulative meta-analysis to update the same meta-analysis continuously.⁸⁸ Furthermore, considering the fast-growing population of PwD and the lack of evidence for caregiver interventions in LMICs, it is also promising to devise a prospective meta-analysis for similar interventions,⁸⁹ which enables patient-level data collection and provides more accurate effect size estimation.^{90,91}

5 CONCLUSIONS

This review shows that caregiver interventions were feasible and effective for improving the health of both PwD caregivers in LMICs, providing evidence for providing caregiver support in LMICs. Future research needs to be done to measure the comparative effectiveness of the

interventions, identify the therapeutic components in each intervention and test the scalability of the interventions to wider populations in LMICs.

6 ACKNOWLEDGEMENT

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APPENDIX 1 PRISMA CHECKLIST

Section	Item #	Checklist item	Location
TITLE			
Title	1	Identify the report as a systematic review.	Page 1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	8
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	8
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	9
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	9
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	39
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	9
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	10
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	10
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	10
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	11
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	15
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	11
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	11
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	11
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	11
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	11
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	11
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	95% CI and PI reported in Section 3.5
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of	12

		records identified in the search to the number of studies included in the review, ideally using a flow diagram.	
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	12
Study characteristics	17	Cite each included study and present its characteristics.	12
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	18
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	See forest plots in Section 3.5 and Appendix 8
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Section 3.5
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Section 3.5 and Appendix 6 and 7
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Appendix 6, Appendix 7
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	Leave-one-out analysis in Section 3.5
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Domain 3 of RoB2 in Appendix 5, Figure 2, Figure 3
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Section 3.5
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	26
	23b	Discuss any limitations of the evidence included in the review.	26
	23c	Discuss any limitations of the review processes used.	26
	23d	Discuss implications of the results for practice, policy, and future research.	26
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	N/A
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	N/A
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	N/A
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	8
Competing interests	26	Declare any competing interests of review authors.	28
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	- Data available in Appendix 4 - Code available in the Attachment

APPENDIX 2 SEARCH STRATEGY

Appendix 2.1 Summary

	Condition	Ovid Medline	Embase	Global Health	APA PsycINFO	EBSCO CINAHL	Total
1	Include dementia-related terms	427238	723805	34331	185529	159287	
2	Include therapy-related terms	13930216	18103622	2046951	2371714	3652534	
3	Include RCT-related terms	5552799	5229127	632419	755859	982366	
4	Include LMIC-related terms	4046862	4706713	1683278	504266	705117	
5	Satisfy Conditions 1, 2, 3 & 4	7917	7967	1911	2359	1917	
6	Nonhuman experiments		31095663				
7	Human experiments		24104793				
8	Condition 6 results excluding Condition 7 results	5049406	6992081		379966	210517	
9	Condition 5 results excluding Condition 8 results	6646	7556		2301	1819	
10	Publication type is Editorial or Letter	2097191	1977082				
11	Condition 9 results excluding Condition 10 results	6594	7540				
12	Year of publication limited to 2019-2020	1156	1247	382	404	448	
	Total results:	1156	1247	382	404	448	
	Duplicates:	(0)	(357)	(143)	(212)	(272)	(984)
	Results for screening:	1156	890	239	192	176	2653

Searches updated and run by Andra Fry on 21/09/2022.

Appendix 2.2 Ovid MEDLINE(R)

Ovid MEDLINE(R) ALL <1946 to September 20, 2022>		
Condition		Hit
1	exp Dementia/ or Pick Disease of the Brain/ or Huntington Disease/ or Lewy Body Disease/ or Cognitive Dysfunction/ or (dement* or amentia* or (major adj3 cognit* adj3 disorder*) or alzheimer* or alzeimer* or (cortical adj4 sclerosis) or ((encephalopath* or cogniti* or neurocogniti*) adj4 (aids or acquired immun?deficiency syndrome* or acquired immun? deficiency syndrome* or hiv or human immun?deficiency virus* or human immun? deficiency virus*)) or pick* disease* or (lobar adj3 atroph* adj3 brain) or Huntington* disease* or Huntington* chorea or (Lewy bod* adj3 disease*) or (cerebr* adj3 deteriorat*) or (cerebr* adj3 insufficien*) or ((frontotemporal or fronto temporal or corticobasal or cortico basal or frontal lobe*) adj4 (degenerati* or dysfunction*)) or ((cognit* or memory or cerebr*) adj3 (declin* or impair* or los* or deteriorat* or degenerat* or insufficen*)) or MCI or (mild adj2 cognit* impair*).ti,ab,kf,kw.	427238
2	exp Cognitive Therapy/ or exp Drug Therapy/ or exp Cholinesterase Inhibitors/ or exp Antipsychotic Agents/ or exp Serotonin Uptake Inhibitors/ or exp Benzodiazepines/ or exp "Hypnotics and Sedatives"/ or exp Exercise/ or exp Exercise Therapy/ or exp Psychotherapy/ or exp Counseling/ or Psychosocial Support Systems/ or exp Complementary Therapies/ or Phototherapy/ or exp Advance Care Planning/ or Case Management/ or Caregivers/ed, px or Self-Help Groups/ or exp Social Support/ or Diagnosis, Computer-Assisted/ or Telemedicine/ or exp Computers, Handheld/ or (intervention* or therap* or treatment* or program* or manage* or prevent* or diagnos* or polic* or (cognit* adj3 (therap* or training or rehab*)) or cognit* psycho therap* or cognit* psychotherap* or drug* or medicine* or pharmacotherap* or pharmaco* therap* or cholinesterase agent* or cholinesterase inhibitor* or (tranquili* adj3 (agent* or drug*)) or antipsychotic* or (neuroleptic adj3 (agent* or drug*)) or serotonin uptake inhibitor* or serotonin reuptake inhibitor* or ssri* or benzodiazepine* or (sedative adj3 (effect* or agent*)) or memantine or donepezil or rivastigmine or galantamine or souvenaid or risperidone or haloperidol or olanzapine or quetiapine or citalopram or dextromethorphan or carbamazepine or mirtazapine or sertraline or moclobemide or trazodone or melatonin or ramelteon or methylphenidate or exercis* or physical activit* or physical training or aerobic* or arobic* or kinesi?therap* or kinesi? therap* or (social adj3 (activit* or engag* or stimul*)) or psycholog* therap* or psychotheran* or ((behavio?* or conditioning) adj3 therap*) or	13930216

	counsel?ing or ((psychosocial or psycho social) adj3 (support or interven* or care)) or ((alternative or compl?ment* or traditional) adj3 (medicine* or therap*)) or acupunct* or (herb* adj3 (tea* or remedy or remedies or medicine*)) or ginkgo or homeopath* or ((music or art or aroma or light or photo or pet or pets) adj3 therap*) or aromatherap* or phototherap* or massage* or (mind adj3 body) or (advance? adj3 (care or medical or healthcare) adj3 plan*) or (decision* adj3 (aid* or support)) or (communicati* adj3 skill* adj3 training) or (dementia care adj3 map*) or ((person* or patient*) adj3 cent* adj3 care) or "Resources for Enhancing Alzheimer's Caregiver Health" or "Strategies for Relatives" or ((caregiver* or carer*) adj3 educat*) or telemedicine or tele medicine or (smart adj2 (phone* or device* or tablet*)) or smartphone* or cognitive aid* or reminder* or robot*).ti,ab,kf,kw.	
3	(Randomized Controlled Trial or Controlled Clinical Trial).pt. or (randomi#ed or placebo or randomly or trial or groups).ti,ab. or drug therapy.fs.	5552799
4	Developing Countries.sh,kf,kw. or (Africa* or Asia* or Caribbean* or West Indies or South America* or Latin America* or Central America* or Afghan* or Albania* or Algeria* or Angola* or Anguilla* or Antigua* or Barbuda* or Argentin* or Armenia* or Azerbaijan* or Azeri or Bangladesh* or Barbad* or Benin* or Byelarus or Byelorussian or Belarus* or Belorussia* or Belize* or Bhutan* or Bolivia* or Bosnia* or Herzegovin* or Hercegovin* or Botswana or Botsuana or Motswana or Batswana or Brasil* or Brazil* or Burkina Faso or Burkina Fasso or Burkina* or Burundi* or Urundi* or Cambodia* or Cameroon* or Cameron* or Cape Verd* or Cabo Verde or Central African Republic or Chad* or Tchad* or Chile* or China or Chinese or Colombia* or Columbia* or Comoros or Comoro Islands or Comores or Comoran or Mayotte or Congo* or Costa Rica* or Cote d'Ivoire or Ivory Coast or Ivorian* or Cook Islands or Cuba* or Croat* or Djibouti* or Dominica* or East Timor or East Timur or Timor Leste or Timorese or Ecuador* or Equador* or Egypt* or El Salvador or Salvadoran or Eritrea* or Ethiopia* or Fiji* or Gabon* or Gambia* or Gaza or Georgia Republic or Georgian or Abkhazia* or Abchasia* or South Ossetia* or Ghana* or Grenada or Grenadian or Guatemala* or Guinea* or Guinea Bissau or Guian* or Guyana or Haiti* or Hondura* or India or Indian or Indonesia* or Iran* or Iraq* or Jamaica* or Jordan* or Kazakhstan* or Kazakh or Kenya* or Kiribati or Korea* or Kosovo or Kosova* or Kyrgyzstan or Kirghizia or Kyrgyz or Kirghiz or Kirgizstan or Lao PDR or Laos or Laotian or Lebanon or Lebanese or Lesotho or Mosotho or Basotho or Liberia* or Libya* or Macedonia* or FYROM or Madagasca* or Malagasy or Malaysia* or Malaya* or Malay or Sabah or Sarawak or Malawi* or Maldives or Maldivan or Mali or Malian or Marshall Islands or Marshallese or Mauritania* or Mauriti* or Agalega Islands or Mexico or Mexican or Micronesia* or Middle East* or Moldova* or Moldovia* or Transnistria* or Mongolia* or Montenegr* or Montserrat* or Morocco* or Mozambique or Mozambican or Myanmar* or Myanma or Burma or Burmese or Namibia* or Nauru* or Niue or Nepal* or Nicaragua* or Niger or Nigerien or Nigeria* or Oman* or Pakistan* or Palau* or Palestine or Palestinian or Panama or Panamanian or Paraguay* or Papua New Guinea* or Peru or Peruvian or Philippines or Philipines or Phillipines or Phillippines or Filipino or Philipino or Philippino or Phillipino or Phillippino or Rwanda* or Ruanda* or Saint Helen* or St Helen* or Saint Kitts or St Kitts or Kittian or Nevis* or Saint Lucia* or St Lucia* or Saint Vincent or St Vincent or Vicentian or Grenadines or Samoa* or Sao Tome* or Senegal* or Serbia* or Seychell* or Sierra Leone* or Sri Lanka* or Ceylon or Solomon Island* or Somali* or South Africa* or Sudan or Sudanese or Surinam* or Swaziland or Swazi or Eswatini or Syria or Syrian or Tajikistan or Tadjhikistan or Tadjikistan or Tajik or Tadjhik or Tanzania* or Thailand or Thai or Togo or Togolese or Tonga* or Tunisia* or Tokelau or Trinidad* or Tobago* or Turkey or Turkish or Turks or Turkmenistan or Turkmen or Tuvalu* or Uganda* or Ukraine or Ukrainian or Uruguay* or Uzbekistan* or Uzbek or Vanuatu or Venezuela* or Vietnam* or Viet Nam or Wallis Futuna or West Bank or Yemen* or Zambia* or Zimbabwe*).hw,kf,kw,ti,ab,cp. or (((developing or less* developed or under developed or underdeveloped or middle income or low* income or underserved or under served or deprived or poor*) adj (count* or nation? or population? or world)) or ((developing or less* developed or under developed or underdeveloped or middle income or low* income) adj (economy or economies)) or (low* adj (gdp or gnp or gross domestic or gross national)) or (low adj3 middle adj3 countr*) or lmic or lmic3 or third world or lami countr* or transitional countr*).ti,ab.	4046862
5	1 and 2 and 3 and 4	7917
6	exp Animals/ not Humans/	5049406
7	5 not 6	6646
8	(Comment or Editorial or Letter).pt.	2097191
9	7 not 8	6594
10	limit 9 to yr="2019 - 2020"	1156

Appendix 2.3 Embase

Embase <1974 to 2022 September 20>	
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	Condition	Hit
1	exp Dementia/ or Cognitive defect/ or (dement* or amentia* or (major adj3 cognit* adj3 disorder*) or alzheimer* or alzeimer* or (cortical adj4 sclerosis) or ((encephalopath* or cogniti* or neurocogniti*) adj4 (aids or acquired immun?deficiency syndrome* or acquired immun? deficiency syndrome* or hiv or human immun?deficiency virus* or human immun? deficiency virus*)) or pick* disease* or (lobar adj3 atroph* adj3 brain) or Huntington* disease* or Huntington* chorea or (Lewy bod* adj3 disease*) or (cerebr* adj3 deteriorat*) or (cerebr* adj3 insufficien*) or ((frontotemporal or fronto temporal or corticobasal or cortico basal or frontal lobe*) adj4 (degenerati* or dysfunction*)) or ((cognit* or memory or cerebr*) adj3 (declin* or impair* or los* or deteriorat* or degenerat* or insufficien*)) or MCI or (mild adj2 cognit* impair*).ti,ab,kf,kw.	723805
2	exp Cognitive Therapy/ or exp Drug Therapy/ or exp Cholinesterase Inhibitors/ or exp Neuroleptic Agent/ or exp Serotonin Uptake Inhibitor/ or exp Benzodiazepine Derivative/ or Sedative Agent/ or exp Exercise/ or exp Kinesiotherapy/ or exp Psychotherapy/ or exp Counseling/ or Psychosocial Care/ or exp Alternative Medicine/ or Phototherapy/ or Advance Care Planning/ or Living Will/ or Case Management/ or Caregiver Support/ or Self Help/ or exp Social Support/ or Computer Assisted Diagnosis/ or Telemedicine/ or Telemonitoring/ or Telerehabilitation/ or Teliagnosis/ or Personal Digital Assistant/ or Smartphone/ or (intervention* or therap* or treatment* or program* or manage* or prevent* or diagnos* or polic* or (cognit* adj3 (therap* or training or rehab*)) or cognit* psycho therap* or cognit* psychotherap* or drug* or medicine* or pharmacotherap* or pharmaco* therap* or cholinesterase agent* or cholinesterase inhibitor* or (tranquili* adj3 (agent* or drug*)) or antipsychotic* or (neuroleptic adj3 (agent* or drug*)) or serotonin uptake inhibitor* or serotonin reuptake inhibitor* or ssri* or benzodiazepine* or (sedative adj3 (effect* or agent*)) or memantine or donepezil or rivastigmine or galantamine or souvenaid or risperidone or haloperidol or olanzapine or quetiapine or citalopram or dextromethorphan or carbamazepine or mirtazapine or sertraline or moclobemide or trazodone or melatonin or ramelteon or methylphenidate or exercis* or physical activit* or physical training or aerobic* or arobic* or kinesio?therap* or kinesio? therap* or (social adj3 (activit* or engag* or stimul*)) or psycholog* therap* or psychotherap* or ((behavio?r* or conditioning) adj3 therap*) or counsel?ing or ((psychosocial or psycho social) adj3 (support or interven* or care)) or ((alternative or compl?ment* or traditional) adj3 (medicine* or therap*)) or acupunct* or (herb* adj3 (tea* or remedy or remedies or medicine*)) or ginkgo or homeopath* or ((music or art or aroma or light or photo or pet or pets) adj3 therap*) or aromatherap* or phototherap* or massage* or (mind adj3 body) or (advance? adj3 (care or medical or healthcare) adj3 plan*) or (decision* adj3 (aid* or support)) or (communicati* adj3 skill* adj3 training) or (dementia care adj3 map*) or ((person* or patient*) adj3 cent* adj3 care) or "Resources for Enhancing Alzheimer's Caregiver Health" or "Strategies for Relatives" or ((caregiver* or carer*) adj3 educat*) or telemedicine or tele medicine or (smart adj2 (phone* or device* or tablet*)) or smartphone* or cognitive aid* or reminder* or robot*).ti,ab,kf,kw.	18103622
3	(Randomized Controlled Trial/ or Controlled Clinical Trial/ or random*.ti,ab. or Randomization/ or Intermethod Comparison/ or placebo.ti,ab. or (compare or compared or comparison).ti. or ((evaluated or evaluate or evaluating or assessed or assess) and (compare or compared or comparing or comparison)).ab. or (open adj label).ti,ab. or ((double or single or doubly or singly) adj (blind or blinded or blindly)).ti,ab. or Double Blind Procedure/ or parallel group*1.ti,ab. or (crossover or cross over).ti,ab. or ((assign* or match or matched or allocation) adj5 (alternate or group*1 or intervention*1 or patient*1 or subject*1 or participant*1)).ti,ab. or (assigned or allocated).ti,ab. or (controlled adj7 (study or design or trial)).ti,ab. or (volunteer or volunteers).ti,ab. or Human Experiment/ or trial.ti.) not (((random* adj sampl* adj7 (cross section* or questionnaire*1 or survey* or database*1)).ti,ab. not (Comparative Study/ or Controlled Study/ or randomi?ed controlled.ti,ab. or randomly assigned.ti,ab.)) or (Cross-Sectional Study/ not (Randomized Controlled Trial/ or Controlled Clinical Study/ or Controlled Study/ or randomi?ed controlled.ti,ab. or control group*1.ti,ab.)) or (((case adj control*) and random*) not randomi?ed controlled).ti,ab. or (systematic review not (trial or study)).ti. or (nonrandom* not random*).ti,ab. or random field*.ti,ab. or (random cluster adj3 sampl*).ti,ab. or ((review.ab. and review.pt.) not trial.ti.) or (we searched.ab. and (review.ti. or review.pt.)) or update review.ab. or (databases adj4 searched).ab. or ((rat or rats or mouse or mice or swine or porcine or murine or sheep or lambs or pigs or piglets or rabbit or rabbits or cat or cats or dog or dogs or cattle or bovine or monkey or monkeys or trout or marmoset*1).ti. and Animal Experiment/) or (Animal Experiment/ not (Human Experiment/ or Human/)))	5229127
4	Developing Countries.sh,kf,kw. or (Africa* or Asia* or Caribbean* or West Indies or South America* or Latin America* or Central America* or Afghan* or Albania* or Algeria* or Angola* or Anguilla* or Antigua* or Barbuda* or Argentina* or Armenia* or Azerbaijan* or Azeri or Bangladesh* or Barbad* or Benin* or Byelarus or Byelorussian or Belarus* or Belorussia* or Belize* or Bhutan* or Bolivia* or Bosnia* or Herzegovin* or Hercegovin* or Botswana or Botsuana or Motswana or Batswana or Brasil* or Brazil* or Burkina Faso or Burkina Fasso or Burkina* or Burundi* or Urundi* or Cambodia* or Cameroon* or Cameron* or Cape Verd* or Cabo Verde or Central African Republic or Chad* or Tchad* or Chile* or China or Chinese or Colombia* or Columbia* or Comoros or Comoro Islands or Comores or Comoran or Mayotte or Congo* or Costa Rica* or Cote d'Ivoire or Ivory Coast or Ivorian* or Cook Islands or Cuba* or Croat* or Djibouti* or	4706713

	Dominica* or East Timor or East Timur or Timor Leste or Timorese or Ecuador* or Equador* or Egypt* or El Salvador or Salvadoran or Eritrea* or Ethiopia* or Fiji* or Gabon* or Gambia* or Gaza or Georgia Republic or Georgian or Abkhazia* or Abchasia* or South Ossetia* or Ghana* or Grenada or Grenadian or Guatemala* or Guinea* or Guinea Bissau or Guian* or Guyana or Haiti* or Hondura* or India or Indian or Indonesia* or Iran* or Iraq* or Jamaica* or Jordan* or Kazakhstan* or Kazakh or Kenya* or Kiribati or Korea* or Kosovo or Kosova* or Kyrgyzstan or Kirghizia or Kyrgyz or Kirghiz or Kirgizstan or Lao PDR or Laos or Laotian or Lebanon or Lebanese or Lesotho or Mosotho or Basotho or Liberia* or Libya* or Macedonia* or FYROM or Madagasca* or Malagasy or Malaysia* or Malaya* or Malay or Sabah or Sarawak or Malawi* or Maldives or Maldivan or Mali or Malian or Marshall Islands or Marshallese or Mauritania* or Mauriti* or Agalega Islands or Mexico or Mexican or Micronesia* or Middle East* or Moldova* or Moldovia* or Transnistria* or Mongolia* or Montenegro* or Montserrat* or Morocco* or Mozambique or Mozambican or Myanmar* or Myanma or Burma or Burmese or Namibia* or Nauru* or Niue or Nepal* or Nicaragua* or Niger or Nigerien or Nigeria* or Oman* or Pakistan* or Palau* or Palestine or Palestinian or Panama or Panamanian or Paraguay* or Papua New Guinea* or Peru or Peruvian or Philippines or Philipines or Phillipines or Phillippines or Filipino or Philipino or Philippino or Phillipino or Phillippino or Rwanda* or Ruanda* or Saint Helen* or St Helen* or Saint Kitts or St Kitts or Kittian or Nevis* or Saint Lucia* or St Lucia* or Saint Vincent or St Vincent or Vicentian or Grenadines or Samoa* or Sao Tome* or Senegal* or Serbia* or Seychell* or Sierra Leone* or Sri Lanka* or Ceylon or Solomon Island* or Somali* or South Africa* or Sudan or Sudanese or Surinam* or Swaziland or Swazi or Eswatini or Syria or Syrian or Tajikistan or Tadjikistan or Tadjikistan or Tajik or Tadjik or Tanzania* or Thailand or Thai or Togo or Togolese or Tonga* or Tunisia* or Tokelau or Trinidad* or Tobago* or Turkey or Turkish or Turks or Turkmenistan or Turkmen or Tuvalu* or Uganda* or Ukraine or Ukrainian or Uruguay* or Uzbekistan* or Uzbek or Vanuatu or Venezuela* or Vietnam* or Viet Nam or Wallis Futuna or West Bank or Yemen* or Zambia* or Zimbabwe*).hw,kf,kw,ti,ab,cp. or (((developing or less* developed or under developed or underdeveloped or middle income or low* income or underserved or under served or deprived or poor*) adj (countr* or nation? or population? or world)) or ((developing or less* developed or under developed or underdeveloped or middle income or low* income) adj (economy or economies)) or (low* adj (gdp or gnp or gross domestic or gross national)) or (low adj3 middle adj3 countr*) or lmic or lmic or third world or lami countr* or transitional countr*).ti,ab.	
5	1 and 2 and 3 and 4	7967
6	exp Animal/ or exp Animal Experiment/ or Nonhuman/	31095663
7	exp Human/ or exp Human Experiment/	24104793
8	6 not 7	6992081
9	5 not 8	7556
10	(Editorial or Letter).pt.	1977082
11	9 not 10	7540
12	limit 11 to yr="2019 - 2020"	1247

Appendix 2.4 Global Health

Global Health <1973 to 2022 Week 37>		
Condition		Hit
1	exp Dementia/ or (dement* or amentia* or (major adj3 cognit* adj3 disorder*) or alzheimer* or alzeimer* or (cortical adj4 sclerosis) or ((encephalopath* or cogniti* or neurocogniti*) adj4 (aids or acquired immun?deficiency syndrome* or acquired immun? deficiency syndrome* or hiv or human immun?deficiency virus* or human immun? deficiency virus*)) or pick* disease* or (lobar adj3 atroph* adj3 brain) or Huntington* disease* or Huntington* chorea or (Lewy bod* adj3 disease*) or (cerebr* adj3 deteriorat*) or (cerebr* adj3 insufficien*) or ((frontotemporal or fronto temporal or corticobasal or cortico basal or frontal lobe*) adj4 (degenerati* or dysfunction*)) or ((cognit* or memory or cerebr*) adj3 (declin* or impair* or los* or deteriorat* or degenerat* or insufficen*)) or MCI or (mild adj2 cognit* impair*).ti,ab.	34331
2	exp Drug Therapy/ or exp Cholinesterase Inhibitors/ or exp Neuroleptics/ or exp Selective Serotonin Reuptake Inhibitors/ or exp Benzodiazepines/ or exp Sedatives/ or Exercise/ or Physical Therapy/ or exp Psychotherapy/ or exp Counselling/ or "Complementary and Alternative Medicine"/ or Phototherapy/ or Self Help/ or Self Care/ or exp E-health/ or Mobile Telephones/ or (intervention* or therap* or treatment* or program* or manage* or prevent* or diagnos* or polic* or (cognit* adj3 (therap* or training* or rehab*)) or cognit* psycho therap* or cognit* psychotherap* or drug* or medicine* or pharmacotherap* or pharmaco therap* or cholinesterase agent* or cholinesterase inhibitor* or (tranquili* adj3 (agent* or drug*)) or antipsychotic* or (neuroleptic adj3 (agent* or	2046951

	drug*)) or serotonin uptake inhibitor* or serotonin reuptake inhibitor* or ssri* or benzodiazepine* or (sedative adj3 (effect* or agent*)) or memantine or donepezil or rivastigmine or galantamine or souvenaid or risperidone or haloperidol or olanzapine or quetiapine or citalopram or dextromethorphan or carbamazepine or mirtazapine or sertraline or moclobemide or trazodone or melatonin or ramelteon or methylphenidate or exercis* or physical activit* or physical training or aerobic* or arobic* or kinesi?therap* or kinesi? therap* or (social adj3 (activit* or engag* or stimul*)) or psycholog* therap* or psychotherap* or ((behavio?r* or conditioning) adj3 therap*) or counsel?ing or ((psychosocial or psycho social) adj3 (support or interven* or care)) or ((alternative or compl?ment* or traditional) adj3 (medicine* or therap*)) or acupunct* or (herb* adj3 (tea* or remedy or remedies or medicine*)) or ginkgo or homeopath* or ((music or art or aroma or light or photo or pet or pets) adj3 therap*) or aromatherap* or phototherap* or massage* or (mind adj3 body) or (advance? adj3 (care or medical or healthcare) adj3 plan*) or (decision* adj3 (aid* or support)) or (communicati* adj3 skill* adj3 training) or (dementia care adj3 map*) or ((person* or patient*) adj3 cent* adj3 care) or "Resources for Enhancing Alzheimer's Caregiver Health" or "Strategies for Relatives" or ((caregiver* or carer*) adj3 educat*) or telemedicine or tele medicine or (smart adj2 (phone* or device* or tablet*)) or smartphone* or cognitive aid* or reminder* or robot*).ti,ab.	
3	Randomized Controlled Trials/ or (randomi#ed or placebo or randomly or trial or groups).ti,ab.	632419
4	(Low Income Countries or Lower-Middle Income Countries or Least Developed Countries).sh. or (Africa* or Asia* or Caribbean* or West Indies or South America* or Latin America* or Central America* or Afghan* or Albania* or Algeria* or Angola* or Anguilla* or Antigua* or Barbuda* or Argentin* or Armenia* or Azerbaijan* or Azeri or Bangladesh* or Barbad* or Benin* or Byelarus or Byelorussian* or Belarus* or Belorussia* or Belize* or Bhutan* or Bolivia* or Bosnia* or Herzegovin* or Hercegovin* or Botswana or Botsuana or Motswana or Batswana or Brasil* or Brazil* or Burkina Faso or Burkina Fasso or Burkina* or Burundi* or Urundi* or Cambodia* or Cameroon* or Cameron* or Cape Verd* or Cabo Verde or Central African Republic or Chad* or Tchad* or Chile* or China or Chinese or Colombia* or Columbia* or Comoros or Comoro Islands or Comores or Comoran or Mayotte or Congo* or Costa Rica* or Cote d'Ivoire or Ivory Coast or Ivorian* or Cook Islands or Cuba* or Croat* or Djibouti* or Dominica* or East Timor or East Timur or Timor Leste or Timorese or Ecuador* or Equador* or Egypt* or El Salvador or Salvadoran or Eritrea* or Ethiopia* or Fiji* or Gabon* or Gambia* or Gaza or Georgia Republic or Georgian or Abkhazia* or Abchasia* or South Ossetia* or Ghana* or Grenada or Grenadian or Guatemala* or Guinea* or Guinea Bissau or Guian* or Guyana or Haiti* or Hondura* or India or Indian or Indonesia* or Iran* or Iraq* or Jamaica* or Jordan* or Kazakhstan* or Kazakh or Kenya* or Kiribati or Korea* or Kosovo or Kosova* or Kyrgyzstan or Kirghizia or Kyrgyz or Kirghiz or Kirgizstan or Lao PDR or Laos or Laotian or Lebanon or Lebanese or Lesotho or Mosotho or Basotho or Liberia* or Libya* or Macedonia* or FYROM or Madagasca* or Malagasy or Malaysia* or Malaya* or Malay or Sabah or Sarawak or Malawi* or Maldives or Maldivan or Mali or Malian or Marshall Islands or Marshallese or Mauritania* or Mauriti* or Agalega Islands or Mexico or Mexican or Micronesia* or Middle East* or Moldova* or Moldovia* or Transnistria* or Mongolia* or Monteneg* or Montserrat* or Morocc* or Mozambique or Mozambican or Myanmar* or Myanma or Burma or Burmese or Namibia* or Nauru* or Niue or Nepal* or Nicaragua* or Niger or Nigerien or Nigeria* or Oman* or Pakistan* or Palau* or Palestine or Palestinian or Panama or Panamanian or Paraguay* or Papua New Guinea* or Peru or Peruvian or Philippines or Philipines or Phillipines or Phillippines or Filipino or Philipino or Philippino or Phillipino or Rwanda* or Ruanda* or Saint Helen* or St Helen* or Saint Kitts or St Kitts or Kittian or Nevis* or Saint Lucia* or St Lucia* or Saint Vincent or St Vincent or Vicentian or Grenadines or Samoa* or Sao Tome* or Senegal* or Serbia* or Seychell* or Sierra Leone* or Sri Lanka* or Ceylon or Solomon Island* or Somali* or South Africa* or Sudan or Sudanese or Surinam* or Swaziland or Swazi or Eswatini or Syria or Syrian or Tajikistan or Tadjhikistan or Tadjikistan or Tajik or Tadjhik or Tanzania* or Thailand or Thai or Togo or Togolese or Tonga* or Tunisia* or Tokelau or Trinidad* or Tobago* or Turkey or Turkish or Turks or Turkmenistan or Turkmen or Tuvalu* or Uganda* or Ukraine or Ukrainian or Uruguay* or Uzbekistan* or Uzbek or Vanuatu or Venezuela* or Vietnam* or Viet Nam or Wallis Futuna or West Bank or Yemen* or Zambia* or Zimbabwe*).hw,ti,ab,cp. or (((developing or less* developed or under developed or underdeveloped or middle income or low* income or underserved or under served or deprived or poor*) adj (countr* or nation? or population? or world)) or ((developing or less* developed or under developed or underdeveloped or middle income or low* income) adj (economy or economies)) or (low* adj (gdp or gnp or gross domestic or gross national)) or (low adj3 middle adj3 countr*)) or lmic or lmic3 or third world or lami countr* or transitional countr*).ti,ab.	1683278
5	1 and 2 and 3 and 4	1911
6	limit 5 to yr="2019 - 2020"	382

Appendix 2.5 APA PsycInfo

APA PsycInfo <1806 to September Week 1 2022>

	Condition	Hit
1	exp Dementia/ or Huntington Disease/ or Cognitive Impairment/ or (dement* or amentia* or (major adj3 cognit* adj3 disorder*) or alzheimer* or alzeimer* or (cortical adj4 sclerosis) or ((encephalopath* or cogniti* or neurocogniti*) adj4 (aids or acquired immun?deficiency syndrome* or acquired immun? deficiency syndrome* or hiv or human immun?deficiency virus* or human immun? deficiency virus*)) or pick* disease* or (lobar adj3 atroph* adj3 brain) or Huntington* disease* or Huntington* chorea or (Lewy bod* adj3 disease*) or (cerebr* adj3 deteriorat*) or (cerebr* adj3 insufficien*) or ((frontotemporal or fronto temporal or corticobasal or cortico basal or frontal lobe*) adj4 (degenerati* or dysfunction*)) or ((cognit* or memory or cerebr*) adj3 (declin* or impair* or los* or deteriorat* or degenerat* or insufficen*)) or MCI or (mild adj2 cognit* impair*)),ti,ab.	185529
2	exp Cognitive Therapy/ or exp Drug Therapy/ or exp Cholinesterase Inhibitors/ or exp Tranquilizing Drugs/ or exp Serotonin Reuptake Inhibitors/ or exp Benzodiazepines/ or exp Sedatives/ or exp Exercise/ or Movement Therapy/ or exp Psychotherapy/ or exp Counseling/ or Social Support/ or exp Alternative Medicine/ or Creative Arts Therapy/ or Mind Body Therapy/ or Phototherapy/ or Advance Directives/ or exp Case Management/ or Support Groups/ or exp Self-Help Techniques/ or exp Social Support/ or Computer Assisted Diagnosis/ or Computer Assisted Therapy/ or Telemedicine/ or exp Mobile Devices/ or (intervention* or therap* or treatment* or program* or manage* or prevent* or diagnos* or polic* or (cognit* adj3 (therap* or training or rehab*)) or cognit* psycho therap* or cognit* psychotherap* or drug* or medicine* or pharmacotherap* or pharmaco* therap* or cholinesterase agent* or cholinesterase inhibitor* or (tranquili* adj3 (agent* or drug*)) or antipsychotic* or (neuroleptic adj3 (agent* or drug*)) or serotonin uptake inhibitor* or serotonin reuptake inhibitor* or ssri* or benzodiazepine* or (sedative adj3 (effect* or agent*)) or memantine or donepezil or rivastigmine or galantamine or souvenaid or risperidone or haloperidol or olanzapine or quetiapine or citalopram or dextromethorphan or carbamazepine or mirtazapine or sertraline or moclobemide or trazodone or melatonin or ramelteon or methylphenidate or exercis* or physical activit* or physical training or aerobic* or arobic* or kinesi?therap* or kinesi? therap* or (social adj3 (activit* or engag* or stimul*)) or psycholog* therap* or psychotherap* or ((behavio?r* or conditioning) adj3 therap*) or counsel?ing or ((psychosocial or psychosocial) adj3 (support or interven* or care)) or ((alternative or compl?ment* or traditional) adj3 (medicine* or therap*)) or acupunct* or (herb* adj3 (tea* or remedy or remedies or medicine*)) or ginkgo or homeopath* or ((music or art or aroma or light or photo or pet or pets) adj3 therap*) or aromatherap* or phototherap* or massage* or (mind adj3 body) or (advance? adj3 (care or medical or healthcare) adj3 plan*) or (decision* adj3 (aid* or support)) or (communicati* adj3 skill* adj3 training) or (dementia care adj3 map*) or ((person* or patient*) adj3 cent* adj3 care) or "Resources for Enhancing Alzheimer's Caregiver Health" or "Strategies for Relatives" or ((caregiver* or carer*) adj3 educat*) or telemedicine or tele medicine or (smart adj2 (phone* or device* or tablet*)) or smartphone* or cognitive aid* or reminder* or robot*),ti,ab.	2371714
3	exp Clinical Trials/ or Treatment Effectiveness Evaluation/ or (randomi#ed or placebo or randomly or trial or groups).ti,ab.	755859
4	Developing Countries.sh. or (Africa* or Asia* or Caribbean* or West Indies or South America* or Latin America* or Central America* or Afghan* or Albania* or Algeria* or Angola* or Anguilla* or Antigua* or Barbuda* or Argentin* or Armenia* or Azerbaijan* or Azeri or Bangladesh* or Barbados* or Benin* or Byelarus or Byelorussian or Belarus* or Belorussia* or Belize* or Bhutan* or Bolivia* or Bosnia* or Herzegovin* or Herzegovin* or Botswana or Botsuana or Motswana or Batswana or Brasil* or Brazil* or Burkina Faso or Burkina Fasso or Burkina* or Burundi* or Urundi* or Cambodia* or Cameroon* or Cameron* or Cape Verd* or Cabo Verde or Central African Republic or Chad* or Tchad* or Chile* or China or Chinese or Colombia* or Columbia* or Comoros or Comoro Islands or Comores or Comoran or Mayotte or Congo* or Costa Rica* or Cote d'Ivoire or Ivory Coast or Ivorian* or Cook Islands or Cuba* or Croat* or Djibouti* or Dominica* or East Timor or East Timur or Timor Leste or Timorese or Ecuador* or Equador* or Egypt* or El Salvador or Salvadoran or Eritrea* or Ethiopia* or Fiji* or Gabon* or Gambia* or Gaza or Georgia Republic or Georgian or Abkhazia* or Abchasia* or South Ossetia* or Ghana* or Grenada or Grenadian or Guatemala* or Guinea* or Guinea Bissau or Guian* or Guyana or Haiti* or Hondura* or India or Indian or Indonesia* or Iran* or Iraq* or Jamaica* or Jordan* or Kazakhstan* or Kazakh or Kenya* or Kiribati or Korea* or Kosovo or Kosova* or Kyrgyzstan or Kirghizia or Kyrgyz or Kirghiz or Kirgizstan or Lao PDR or Laos or Laotian or Lebanon or Lebanese or Lesotho or Mosotho or Basotho or Liberia* or Libya* or Macedonia* or FYROM or Madagasca* or Malagasy or Malaysia* or Malaya* or Malay or Sabah or Sarawak or Malawi* or Maldives or Maldivan or Mali or Malian or Marshall Islands or Marshallese or Mauritania* or Mauriti* or Agalega Islands or Mexico or Mexican or Micronesia* or Middle East* or Moldova* or Moldovia* or Transnistria* or Mongolia* or Montenegr* or Montserrat* or Morocc* or Mozambique or Mozambican or Myanmar* or Myanma or Burma or Burmese or Namibia* or Nauru* or Niue or Nepal* or Nicaragua* or Niger or Nigerien or Nigeria* or Oman* or Pakistan* or Palau* or Palestine or Palestinian or Panama or Panamanian or Paraguay* or Papua New Guinea* or Peru or Peruvian or Philippines or Philipines or Phillipines or Phillippines or Filipino or Philipino or Philippino or Phillipino or Phillippino or Rwanda* or Ruanda* or Saint	504266

	Helen* or St Helen* or Saint Kitts or St Kitts or Kittian or Nevis* or Saint Lucia* or St Lucia* or Saint Vincent or St Vincent or Vicentian or Grenadines or Samoa* or Sao Tome* or Senegal* or Serbia* or Seychell* or Sierra Leone* or Sri Lanka* or Ceylon or Solomon Island* or Somali* or South Africa* or Sudan or Sudanese or Surinam* or Swaziland or Swazi or Eswatini or Syria or Syrian or Tajikistan or Tadjikistan or Tadjikistan or Tajik or Tadjik or Tanzania* or Thailand or Thai or Togo or Togolese or Tonga* or Tunisia* or Tokelau or Trinidad* or Tobago* or Turkey or Turkish or Turks or Turkmenistan or Turkmen or Tuvalu* or Uganda* or Ukraine or Ukrainian or Uruguay* or Uzbekistan* or Uzbek or Vanuatu or Venezuela* or Vietnam* or Viet Nam or Wallis Futuna or West Bank or Yemen* or Zambia* or Zimbabwe*).hw,ti,ab,lo. or (((developing or less* developed or under developed or underdeveloped or middle income or low* income or underserved or under served or deprived or poor*) adj (countr* or nation? or population? or world)) or ((developing or less* developed or under developed or underdeveloped or middle income or low* income) adj (economy or economies)) or (low* adj (gdp or gnp or gross domestic or gross national)) or (low adj3 middle adj3 countr*) or lmic or lmics or third world or lami countr* or transitional countr*).ti,ab.	
5	1 and 2 and 3 and 4	2359
6	(animal not human).po.	379966
7	5 not 6	2301
8	limit 7 to yr="2019 - 2020"	404

Appendix 2.6 CINAHL EBSCO

CINAHL EBSCO		
Condition		Hit
S1	(MH "Dementia+") or (MH "Huntington's Disease") or (MH "Cognition Disorders") or TI ((dement* or amentia* or (major N2 cognit* N2 disorder*) or alzheimer* or alzeimer* or (cortical N3 sclerosis) or ((encephalopath* or cogniti* or neurocogniti*) N3 (aids or "acquired immun#deficiency syndrome*" or "acquired immun# deficiency syndrome*" or hiv or "human immun#deficiency virus*" or "human immun# deficiency virus*") or "pick* disease*" or (lobar N2 atroph* N2 brain) or "Huntington* disease*" or "Huntington* chorea" or ("Lewy bod*" N2 disease*) or (cerebr* N2 deteriorat*) or (cerebr* N2 insufficien*) or ((frontotemporal or "fronto temporal" or corticobasal or "cortico basal" or "frontal lobe*") N3 (degenerati* or dysfunction*)) or ((cognit* or memory or cerebr*) N2 (declin* or impair* or los* or deteriorat* or degenerat* or insufficien*)) or MCI or (mild N1 "cognit* impair*"))) OR AB ((dement* or amentia* or (major N2 cognit* N2 disorder*) or alzheimer* or alzeimer* or (cortical N3 sclerosis) or ((encephalopath* or cogniti* or neurocogniti*) N3 (aids or "acquired immun#deficiency syndrome*" or "acquired immun# deficiency syndrome*" or hiv or "human immun#deficiency virus*" or "human immun# deficiency virus*") or "pick* disease*" or (lobar N2 atroph* N2 brain) or "Huntington* disease*" or "Huntington* chorea" or ("Lewy bod*" N2 disease*) or (cerebr* N2 deteriorat*) or (cerebr* N2 insufficien*) or ((frontotemporal or "fronto temporal" or corticobasal or "cortico basal" or "frontal lobe*") N3 (degenerati* or dysfunction*)) or ((cognit* or memory or cerebr*) N2 (declin* or impair* or los* or deteriorat* or degenerat* or insufficien*)) or MCI or (mild N1 "cognit* impair*")))	159287
S2	(MH "Cognitive Therapy+") or (MH "Drug Therapy+") or (MH "Cholinesterase Inhibitors+") or (MH "Antipsychotic Agents+") or (MH "Serotonin Uptake Inhibitors+") or (MH "Antianxiety Agents, Benzodiazepine+") or (MH "Hypnotics and Sedatives+") or (MH "Exercise+") or (MH "Therapeutic Exercise+") or (MH "Psychotherapy+") or (MH "Counseling+") or (MH "Support, Psychosocial+") or (MH "Alternative Therapies+") or (MH "Advance Care Planning") or (MH "Advance Directives+") or (MH "Case Management") or (MH "Patient Care Plans+") or (MH "Caregivers/ED/PF") or (MH "Support Groups") or (MH "Diagnosis, Computer Assisted") or (MH "Telehealth+") or (MH "Computers, Hand-Held+") or TI ((intervention* or therap* or treatment* or program* or manage* or prevent* or diagnos* or polic* or (cognit* N2 (therap* or training or rehab*)) or "cognit* psycho therap*" or "cognit* psychotherap*" or drug* or medicine* or pharmacotherap* or "pharmaco* therap*" or "cholinesterase agent*" or "cholinesterase inhibitor*" or (tranquili* N2 (agent* or drug*)) or antipsychotic* or (neuroleptic N2 (agent* or drug*)) or "serotonin uptake inhibitor*" or "serotonin reuptake inhibitor*" or ssri* or benzodiazepine* or (sedative N2 (effect* or agent*)) or memantine or donepezil or rivastigmine or galantamine or souvenaid or risperidone or haloperidol or olanzapine or quetiapine or citalopram or dextromethorphan or carbamazepine or mirtazapine or sertraline or moclobemide or trazodone or melatonin or ramelteon or methylphenidate or exercis* or "physical activit*" or "physical training" or aerobic* or arobic* or kinesi#therap* or "kinesi# therap*" or (social N2 (activit* or engag* or stimul*)) or "psycholog* therap*" or psychotherap* or ((behavio#r* or conditioning) N2 therap*) or counsel#ing or ((psychosocial or "psycho social") N2 (support or interven* or care)) or ((alternative or compl#ment* or traditional) N2 (medicine* or therap*)) or acupunct* or (herb* N2 (tea* or remedy or remedies or medicine*)) or ginkgo or homeopath* or ((music or art or aroma or light or photo or pet or pets) N2 therap*) or aromatherap* or phototherap*	3652534

	or massage* or (mind N2 body) or (advance# N2 (care or medical or healthcare) N2 plan*) or (decision* N2 (aid* or support)) or (communicati* N2 skill* N2 training) or ("dementia care" N2 map*) or ((person* or patient*) N2 cent* N2 care) or "Resources for Enhancing Alzheimer's Caregiver Health" or "Strategies for Relatives" or ((caregiver* or carer*) N2 educat*) or telemedicine or "tele medicine" or (smart N2 (phone* or device* or tablet*)) or smartphone* or "cognitive aid*" or reminder* or robot*)) OR AB ((intervention* or therap* or treatment* or program* or manage* or prevent* or diagnos* or polic* or (cognit* N2 (therap* or training or rehab*)) or "cognit* psycho therap*" or "cognit* psychotherap*" or drug* or medicine* or pharmacotherap* or "pharmaco* therap*" or "cholinesterase agent*" or "cholinesterase inhibitor*" or (tranquili* N2 (agent* or drug*)) or antipsychotic* or (neuroleptic N2 (agent* or drug*)) or "serotonin uptake inhibitor*" or "serotonin reuptake inhibitor*" or ssri* or benzodiazepine* or (sedative N2 (effect* or agent*)) or memantine or donepezil or rivastigmine or galantamine or souvenaid or risperidone or haloperidol or olanzapine or quetiapine or citalopram or dextromethorphan or carbamazepine or mirtazapine or sertraline or moclobemide or trazodone or melatonin or ramelteon or methylphenidate or exercis* or "physical activit*" or "physical training" or aerobic* or arobic* or kinesi#therap* or "kinesi# therap*" or (social N2 (activit* or engag* or stimul*)) or "psycholog* therap*" or psychotherap* or ((behavio#i* or conditioning) N2 therap*) or counsel#ing or ((psychosocial or "psycho social") N2 (support or interven* or care)) or ((alternative or compl#ment* or traditional) N2 (medicine* or therap*)) or acupunct* or (herb* N2 (tea* or remedy or remedies or medicine*)) or ginkgo or homeopath* or ((music or art or aroma or light or photo or pet or pets) N2 therap*) or aromatherap* or phototherap* or massage* or (mind N2 body) or (advance# N2 (care or medical or healthcare) N2 plan*) or (decision* N2 (aid* or support)) or (communicati* N2 skill* N2 training) or ("dementia care" N2 map*) or ((person* or patient*) N2 cent* N2 care) or "Resources for Enhancing Alzheimer's Caregiver Health" or "Strategies for Relatives" or ((caregiver* or carer*) N2 educat*) or telemedicine or "tele medicine" or (smart N2 (phone* or device* or tablet*)) or smartphone* or "cognitive aid*" or reminder* or robot*))	
S3	(MH "Randomized Controlled Trials") or (MH "Double-Blind Studies") or (MH "Single-Blind Studies") or (MH "Random Assignment") or (MH "Pretest-Posttest Design+") or (MH "Cluster Sample+") or TI (randomised or randomized) or AB random* or TI trial or (MH "Sample Size" and AB (assigned OR allocated OR control)) or (MH "Placebos") or (MH "Crossover Design") or (MH "Comparative Studies") or PT "randomized controlled trial" or AB ((control W5 group) or (cluster W3 RCT))	982366
S4	(MH "Developing Countries")	19971
S5	TI (Africa* or Asia* or Caribbean* or "West Indies" or "South America*" or "Latin America*" or "Central America*" or Afghan* or Albania* or Algeria* or Angola* or Anguilla* or Antigua* or Barbuda* or Argentin* or Armenia* or Azerbaijan* or Azeri or Bangladesh* or Barbud* or Benin* or Byelarus or Byelorussian or Belarus* or Belorussia* or Belize* or Bhutan* or Bolivia* or Bosnia* or Herzegovin* or Hercegovin* or Botswana or Botswana or Motswana or Batswana or Brasil* or Brazil* or "Burkina Faso" or "Burkina Fasso" or Burkina* or Burundi* or Urundi* or Cambodia* or Cameroon* or Cameron* or "Cape Verd*" or "Cabo Verde" or "Central African Republic" or Chad* or Tchad* or Chile* or China or Chinese or Colombia* or Columbia* or Comoros or "Comoro Islands" or Comores or Comoran or Mayotte or Congo* or "Costa Rica*" or "Cote d'Ivoire" or "Ivory Coast" or Ivorian* or "Cook Islands" or Cuba* or Croat* or Djibouti* or Dominica* or "East Timor" or "East Timur" or "Timor Leste" or Timorese or Ecuador* or Equador* or Egypt* or "El Salvador" or Salvadoran or Eritrea* or Ethiopia* or Fiji* or Gabon* or Gambia* or Gaza or "Georgia Republic" or Georgian or Abkhazia* or Abchasia* or "South Ossetia*" or Ghana* or Grenada or Grenadian or Guatemala* or Guinea* or "Guinea Bissau" or Guian* or Guyana or Haiti* or Hondura* or India or Indian or Indonesia* or Iran* or Iraq* or Jamaica* or Jordan* or Kazakhstan* or Kazakh or Kenya* or Kiribati or Korea* or Kosovo or Kosova* or Kyrgyzstan or Kirghizia or Kyrgyz or Kirghiz or Kirgizstan or "Lao PDR" or Laos or Laotian or Lebanon or Lebanese or Lesotho or Mosotho or Basotho or Liberia* or Libya* or Macedonia* or FYROM or Madagasca* or Malagasy or Malaysia* or Malaya* or Malay or Sabah or Sarawak or Malawi* or Maldives or Maldivan or Mali or Malian or "Marshall Islands" or Marshallese or Mauritania* or Mauriti* or "Agalega Islands" or Mexico or Mexican or Micronesia* or "Middle East*" or Moldova* or Moldovia* or Transnistria* or Mongolia* or Montenegri* or Montserrat* or Morocc* or Mozambique or Mozambican or Myanmar* or Myanma or Burma or Burmese or Namibia* or Nauru* or Niue or Nepal* or Nicaragua* or Niger or Nigerien or Nigeria* or Oman* or Pakistan* or Palau* or Palestine or Palestinian or Panama or Panamanian or Paraguay* or "Papua New Guinea*" or Peru or Peruvian or Philippines or Philipines or Philipines or Philippines or Filipino or Philipino or Philippino or Phillipino or Rwanda* or "Saint Helen*" or "St Helen*" or "Saint Kitts" or "St Kitts" or Kittian or Nevis* or "Saint Lucia*" or "St Lucia*" or "Saint Vincent" or "St Vincent" or Vicentian or Grenadines or Samoa* or "Sao Tome*" or Senegal* or Serbia* or Seychell* or "Sierra Leone*" or "Sri Lanka*" or Ceylon or "Solomon Island*" or Somali* or "South Africa*" or Sudan or Sudanese or Surinam* or Swaziland or Swazi or Eswatini or Syria or Syrian or Tajikistan or Tadjhikistan or Tadjikistan or Tajik or Tadjhik or Tanzania* or Thailand or Thai or Togo or Togolese or Tonga* or Tunisia* or Tokelau or Trinidad* or Tobago* or Turkey or Turkish or Turks or Turkmenistan or Turkmen or Tuvalu* or Uganda* or	679546

Ukraine or Ukrainian or Uruguay* or Uzbekistan* or Uzbek or Vanuatu or Venezuela* or Vietnam* or "Viet Nam" or "Wallis Futuna" or "West Bank" or Yemen* or Zambia* or Zimbabwe*) OR AB (Africa* or Asia* or Caribbean* or "West Indies" or "South America*" or "Latin America*" or "Central America*" or Afghan* or Albania* or Algeria* or Angola* or Anguilla* or Antigua* or Barbuda* or Argentina* or Armenia* or Azerbaijan* or Azeri or Bangladesh* or Barbados* or Benin* or Byelorussia* or Byelorussian or Belarus* or Belorussia* or Belize* or Bhutan* or Bolivia* or Bosnia* or Herzegovina* or Hercegovin* or Botswana or Botsuana or Motswana or Batswana or Brasil* or Brazil* or "Burkina Faso" or "Burkina Fasso" or Burkina* or Burundi* or Urundi* or Cambodia* or Cameroon* or Cameron* or "Cape Verde*" or "Cabo Verde" or "Central African Republic" or Chad* or Tchad* or Chile* or China or Chinese or Colombia* or Columbia* or Comoros or "Comoro Islands" or Comores or Comoran or Mayotte or Congo* or "Costa Rica*" or "Cote d'Ivoire" or "Ivory Coast" or Ivorian* or "Cook Islands" or Cuba* or Croatia* or Djibouti* or Dominica* or "East Timor" or "East Timor*" or "Timor Leste" or Timorese or Ecuador* or Equador* or Egypt* or "El Salvador" or Salvadoran or Eritrea* or Ethiopia* or Fiji* or Gabon* or Gambia* or Gaza or "Georgia Republic" or Georgian or Abkhazia* or Abchasia* or "South Ossetia*" or Ghana* or Grenada or Grenadian or Guatemala* or Guinea* or "Guinea Bissau" or Guian* or Guyana or Haiti* or Honduras* or India or Indian or Indonesia* or Iran* or Iraq* or Jamaica* or Jordan* or Kazakhstan* or Kazakh or Kenya* or Kiribati or Korea* or Kosovo or Kosova* or Kyrgyzstan or Kirghizia or Kyrgyz or Kirghiz or Kirgizstan or "Lao PDR" or Laos or Laotian or Lebanon or Lebanese or Lesotho or Mosotho or Basotho or Liberia* or Libya* or Macedonia* or FYROM or Madagaska* or Malagasy or Malaysia* or Malaya* or Malay or Sabah or Sarawak or Malawi* or Maldives or Maldivan or Mali or Malian or "Marshall Islands" or Marshallese or Mauritania* or Mauriti* or "Agalega Islands" or Mexico or Mexican or Micronesia* or "Middle East*" or Moldova* or Moldovia* or Transnistria* or Mongolia* or Montenegro* or Montserrat* or Morocco* or Mozambique or Mozambican or Myanmar* or Myanma or Burma or Burmese or Namibia* or Nauru* or Niue or Nepal* or Nicaragua* or Niger or Nigerien or Nigeria* or Oman* or Pakistan* or Palau* or Palestine or Palestinian or Panama or Panamanian or Paraguay* or "Papua New Guinea*" or Peru or Peruvian or Philippines or Philipines or Phillipines or Philippines or Filipino or Philipino or Philippino or Phillipino or Philippino or Rwanda* or Ruanda* or "Saint Helen*" or "St Helen*" or "Saint Kitts" or "St Kitts" or Kittian or Nevis* or "Saint Lucia*" or "St Lucia*" or "Saint Vincent" or "St Vincent" or Vicentian or Grenadines or Samoa* or "Sao Tome*" or Senegal* or Serbia* or Seychell* or "Sierra Leone*" or "Sri Lanka*" or Ceylon or "Solomon Island*" or Somali* or "South Africa*" or Sudan or Sudanese or Surinam* or Swaziland or Swazi or Eswatini or Syria or Syrian or Tajikistan or Tadjikistan or Tadjikistan or Tajik or Tadjik or Tanzania* or Thailand or Thai or Togo or Togolese or Tonga* or Tunisia* or Tokelau or Trinidad* or Tobago* or Turkey or Turkish or Turks or Turkmenistan or Turkmen or Tuvalu* or Uganda* or Ukraine or Ukrainian or Uruguay* or Uzbekistan* or Uzbek or Vanuatu or Venezuela* or Vietnam* or "Viet Nam" or "Wallis Futuna" or "West Bank" or Yemen* or Zambia* or Zimbabwe*) OR MH (Africa* or Asia* or Caribbean* or "West Indies" or "South America*" or "Latin America*" or "Central America*" or Afghan* or Albania* or Algeria* or Angola* or Anguilla* or Antigua* or Barbuda* or Argentina* or Armenia* or Azerbaijan* or Azeri or Bangladesh* or Barbados* or Benin* or Byelorussia* or Byelorussian or Belarus* or Belorussia* or Belize* or Bhutan* or Bolivia* or Bosnia* or Herzegovina* or Hercegovin* or Botswana or Botsuana or Motswana or Batswana or Brasil* or Brazil* or "Burkina Faso" or "Burkina Fasso" or Burkina* or Burundi* or Urundi* or Cambodia* or Cameroon* or Cameron* or "Cape Verde*" or "Cabo Verde" or "Central African Republic" or Chad* or Tchad* or Chile* or China or Chinese or Colombia* or Columbia* or Comoros or "Comoro Islands" or Comores or Comoran or Mayotte or Congo* or "Costa Rica*" or "Cote d'Ivoire" or "Ivory Coast" or Ivorian* or "Cook Islands" or Cuba* or Croatia* or Djibouti* or Dominica* or "East Timor" or "East Timor*" or "Timor Leste" or Timorese or Ecuador* or Equador* or Egypt* or "El Salvador" or Salvadoran or Eritrea* or Ethiopia* or Fiji* or Gabon* or Gambia* or Gaza or "Georgia Republic" or Georgian or Abkhazia* or Abchasia* or "South Ossetia*" or Ghana* or Grenada or Grenadian or Guatemala* or Guinea* or "Guinea Bissau" or Guian* or Guyana or Haiti* or Honduras* or India or Indian or Indonesia* or Iran* or Iraq* or Jamaica* or Jordan* or Kazakhstan* or Kazakh or Kenya* or Kiribati or Korea* or Kosovo or Kosova* or Kyrgyzstan or Kirghizia or Kyrgyz or Kirghiz or Kirgizstan or "Lao PDR" or Laos or Laotian or Lebanon or Lebanese or Lesotho or Mosotho or Basotho or Liberia* or Libya* or Macedonia* or FYROM or Madagaska* or Malagasy or Malaysia* or Malaya* or Malay or Sabah or Sarawak or Malawi* or Maldives or Maldivan or Mali or Malian or "Marshall Islands" or Marshallese or Mauritania* or Mauriti* or "Agalega Islands" or Mexico or Mexican or Micronesia* or "Middle East*" or Moldova* or Moldovia* or Transnistria* or Mongolia* or Montenegro* or Montserrat* or Morocco* or Mozambique or Mozambican or Myanmar* or Myanma or Burma or Burmese or Namibia* or Nauru* or Niue or Nepal* or Nicaragua* or Niger or Nigerien or Nigeria* or Oman* or Pakistan* or Palau* or Palestine or Palestinian or Panama or Panamanian or Paraguay* or "Papua New Guinea*" or Peru or Peruvian or Philippines or Philipines or Phillipines or Philippines or Filipino or Philipino or Philippino or Phillipino or Philippino or Rwanda* or Ruanda* or "Saint Helen*" or "St Helen*" or "Saint Kitts" or "St Kitts" or Kittian or Nevis* or "Saint Lucia*" or "St Lucia*" or "Saint Vincent" or "St Vincent" or Vicentian or Grenadines or Samoa* or "Sao Tome*" or Senegal* or Serbia* or Seychell* or "Sierra Leone*" or "Sri Lanka*" or Ceylon or "Solomon Island*" or Somali* or "South Africa*" or Sudan or Sudanese or Surinam* or Swaziland or	
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	Swazi or Eswatini or Syria or Syrian or Tajikistan or Tadjikistan or Tadjikistan or Tajik or Tadjik or Tanzania* or Thailand or Thai or Togo or Togolese or Tonga* or Tunisia* or Tokelau or Trinidad* or Tobago* or Turkey or Turkish or Turks or Turkmenistan or Turkmen or Tuvalu* or Uganda* or Ukraine or Ukrainian or Uruguay* or Uzbekistan* or Uzbek or Vanuatu or Venezuela* or Vietnam* or "Viet Nam" or "Wallis Futuna" or "West Bank" or Yemen* or Zambia* or Zimbabwe*)	
S6	TI ((developing or "less* developed" or "under developed" or underdeveloped or "middle income" or "low* income" or underserved or "under served" or deprived or poor*) N1 (countr* or nation# or population# or world)) or ((developing or "less* developed" or "under developed" or underdeveloped or "middle income" or "low* income") N1 (economy or economies)) or (low* N1 (gdp or gnp or "gross domestic" or "gross national")) or (low N2 middle N2 countr*) or lmic or lmic or "third world" or "lami countr*" or "transitional countr*") OR AB ((developing or "less* developed" or "under developed" or underdeveloped or "middle income" or "low* income" or underserved or "under served" or deprived or poor*) N1 (countr* or nation# or population# or world)) or ((developing or "less* developed" or "under developed" or underdeveloped or "middle income" or "low* income") N1 (economy or economies)) or (low* N1 (gdp or gnp or "gross domestic" or "gross national")) or (low N2 middle N2 countr*) or lmic or lmic or "third world" or "lami countr*" or "transitional countr*")	46191
S7	S4 or S5 or S6	705117
S8	S1 AND S2 AND S3 AND S7	1917
S9	(MH "Animals+" OR MH "Animal Studies" OR TI "animal model*") NOT MH "Human"	210517
S10	S8 NOT S9	1819
S11	Limiters - Published Date: 20190101-20211231	448

APPENDIX 3 R PACKAGES FOR STATISTICAL ANALYSIS

We followed the book *Doing Meta-Analysis with R: A Hands-On Guide* by Harrer et al. (2021) to perform meta-analysis with the R packages described in the book.⁹² We considered the methods in the book well validated because it was one of the most popular and up-to-date books for meta-analysis with R, which had been critically reviewed by Lortie (2022),⁹³ who praised it as a "compelling and balanced offering that provides a reasoned and replicable approach to meta-analysis that is exceptionally well grounded in theory and statistics."

To convert unstandardised effect measures into SMD, as suggested by Harrer et al. (2021), we used the R package 'esc', which was an R implementation of the web-based effect size calculator from the Campbell Collaboration. The calculator was an implementation of the formula in Lipsey and Wilson (2001).³⁷ First, calculating results with a popular method could lead to more comparability with other studies using the same methods. Second, the package and the calculators not only allowed us to calculate SMD from all 12 collected outcome data items, i.e. mean, SD, and sample sizes of the intervention and control group, before and after the intervention, with the function 'esc_mean_gain' but it also allowed us to calculate SMD from other types of data, such as regression coefficients and Cohen's d.^{94,95}

As suggested by Harrer et al. (2021), we used the R package 'meta' to perform meta-analysis, subgroup analysis and meta-regression, which had been compared with other R packages for meta-analysis by Polanin et al. (2016).⁹⁶ Compared with 'metafor', the most used R package for meta-analysis, 'meta' offered comprehensive functionality for meta-analysis but could not handle dependent effect sizes. Yet, we noted that in our case, 'meta' and 'metafor' yielded the same results, but 'meta' provided simpler options to display visual elements in the forest plot, especially PI. We used the R package 'dmetar',⁹⁷ the companion R package of Harrer et al. (2021) to perform outlier and influence diagnostics,⁹⁸ including leave-one-out analysis.⁶⁴

APPENDIX 4 DATA EXTRACTION TABLE

xa	Intervention	Effect measure	Instrument	Data type	Pre_mean	Pre_N	Pre_SD	Post_mean	Post_N	Post_SD	Attrition	Attrition.percent	Measurement point	Treatment duration
Amit 2008	MCI	Patient ADL	EASI	Beta				-0.36	33	1.27	8	19.51	180	180
Amit 2008	NULL	Patient ADL	EASI	Beta					26		14	35.00	180	180
Amit 2008	MCI	Caregiver mental health	GHQ-12	Beta				-1.12	33	1.81	8	19.51	180	180
Amit 2008	NULL	Caregiver mental health	GHQ-12	Beta					26		14	35.00	180	180
Amit 2008	MCI	Caregiver distress	NPI-D	Beta				-1.96	33	2.95	8	19.51	180	180
Amit 2008	NULL	Caregiver distress	NPI-D	Beta					26		14	35.00	180	180
Amit 2008	MCI	Patient neuropsychiatric symptoms	NPI-S	Beta				-1.19	33	3.13	8	19.51	180	180
Amit 2008	NULL	Patient neuropsychiatric symptoms	NPI-S	Beta					26		14	35.00	180	180
Amit 2008	MCI	Caregiver burden	ZBI	Beta				-3.29	33	6.65	8	19.51	180	180
Amit 2008	NULL	Caregiver burden	ZBI	Beta					26		14	35.00	180	180
Chen 2020	CCI	Patient ADL	BADL	Raw	14.53	51	1.77	11.41	51	1.08			365	365
Chen 2020	NULL	Patient ADL	BADL	Raw	15.02	51	1.82	9.31	51	1.52			365	365
Chen 2020	CCI	Patient neuropsychiatric symptoms	BEHAVE-AD	Raw	19.32	51	3.53	7.21	51	2.88			365	365
Chen 2020	NULL	Patient neuropsychiatric symptoms	BEHAVE-AD	Raw	19.47	51	3.37	10.37	51	2.17			365	365
Chen 2020	CCI	Patient cognitive function	MoCA	Raw	17.76	51	3.68	14.64	51	2.38			365	365
Chen 2020	NULL	Patient cognitive function	MoCA	Raw	17.04	51	3.52	12.59	51	2.70			365	365
Ghaffari 2019	CBT	Caregiver mental health	GHQ-28	Raw	39.90	27	8.40	16.08	25	3.70	2	7.41	91	56
Ghaffari 2019	CBT	Caregiver anxiety	GHQ-28	Raw	11.60	27	2.80	3.40	25	1.50	2	7.41	91	56
Ghaffari 2019	CBT	Caregiver depression	GHQ-28	Raw	4.50	27	3.04	0.40	25	0.70	2	7.41	91	56
Ghaffari 2019	NULL	Caregiver mental health	GHQ-28	Raw	32.04	27	7.02	32.90	25	7.20	2	7.41	91	56
Ghaffari 2019	NULL	Caregiver anxiety	GHQ-28	Raw	10.04	27	4.02	10.50	25	3.70	2	7.41	91	56
Ghaffari 2019	NULL	Caregiver depression	GHQ-28	Raw	4.80	27	3.60	5.08	25	3.70	2	7.41	91	56
Govindakumari 2020	NULL	Caregiver burden	Unknown	Raw	55.36	157	5.64	53.87	157	6.14	0	0.00	30	30
Govindakumari 2020	PES	Caregiver burden	Unknown	Raw	54.14	157	5.39	19.61	157	3.63	0	0.00	30	30
Govindakumari 2020	NULL	Patient QoL	Unknown	Raw	180.88	157	20.93	182.36	157	22.80	0	0.00	30	30
Govindakumari 2020	PES	Patient QoL	Unknown	Raw	180.24	157	20.55	279.73	157	17.17	0	0.00	30	30
Guerra 2011	PEI	Patient QoL	DEMQOL	Cohen's d		29		0.32	27	2.16	2	6.90	180	35
Guerra 2011	NULL	Patient QoL	DEMQOL	SMD		29			29		0	0.00	180	35
Guerra 2011	PEI	Caregiver distress	NPI-D	Cohen's d		29		-0.09	27	1.04	2	6.90	180	35
Guerra 2011	NULL	Caregiver distress	NPI-D	SMD		29			29		0	0.00	180	35
Guerra 2011	PEI	Patient neuropsychiatric symptoms	NPI-S	Cohen's d		29		-0.10	27	1.06	2	6.90	180	35
Guerra 2011	NULL	Patient neuropsychiatric symptoms	NPI-S	SMD		29			29		0	0.00	180	35
Guerra 2011	PEI	Caregiver distress	SRQ-20	Cohen's d		29		-0.14	27	0.99	2	6.90	180	35
Guerra 2011	NULL	Caregiver distress	SRQ-20	SMD		29			29		0	0.00	180	35
Guerra 2011	PEI	Caregiver environmental QoL	WHO-QoL-Bref	Cohen's d		29		-0.22	27	0.98	2	6.90	180	35
Guerra 2011	PEI	Caregiver physical quality of life	WHO-QoL-Bref	Cohen's d		29		0.49	27	0.92	2	6.90	180	35
Guerra 2011	PEI	Caregiver psychological QoL	WHO-QoL-Bref	Cohen's d		29		0.10	27	1.07	2	6.90	180	35
Guerra 2011	PEI	Caregiver QoL	WHO-QoL-Bref	Cohen's d		29		0.39	27	1.01	2	6.90	180	35

Guerra 2011	NULL	Caregiver environmental QoL	WHO-QoL-Bref	SMD		29			29		0	0.00	180	35
Guerra 2011	NULL	Caregiver physical QoL	WHO-QoL-Bref	SMD		29			29		0	0.00	180	35
Guerra 2011	NULL	Caregiver psychological QoL	WHO-QoL-Bref	SMD		29			29		0	0.00	180	35
Guerra 2011	NULL	Caregiver QoL	WHO-QoL-Bref	SMD		29			29		0	0.00	180	35
Guerra 2011	PEI	Caregiver burden	ZBI	Cohen's d		29		-1.02	27	0.95	2	6.90	180	35
Guerra 2011	NULL	Caregiver burden	ZBI	SMD		29			29		0	0.00	180	35
Hinton 2020	MCI	Caregiver mental health	PHQ-4	Raw	4.10	25	3.00	0.90	25	1.00	5	16.67	90	35
Hinton 2020	NULL	Caregiver mental health	PHQ-4	Raw	4.60	26	2.80	2.90	26	2.80	4	13.33	90	35
Hinton 2020	MCI	Caregiver burden	ZBI	Raw	6.70	25	2.10	2.90	25	2.40	5	16.67	90	35
Hinton 2020	NULL	Caregiver burden	ZBI	Raw	6.80	26	2.30	5.50	26	3.40	4	13.33	90	35
JIANG 2012	CCI	Patient cognitive function	ADAS-cog	Raw	29.38	48	7.23	23.02	46	6.15	2	4.17	365	365
JIANG 2012	NULL	Patient cognitive function	ADAS-cog	Raw	28.96	48	6.80	25.92	48	6.72	0	0.00	365	365
JIANG 2012	CCI	Patient ADL	ADL	Raw	35.46	48	7.46	28.65	46	7.20	2	4.17	365	365
JIANG 2012	NULL	Patient ADL	ADL	Raw	34.96	48	6.80	32.04	48	6.79	0	0.00	365	365
JIANG 2012	CCI	Patient neuropsychiatric symptoms	Global Deterioration Scale	Raw	3.88	48	0.82	3.00	46	1.07	2	4.17	365	365
JIANG 2012	NULL	Patient neuropsychiatric symptoms	Global Deterioration Scale	Raw	3.85	48	0.77	3.48	48	0.95	0	0.00	365	365
JIANG 2012	CCI	Patient cognitive function	MMSE	Raw	18.31	48	3.63	22.08	46	4.17	2	4.17	365	365
JIANG 2012	NULL	Patient cognitive function	MMSE	Raw	18.52	48	3.40	20.33	48	3.49	0	0.00	365	365
JIANG 2012	CCI	Patient neuropsychiatric symptoms	NPI	Raw	13.90	48	4.04	9.31	46	3.67	2	4.17	365	365
JIANG 2012	NULL	Patient neuropsychiatric symptoms	NPI	Raw	13.52	48	4.74	11.21	48	4.82	0	0.00	365	365
Liu 2017	CCI	Caregiver burden	CBI	Raw	53.51	39	16.10	33.64	39	15.49	0	0.00	90	90
Liu 2017	NULL	Caregiver burden	CBI	Raw	56.46	39	9.18	58.26	39	10.60	0	0.00	90	90
Liu 2017	CCI	Caregiver QoL	WHO-QoL-Bref	Raw	48.82	39	7.83	55.04	39	5.88	0	0.00	90	90
Liu 2017	NULL	Caregiver QoL	WHO-QoL-Bref	Raw	48.67	39	4.20	46.77	39	4.00	0	0.00	90	90
Lök 2017	PEI	Caregiver feeding	Healthy Life Style Behavior Scale II	Raw	18.10	20	3.09	24.75	20	1.25			49	49
Lök 2017	PEI	Caregiver healthy responsibility	Healthy Life Style Behavior Scale II	Raw	18.75	20	3.71	23.20	20	2.01			49	49
Lök 2017	PEI	Caregiver interpersonal communication	Healthy Life Style Behavior Scale II	Raw	18.00	20	3.92	23.75	20	1.40			49	49
Lök 2017	PEI	Caregiver moral development	Healthy Life Style Behavior Scale II	Raw	18.15	20	3.88	22.10	20	1.02			49	49
Lök 2017	PEI	Caregiver physical activity	Healthy Life Style Behavior Scale II	Raw	16.35	20	3.06	24.20	20	1.47			49	49
Lök 2017	PEI	Caregiver stress management	Healthy Life Style Behavior Scale II	Raw	16.25	20	3.40	22.05	20	1.53			49	49
Lök 2017	NULL	Caregiver feeding	Healthy Life Style Behavior Scale II	Raw	18.15	20	1.18	17.09	20	1.25			49	49
Lök 2017	NULL	Caregiver healthy responsibility	Healthy Life Style Behavior Scale II	Raw	17.30	20	1.55	17.05	20	2.13			49	49
Lök 2017	NULL	Caregiver interpersonal communication	Healthy Life Style Behavior Scale II	Raw	19.55	20	1.35	18.65	20	1.34			49	49
Lök 2017	NULL	Caregiver moral development	Healthy Life Style Behavior Scale II	Raw	17.05	20	0.94	16.20	20	1.05			49	49
Lök 2017	NULL	Caregiver physical activity	Healthy Life Style Behavior Scale II	Raw	16.70	20	1.34	15.07	20	1.48			49	49
Lök 2017	NULL	Caregiver stress management	Healthy Life Style Behavior Scale II	Raw	16.85	20	1.98	16.65	20	1.34			49	49
Lök 2017	PEI	Caregiver burden	ZBI	Raw	78.30	20	4.46	67.10	20	7.18			49	49
Lök 2017	NULL	Caregiver burden	ZBI	Raw	78.45	20	3.26	79.30	20	3.79			49	49
Mahdavi 2017	GS	Caregiver burden	CSI	Raw	36.75	32	1.35	36.64	28	1.28	4	12.50	35	35
Mahdavi 2017	Misc.	Caregiver burden	CSI	Raw	37.16	33	1.26	32.43	30	2.73	3	9.09	35	35

Mahdavi 2017	NULL	Caregiver burden	CSI	Raw	37.58	35	1.64	37.19	31	1.53	4	11.43	35	35
Mercedeh 2014	CBT	Caregiver burden	CBI	Raw	31.60	10	16.36	24.00	10	14.25			35	35
Mercedeh 2014	NULL	Caregiver burden	CBI	Raw	25.30	10	7.33	44.70	10	13.71			35	35
Mercedeh 2014	CBT	Caregiver depression	Ham-D	Raw	20.00	10	6.16	6.70	10	2.54			35	35
Mercedeh 2014	NULL	Caregiver depression	Ham-D	Raw	12.70	10	4.57	19.40	10	9.21			35	35
Mercedeh 2014	CBT	Caregiver QoL	SF36	Raw	58.01	10	24.01	70.98	10	11.11			35	35
Mercedeh 2014	NULL	Caregiver QoL	SF36	Raw	47.61	10	18.32	37.57	10	12.80			35	35
Orawan 2018	MCI	Caregiver positive experience	PACQ	Raw	37.11	36	5.80	42.58	36	3.20	4	10.00	56	35
Orawan 2018	MCI	Caregiver positive experience	PACQ	Raw	37.11	36	5.80	43.27	36	2.47	4	10.00	84	35
Orawan 2018	MCI	Caregiver positive experience	PACQ	Raw	37.11	36	5.80	44.08	36	1.13	4	10.00	140	35
Orawan 2018	NULL	Caregiver positive experience	PACQ	Raw	37.19	36	6.03	37.36	36	5.89	4	10.00	84	35
Orawan 2018	NULL	Caregiver positive experience	PACQ	Raw	37.19	36	6.03	39.19	36	4.52	4	10.00	56	35
Orawan 2018	NULL	Caregiver positive experience	PACQ	Raw	37.19	36	6.03	37.75	36	3.86	4	10.00	140	35
Orawan 2018	MCI	Caregiver subjective wellbeing	TSWBS	Raw	79.44	36	11.19	98.61	36	10.80	4	10.00	56	35
Orawan 2018	MCI	Caregiver subjective wellbeing	TSWBS	Raw	79.44	36	11.19	98.91	36	10.05	4	10.00	84	35
Orawan 2018	MCI	Caregiver subjective wellbeing	TSWBS	Raw	79.44	36	11.19	100.27	36	9.56	4	10.00	140	35
Orawan 2018	NULL	Caregiver subjective wellbeing	TSWBS	Raw	79.77	36	11.02	99.80	36	11.25	4	10.00	84	35
Orawan 2018	NULL	Caregiver subjective wellbeing	TSWBS	Raw	79.77	36	11.02	98.91	36	11.23	4	10.00	140	35
Orawan 2018	NULL	Caregiver subjective wellbeing	TSWBS	Raw	79.77	36	11.02	100.11	36	10.49	4	10.00	56	35
Pahlavanzadeh 2010	PEI	Caregiver burden	ZBI	Raw	42.00	25	65.17	44.35	25	65.15	5	16.67	35	35
Pahlavanzadeh 2010	PEI	Caregiver burden	ZBI	Raw	42.00	25	65.17	56.33	25	54.14	5	16.67	65	35
Pahlavanzadeh 2010	NULL	Caregiver burden	ZBI	Raw	43.28	25	13.08	64.50	25	25.13	5	16.67	65	35
Pahlavanzadeh 2010	NULL	Caregiver burden	ZBI	Raw	43.28	25	13.08	46.80	25	3.13	5	16.67	35	35
Pan 2019	CBT	Caregiver depression	CES-D	Raw	13.93	56	3.45	9.40	54	3.39	2	3.57	150	150
Pan 2019	CBT	Caregiver depression	CES-D	Raw	13.93	56	3.45	10.19	47	2.78	9	16.07	210	150
Pan 2019	NULL	Caregiver depression	CES-D	Raw	13.21	56	3.11	12.21	35	3.56	21	37.50	210	150
Pan 2019	NULL	Caregiver depression	CES-D	Raw	13.21	56	3.11	11.96	45	3.40	11	19.64	150	150
Pan 2019	CBT	Caregiver mutuality	Mutuality Scale	Raw	1.15	56	0.40	1.33	54	0.59	2	3.57	150	150
Pan 2019	CBT	Caregiver mutuality	Mutuality Scale	Raw	1.15	56	0.40	1.27	47	0.54	9	16.07	210	150
Pan 2019	NULL	Caregiver mutuality	Mutuality Scale	Raw	1.15	56	0.51	1.13	35	0.50	21	37.50	210	150
Pan 2019	NULL	Caregiver mutuality	Mutuality Scale	Raw	1.15	56	0.51	1.19	45	0.42	11	19.64	150	150
Pan 2019	CBT	Caregiver active coping	Simplified Coping Scale	Raw	1.15	56	0.40	1.45	47	0.43	9	16.07	210	150
Pan 2019	CBT	Caregiver active coping	Simplified Coping Scale	Raw	1.15	56	0.40	1.45	54	0.43	2	3.57	150	150
Pan 2019	CBT	Caregiver passive coping	Simplified Coping Scale	Raw	1.38	56	0.42	1.13	47	0.39	9	16.07	210	150
Pan 2019	CBT	Caregiver passive coping	Simplified Coping Scale	Raw	1.38	56	0.42	1.24	54	0.38	2	3.57	150	150
Pan 2019	NULL	Caregiver passive coping	Simplified Coping Scale	Raw	1.39	56	0.35	1.13	35	0.39	21	37.50	210	150
Pan 2019	NULL	Caregiver active coping	Simplified Coping Scale	Raw	1.18	56	0.37	1.28	45	0.38	11	19.64	150	150
Pan 2019	NULL	Caregiver active coping	Simplified Coping Scale	Raw	1.18	56	0.37	1.24	35	0.32	21	37.50	210	150
Pan 2019	NULL	Caregiver passive coping	Simplified Coping Scale	Raw	1.39	56	0.35	1.29	45	0.32	11	19.64	150	150
Salamizadeh 2017	Misc.	Caregiver self-efficacy	GSE-10	Raw	29.80	26	4.80	32.73	26	4.75	4	13.33	56	35
Salamizadeh 2017	NULL	Caregiver self-efficacy	GSE-10	Raw	28.39	28	6.41	27.85	28	5.98	2	6.67	56	35

Shata 2017	MCI	Caregiver anxiety	TMAS	MD		55		-14.72	55	5.68	5	8.33	146	56
Shata 2017	MCI	Caregiver anxiety	TMAS	MD		55		-14.21	55	5.51	5	8.33	56	56
Shata 2017	NULL	Caregiver anxiety	TMAS	MD		59		1.13	59	1.04	1	1.67	146	56
Shata 2017	NULL	Caregiver anxiety	TMAS	MD		59		0.49	59	0.67	1	1.67	56	56
Shata 2017	MCI	Caregiver depression	Ham-D	MD		55		-9.87	55	4.05	5	8.33	146	56
Shata 2017	MCI	Caregiver depression	Ham-D	MD		55		-9.76	55	4.02	5	8.33	56	56
Shata 2017	NULL	Caregiver depression	Ham-D	MD		59		0.49	59	0.67	1	1.67	146	56
Shata 2017	NULL	Caregiver depression	Ham-D	MD		59		0.08	59	0.28	1	1.67	56	56
Shata 2017	MCI	Caregiver burden	ZBI	MD		55		-17.40	55	8.38	5	8.33	146	56
Shata 2017	MCI	Caregiver burden	ZBI	MD		55		-17.36	55	8.34	5	8.33	56	56
Shata 2017	NULL	Caregiver burden	ZBI	MD		59		0.74	59	0.84	1	1.67	146	56
Shata 2017	NULL	Caregiver burden	ZBI	MD		59		0.15	59	0.40	1	1.67	56	56
Söylemez 2016	PEI	Caregiver depression	BDI	Raw	9.74	35	4.43	7.46	35	3.42	0	0.00	90	90
Söylemez 2016	PEI	Caregiver depression	BDI	Raw	9.74	35	4.43	7.97	33	3.42	2	5.71	180	90
Söylemez 2016	NULL	Caregiver depression	BDI	Raw	8.63	35	3.18	9.86	32	3.47	3	8.57	180	90
Söylemez 2016	NULL	Caregiver depression	BDI	Raw	8.63	35	3.18	8.83	34	3.34	1	2.86	90	90
Söylemez 2016	PEI	Caregiver burden	CBI	Raw	38.02	35	21.49	34.71	33	18.82	2	5.71	180	90
Söylemez 2016	PEI	Caregiver burden	CBI	Raw	38.02	35	21.49	33.33	35	18.37	0	0.00	90	90
Söylemez 2016	NULL	Caregiver burden	CBI	Raw	31.19	35	16.54	35.66	32	17.07	3	8.57	180	90
Söylemez 2016	NULL	Caregiver burden	CBI	Raw	31.19	35	16.54	32.86	34	16.10	3	8.57	90	90
Söylemez 2016	PEI	Patient QoL	QoL-AD	Raw	31.60	35	4.45	33.60	35	4.72	0	0.00	90	90
Söylemez 2016	PEI	Patient QoL	QoL-AD	Raw	31.60	35	4.45	33.48	33	4.69	2	5.71	180	90
Söylemez 2016	NULL	Patient QoL	QoL-AD	Raw	32.83	35	4.69	32.63	34	4.64	1	2.86	90	90
Söylemez 2016	NULL	Patient QoL	QoL-AD	Raw	32.83	35	4.69	31.91	32	4.30	3	8.57	180	90
Söylemez 2016	PEI	Caregiver QoL	WHO-QoL-Bref	Raw	86.00	35	9.15	89.91	35	8.75	0	0.00	90	90
Söylemez 2016	PEI	Caregiver QoL	WHO-QoL-Bref	Raw	86.00	35	9.15	88.51	33	8.28	2	5.71	180	90
Söylemez 2016	NULL	Caregiver QoL	WHO-QoL-Bref	Raw	87.20	35	5.69	85.23	32	5.68	3	8.57	180	90
Söylemez 2016	NULL	Caregiver QoL	WHO-QoL-Bref	Raw	87.20	35	5.69	86.20	34	5.21	1	2.86	90	90
SU 2012	CCI	Patient ADL	ADL	Raw	23.80	20	3.30	20.80	20	2.40			194	14
SU 2012	NULL	Patient ADL	ADL	Raw	23.60	20	3.30	22.20	20	2.90			194	14
SU 2012	CCI	Patient cognitive function	MMSE	Raw	18.36	20	3.19	19.58	20	4.20			194	14
SU 2012	NULL	Patient cognitive function	MMSE	Raw	18.56	20	3.67	19.33	20	4.15			194	14
SUN 2010	CCI	Patient QoL	SF36	Raw	448.10	19	132.40	570.10	19	141.90			365	365
SUN 2010	NULL	Patient QoL	SF36	Raw	450.10	19	151.00	405.10	19	146.40			365	365
TAN 2010	CCI	Patient ADL	ADL	Raw	14.61	26	4.61	15.86	26	3.81			365	365
TAN 2010	NULL	Patient ADL	ADL	Raw	15.94	26	3.91	12.61	26	4.96			365	365
TAN 2010	CCI	Patient cognitive function	MMSE	Raw	15.10	26	2.56	16.12	26	2.84			365	365
TAN 2010	NULL	Patient cognitive function	MMSE	Raw	15.32	26	1.98	14.07	26	1.63			365	365
Uyar 2019	MCI	Caregiver anxiety	BAI	Raw	10.06	30	8.89	6.18	26	7.26	4	13.33	112	112
Uyar 2019	NULL	Caregiver anxiety	BAI	Raw	12.23	31	8.05	15.47	28	9.28	3	9.68	112	112
Uyar 2019	MCI	Caregiver depression	BDI	Raw	11.62	30	8.34	9.68	26	8.45	4	13.33	112	112
Uyar 2019	NULL	Caregiver depression	BDI	Raw	15.23	31	7.70	20.64	28	8.94	3	9.68	112	112
Uyar 2019	MCI	Caregiver distress	NPI-D	Raw	11.87	30	9.73	10.50	26	8.70	4	13.33	112	112
Uyar 2019	NULL	Caregiver distress	NPI-D	Raw	9.35	31	8.06	14.64	28	10.06	3	9.68	112	112

Uyar 2019	MCI	Patient neuropsychiatric symptoms	NPI-S	Raw	31.18	30	23.79	26.50	26	21.99	4	13.33	112	112
Uyar 2019	NULL	Patient neuropsychiatric symptoms	NPI-S	Raw	19.41	31	16.09	30.70	28	23.78	3	9.68	112	112
Uyar 2019	MCI	Patient QoL	QoL-Ada	Raw	31.18	22	6.14	31.79	17	5.41	5	22.73	112	112
Uyar 2019	NULL	Patient QoL	QoL-Ada	Raw	33.29	20	4.96	32.39	17	4.33	3	15.00	112	112
Uyar 2019	MCI	Caregiver psychological quality of life	SF36 mental health	Raw	41.79	30	9.02	44.87	26	8.94	4	13.33	112	112
Uyar 2019	NULL	Caregiver psychological quality of life	SF36 mental health	Raw	42.18	31	10.52	39.47	28	9.39	3	9.68	112	112
Uyar 2019	MCI	Caregiver physical quality of life	SF36 Physical health	Raw	41.75	30	10.00	45.41	26	7.94	4	13.33	112	112
Uyar 2019	NULL	Caregiver physical quality of life	SF36 Physical health	Raw	41.59	31	9.78	37.74	28	10.77	3	9.68	112	112
Uyar 2019	MCI	Caregiver burden	ZBI	Raw	31.43	30	18.18	27.87	26	18.39	4	13.33	112	112
Uyar 2019	NULL	Caregiver burden	ZBI	Raw	40.17	31	18.10	40.70	28	17.66	3	9.68	112	112
Wang 2010b	CCI	Patient ADL	ADL	Raw	20.90	40	1.90	19.40	40	1.80			365	365
Wang 2010b	NULL	Patient ADL	ADL	Raw	20.60	40	1.80	20.30	40	1.70			365	365
Wang 2010b	CCI	Patient cognitive function	MMSE	Raw	19.21	40	3.93	21.02	40	3.88			365	365
Wang 2010b	NULL	Patient cognitive function	MMSE	Raw	19.48	40	2.84	18.50	40	3.10			365	365
Wang 2012e	GS	Caregiver distress	NPI-D	Raw	46.84	39	8.11	37.47	39	9.68			198	168
Wang 2012e	NULL	Caregiver distress	NPI-D	Raw	46.91	39	9.68	43.88	39	13.56			198	168
Wang 2012e	GS	Caregiver QoL	WHO-QoL-Bref	Raw	97.89	39	14.11	114.02	39	13.98			198	168
Wang 2012e	NULL	Caregiver QoL	WHO-QoL-Bref	Raw	103.71	39	9.28	88.19	39	7.56			198	168
Wang 2014a	CCI	Patient ADL	ADL	Raw	34.14	62	5.35	32.99	62	6.35	2	3.13	365	365
Wang 2014a	CCI	Patient ADL	ADL	Raw	34.14	62	5.35	33.16	62	5.29	2	3.13	180	365
Wang 2014a	NULL	Patient ADL	ADL	Raw	34.21	61	5.74	44.53	61	7.56	3	4.69	365	365
Wang 2014a	NULL	Patient ADL	ADL	Raw	34.21	61	5.74	38.95	61	6.57	3	4.69	180	365
Wang 2014a	CCI	Patient cognitive function	MMSE	Raw	14.19	62	3.12	15.98	62	5.49	2	3.13	365	365
Wang 2014a	CCI	Patient cognitive function	MMSE	Raw	14.19	62	3.12	14.29	62	4.41	2	3.13	180	365
Wang 2014a	NULL	Patient cognitive function	MMSE	Raw	14.12	61	3.12	11.99	61	6.39	3	4.69	365	365
Wang 2014a	NULL	Patient cognitive function	MMSE	Raw	14.12	61	3.12	13.23	61	4.49	3	4.69	180	365
Yang 2017a	CCI	Patient ADL	ADL	Raw	38.56	71	8.35	30.85	71	11.14			365	365
Yang 2017a	CCI	Patient ADL	ADL	Raw	38.56	71	8.35	33.92	71	8.04			180	365
Yang 2017a	NULL	Patient ADL	ADL	Raw	37.21	70	5.93	34.09	70	6.17			365	365
Yang 2017a	NULL	Patient ADL	ADL	Raw	37.21	70	5.93	34.69	70	6.11			180	365
Yang 2017a	CCI	Patient neuropsychiatric symptoms	BEHAVE-AD	Raw	14.19	71	2.90	7.76	71	3.53			180	365
Yang 2017a	CCI	Patient neuropsychiatric symptoms	BEHAVE-AD	Raw	14.19	71	2.90	5.87	71	2.87			365	365
Yang 2017a	NULL	Patient neuropsychiatric symptoms	BEHAVE-AD	Raw	14.15	70	2.77	8.62	70	3.94			180	365
Yang 2017a	NULL	Patient neuropsychiatric symptoms	BEHAVE-AD	Raw	14.15	70	2.77	7.74	70	3.56			365	365
Yang 2017a	CCI	Caregiver QoL	GQOLI-74	Raw	206.73	71	34.75	236.87	71	42.99			365	365
Yang 2017a	CCI	Caregiver QoL	GQOLI-74	Raw	206.73	71	34.75	216.61	71	36.10			180	365
Yang 2017a	NULL	Caregiver QoL	GQOLI-74	Raw	206.64	70	33.79	219.51	70	39.32			365	365
Yang 2017a	NULL	Caregiver QoL	GQOLI-74	Raw	206.64	70	33.79	210.84	70	30.58			180	365
Yang 2017a	CCI	Patient cognitive function	MMSE	Raw	9.88	71	2.33	14.97	71	3.03			365	365
Yang 2017a	CCI	Patient cognitive function	MMSE	Raw	9.88	71	2.33	12.31	71	2.43			180	365
Yang 2017a	NULL	Patient cognitive function	MMSE	Raw	9.85	70	1.89	11.32	70	3.21			180	365
Yang 2017a	NULL	Patient cognitive function	MMSE	Raw	9.85	70	1.89	13.27	70	3.02			365	365

Yang 2017a	CCI	Patient QoL	QoL-AD	Raw	33.17	71	8.85	39.13	71	8.08			365	365
Yang 2017a	CCI	Patient QoL	QoL-AD	Raw	33.17	71	8.85	35.39	71	7.90			180	365
Yang 2017a	NULL	Patient QoL	QoL-AD	Raw	33.09	70	6.05	34.49	70	6.96			365	365
Yang 2017a	NULL	Patient QoL	QoL-AD	Raw	33.09	70	6.05	33.40	70	6.02			180	365
Zarepour 2020	PEI	Caregiver anxiety	SQ	Raw	74.11	35	4.73	70.51	35	3.78	0	0.00	58	28
Zarepour 2020	PEI	Caregiver anxiety	SQ	Raw	74.11	35	4.73	70.31	35	3.78	0	0.00	118	28
Zarepour 2020	NULL	Caregiver anxiety	SQ	Raw	75.63	35	3.08	76.22	35	5.08	0	0.00	118	28
Zarepour 2020	NULL	Caregiver anxiety	SQ	Raw	75.63	35	3.08	76.45	35	3.45	0	0.00	58	28
ZHAO 2010	CCI	Patient QoL	SF36	Raw	431.40	25	126.50	524.70	25	128.40			365	365
ZHAO 2010	NULL	Patient QoL	SF36	Raw	415.30	25	140.80	370.80	25	132.60			365	365

Note: To calculate intervention duration and measurement points from the start of trial in days, we defined that a year equates 365 days, a month equates 30 days, and a week equates 7 days. For regular intervention sessions, we ignore any absence from the sessions and interruption between the sessions and calculate the shortest duration to complete the session if the authors do not specific the total duration of the intervention sessions. For example, for an intervention that consists of five weekly sessions, the intervention duration is 5 weeks, which is 35 days.

APPENDIX 5 RISK OF BIAS OF INCLUDED STUDIES

The risk of bias information of the 20 studies from the in-house dataset were obtained from the dataset and was visualised in the traffic light plot below:

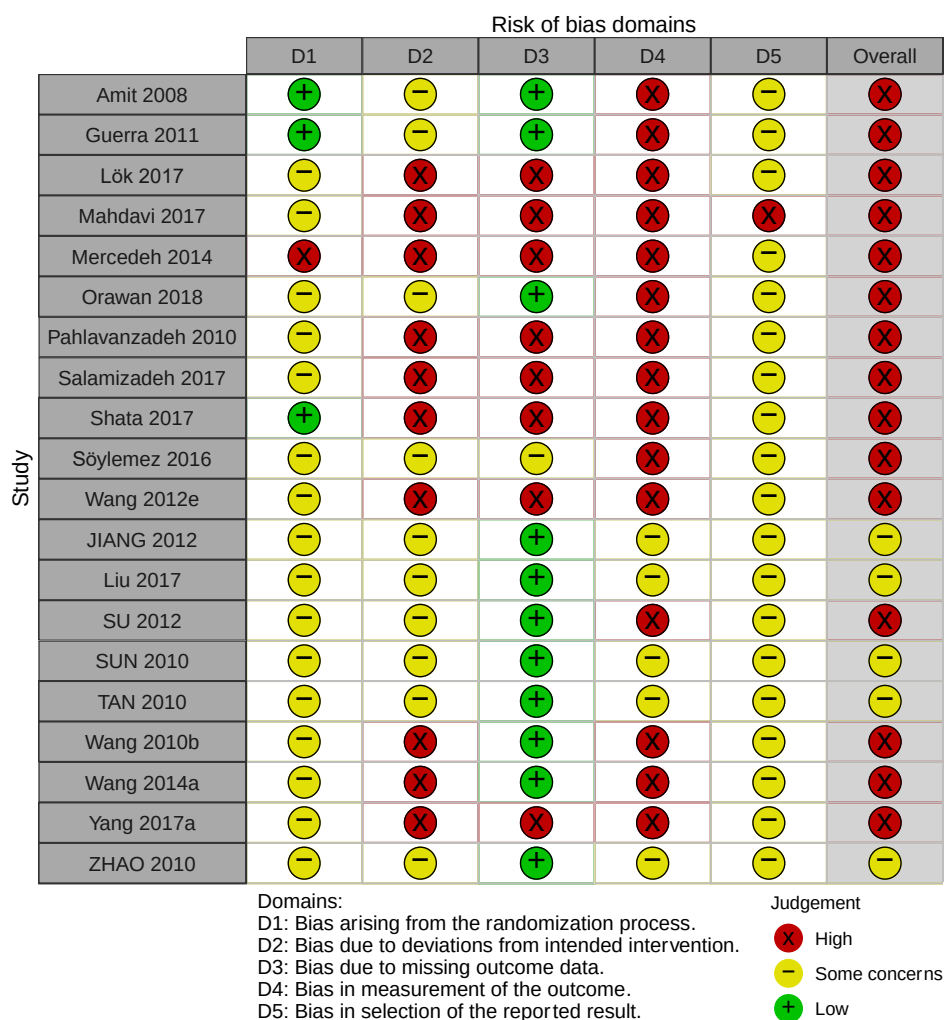


Figure 12 Traffic light plot of risk of bias of 20 previously included studies

For new studies, the risk of bias was assessed in the following domains according to the RoB 2 tool:

- Domain 1: Bias arising from the randomization process
- Domain 2: Bias due to deviations from intended intervention
- Domain 3: Bias due to missing outcome data
- Domain 4: Bias in measurement of outcome
- Domain 5: Bias in selection of the reported result

We followed the suggested algorithms in the RoB 2 tool to perform the assess the risk of bias. An example of the suggested algorithms is shown below:

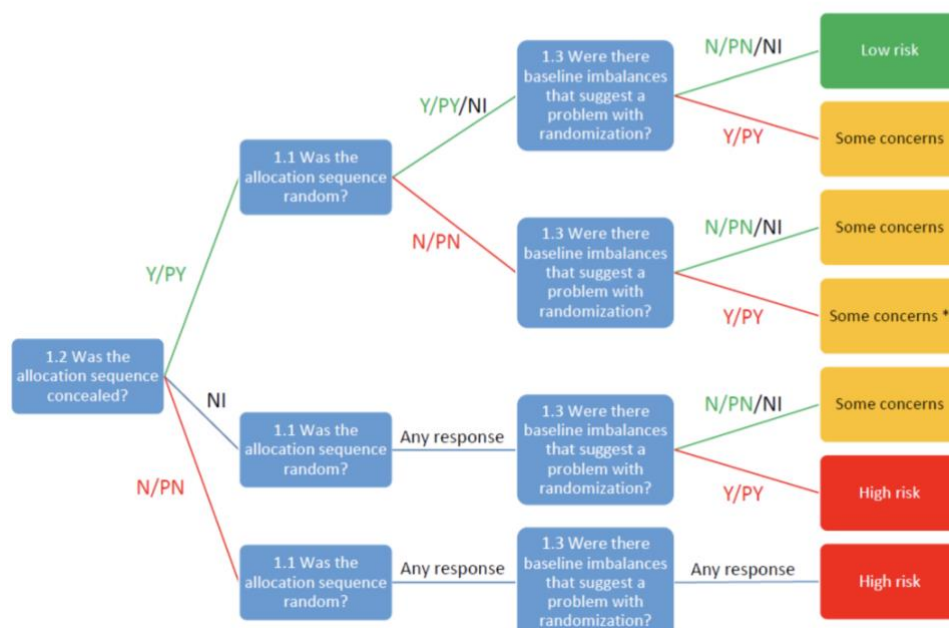


Figure 13 Example of risk of bias decision derived from pre-defined algorithms

Four reviewers jointly discussed the answers and rationales for each question in RoB2. Each group member shared their comments and rationale on risk of bias through a shared Google Drive folder to ensure the judgements were based on the same evidence. Thus, we generated an algorithm result for each domain and each study, which were reviewed by a reviewer to make the final judgement.

Appendix 5.1 Chen 2020

Domain	Judgement	Rationale
Domain 1: Bias arising from the randomization process	Some concerns	Q1.2 Was the allocation sequence concealed? NI Not mentioned in the article Q1.1 Randomisation? Y “According to the random number method, patients were divided into the control group and observation group, 51 cases each.” Q1.3 Baseline differences? N “In the control group, there were 23 males and 28 females (aged 63-82 y, with a mean age of 71.43 ± 5.14 y), the course of disease was from 4 to 12 y (mean duration of 7.44 ± 2.15 y). In the observation group, there were 24 males and 27 females (aged 62-84 y, with a mean age of 72.38 ± 4.27 y); the course of disease was from 5 to 10 y (mean duration of 7.13 ± 1.64 y). There was no statistical difference in the baseline data between these two groups.” Algorithm judgement: Some concerns Reviewer judgement: Some concerns (due to insufficient details about randomisation and allocation concealment)
Domain 2: Bias due to deviations from intended intervention	Some concerns	Q2.1 Participant aware of intervention? Y The “families voluntarily participated in the study and signed an informed consent.” Thus, the participants were aware of the experiment. “In addition to the routine nursing measures of the control group, the following extended nursing measures were taken.” Thus, the participants knew which group they were in because the control group only received routine care, which was the same as the care before the trial. The change of care methods enabled the participants to know which group they were in. Q2.2 Personnel aware of intervention? Y According to the authors, the control group only received routine care, while the interventionists “participated in the study actively and were trained with the extended nursing scheme.” Thus, they should know that they were in the intervention group. Q2.3 Deviation from intervention due to trial context? PN The authors mentioned that “Enhancement of communication among doctors, nurses and patients and if the patients or nursing staff do not appear in group chat for a month, the nursing team members inquired the reasons, informed the significance of active participation in group chat and recommended participation.” Thus, there could be deviations during the study. However, this could happen if there was not any trial in place. Q2.5 Appropriate analysis to estimate the effects of assignment? Y The authors did not report attrition in the main text. Tables 1-3 show that the number of participants one year later was the same as the number of participants randomised. Thus, we assumed there was no attrition due to the intervention. In this case, we assumed all the participants randomised were included and were analysed in their group correctly. Algorithm judgement: Low risk Reviewer judgement: Some concerns (The authors did not explicitly mention that they did not have dropouts. Clarification is needed.)

Domain 3: Bias due to missing outcome data	Some concerns	Q3.1 Were outcome data available for all or nearly all participants randomised? Y The authors did not report any missing data or describe any dropout. The analysed sample sizes in Table 1, Table 2 and Table 3 were all consistent with the initial sample size (n = 51), which also suggests no missing data. Thus, the data were available for all participants randomised. Algorithm judgement: Low risk Reviewer judgement: Some concerns (The authors did not explicitly mention that they did not have dropouts. Clarification is needed.)
Domain 4: Bias in measurement of the outcome	Some concerns	Q4.1 Methods of measuring the outcome inappropriate? N Q4.2 Measurement or ascertainment of outcome differ between groups? N “The behavioral pathology assessment scale of Alzheimer’s disease (BEHAVE-AD) [8] was used in this study. The behavioral and psychological symptoms of the patients were scored by the doctors. [...] Montreal Cognitive Assessment Scale (MoCA)[9] and mini mental state examination scale (MMSE) were used to evaluate the cognitive function of the 2 groups.” All the scales used are well-established methods of outcome measurement. Both groups used the same scales. Q4.3 Outcome assessors aware of intervention received? NI Q4.4 Could assessments have been influenced by knowledge of intervention? NI Q4.5 Likely that assessments have been influenced by knowledge of intervention? NI The authors did not mention who performed the measurement and whether they were masked. There is insufficient information. BEHAVE-AD, MoCA and MMSE were likely to be performed by a doctor, but it was unclear whether it might be an interventionist as well. Algorithm judgement: High risk Reviewer judgement: Some concerns (There was insufficient information to show low risk. However, there was also no sufficient information to indicate obvious flaws or bias.)
Domain 5: Bias in selection of the reported result	Some concerns	Q 5.2 Selected from multiple eligible outcome measurements? NI We didn’t find any pre-specified protocol or analysis plan for the study. Thus, we did not know what outcomes were originally planned to be reported and unable to tell whether there might be selective reporting of outcome measurements. Q 5.3 Selected from multiple analyses data? N Q 5.1 Consistent with pre-specified analysis plan? NI We didn’t find any pre-specified protocol or analysis plan for the study. Analysis intentions were not described in this study. Thus, there is not enough information to enable judgement. Algorithm judgement: Some concerns Reviewer judgement: Some concerns
Overall	Some concerns	Algorithm judgement: Some concerns/High risk Reviewer judgement: Some concerns (All five domains were judged to be some concerns, largely due to insufficient information rather than obvious flaws.)

Appendix 5.2 Ghaffari 2019

Domain	Judgement	Rationale
Domain 1: Bias arising from the randomization process	High risk	Q1.2 Was the allocation sequence concealed? NI Q1.1 Randomisation? NI The eligible participants were “grouped into intervention and control groups by a block-randomized allocation method”, yet more details may be required. There is no clear mentioning of a central computerised programme for randomisation, so the randomisation may not be consistent with the best practice. Also, it is not clear whether the randomisation was concealed. Q1.3 Baseline differences? Y The authors used block randomisation with additional adjustment for age, sex, marital status, occupation, education level, income level, caring duration, smoking situation, and family relationship. - Reviewer 1: when using chi-square and Fisher’s exact tests to recalculate the result from Table 2, it was not possible to get the same result. - Reviewer 2: when visually inspecting the baseline demographic data, one could find major differences in variables including the income group. To be more specific, of 16 high-income participants, 11 were in the intervention group. Algorithm judgement: High risk Reviewer judgement: High risk (unclear allocation sequence concealment + not enough information for randomisation + obvious baseline differences that favours the intervention)
Domain 2: Bias due to deviations from intended intervention	Some concerns	Q2.1 Participant aware of intervention? Y Q2.2 Personnel aware of intervention? Y The participants were aware of the assigned intervention, as they had written consent of the experiment and the control group had no intervention at all. Same case for the people delivering the intervention. Q2.3 Deviation from intervention due to trial context? NI The authors did not mention any deviation from planned interventions or any measures to avoid or correct for the deviation. Thus, there is insufficient information. Q2.5 Appropriate analysis to estimate the effects of assignment? NI There is insufficient information. Although we assume that the authors used ITT analysis, it is unclear how the authors dealt with participants with missing data. Algorithm judgement: High risk Reviewer judgement: Some concerns (There was insufficient information to show low risk. However, there was also no sufficient information to indicate that the authors excluded the patients improperly or that the participants declined because of the true effect.)
Domain 3: Bias due to missing outcome data	Some concerns	Q3.1 Were outcome data available for all or nearly all participants randomised? N “In total, 64 family caregivers of older adults with Alzheimer’s disease completed an initial eligibility assessment. However, 10 family caregivers were excluded from the study based on the exclusion criteria (n = 4) or due to declining to participate (n = 6). [...] Two participants in the intervention group missed two or more program sessions and two participants in the control group missed assessments due to the death of their patients during the program. Therefore, of 54 family caregivers, 50 (92.6%)

		completed all follow-up assessments (25 in the intervention group and 25 in the control group).” Q3.2 Evidence that result is not biased? N The authors did not mention such evidence. Q3.3 Missingness could depend on true value? NI Q3.4 Likely that missingness depended on true value? NI “10 family caregivers were excluded from the study based on the exclusion criteria (n = 4) or due to declining to participate (n = 6).” This needs clarification about the exact reasons why the participants were excluded or declined to participate. “Two participants in the intervention group missed two or more program sessions and two participants in the control group missed assessments due to the death of their patients during the program.” This seems to be irrelevant to the true value. Algorithm judgement: High risk Reviewer judgement: Some concerns (There was insufficient information to show low risk. However, there was also no sufficient information to indicate that the authors excluded the patients improperly or that the participants declined because of the true effect.)
Domain 4: Bias in measurement of the outcome	High risk	Q4.1 Methods of measuring the outcome inappropriate? N Q4.2 Measurement or ascertainment of outcome differ between groups? N “1. The demographic characteristics questionnaire (including age, sex, marital status, occupation, education level, income level, caring duration, smoking habits, and family relationship with the patient). 2. Connor-Davidson Resilience Scale: This instrument is based on a five-point Likert scale (from 0 = absolutely correct to 4 = always correct). The reliability and validity of this questionnaire were approved in Besharat’s research (11) with a Cronbach’s alpha of 0.84. The questionnaire was sent to 10 faculty members for validity assessment and administered to 30 caregivers for reliability assessment. Internal consistency was approved by Cronbach’s alpha of 0.87 and stability with test-retest method (r = 0.92). The demographic characteristics and resilience questionnaires were completed before the intervention in both groups. 3. General Health Questionnaire (GHQ): The GHQ consists of 28 items and four subscales (somatic symptoms, anxiety/insomnia, social dysfunction, and severe depression). Participants receiving a cutoff score of 23 or higher were considered to have probable psychiatric distress (11, 12). The GHQ was completed before and 35 - 40 days after the intervention in both groups.” Q4.3 Outcome assessors aware of intervention received? PY Q4.4 Could assessments have been influenced by knowledge of intervention? PY Connor-Davidson Resilience Scale and General Health Questionnaire are self-reported questionnaires. The self-assessments may be compromised as the patient knew that they were in the intervention group and thus had a positive psychological suggestion. Q4.5 Likely that assessments have been influenced by knowledge of intervention? PY There is no direct evidence of such influence. However, according to the guideline of RoB 2 tool, “when there are strong levels of belief in either beneficial or harmful effects of the intervention, it is more likely that the outcome was influenced by knowledge of the intervention received.” The authors mentioned that “Many resilience studies explore personal characteristics that may affect coping or resilience and focus on individuals experiencing specific adverse circumstances (e.g., illness, bereavement, abuse, etc.). Researchers have

		demonstrated the effectiveness of interventions that assist family caregivers to provide support for patients with dementia (8-10).” Also, intuitively, the intervention tends to be effective. Thus, it was likely that the knowledge affected the assessment. Also, in the Cochrane Handbook, it is mentioned that “Furthermore, risk of bias may be high for some outcomes and low for others, even if the same people were unblinded in the study. For example, knowledge of the assigned intervention may impact on patient-reported outcomes (such as level of pain), while not impacting on other outcomes such as mortality.” Algorithm judgement: High risk Reviewer judgement: High risk (only contained self-reported outcomes, with no participant blinded from the interventions)
Domain 5: Bias in selection of the reported result	Some concerns	Q 5.2 Selected from multiple eligible outcome measurements? NI The study reported caregiver mental health but not caregiver resilience scores, while both were listed as outcomes in the study protocol available at https://www.irct.ir/trial/23558. However, the trial was updated retrospectively. Thus, it is unclear whether the study protocol was the same as the original protocol. Q 5.3 Selected from multiple analyses data? NI The analysis intentions or plan were not reported. There is no study protocol or analysis plan available. Thus, we don’t know whether the selection happened. Algorithm judgement: Some concerns Reviewer judgement: Some concerns (due to insufficient information)
Overall	High risk	Algorithm judgement: High risk Reviewer judgement: High risk (significant baseline differences + bias due to self-reported outcomes + some concerns in three domains)

Appendix 5.3 Govindakumari 2020

Domain	Judgement	Rationale
Domain 1: Bias arising from the randomization process	Some concerns	Q1.2 Was the allocation sequence concealed? NI Q1.1 Randomisation? NI “After recruiting, the participants were randomly grouped into two groups that is control and intervention groups with 157 participants in each group.” The authors claimed that the study was randomised, but no further information about randomisation was given. Q1.3 Baseline differences? NI Only outcomes were measured at baseline (Table 1), which showed no significant difference. However, other relevant information, e.g. demographic information, was lacking. Algorithm judgement: Some concerns Reviewer judgement: Some concerns (due to insufficient details about randomisation, allocation concealment, as well as the baseline data)
Domain 2: Bias due to deviations from intended intervention	Some concerns	Q2.1 Participant aware of intervention? NI Q2.2 Personnel aware of intervention? NI The authors only mentioned that “cognitive training was given to all eligible subjects individually two sessions weekly continuously for a period of 1 month i.e. a total of 8 sessions. Each session will take approximately 1 hour.” They did not mention any placebo or sham intervention or describe what they did to the control group. We are unclear whether the participants or interventionists were blinded from. Q2.3 Deviations from intervention due to trial context? NI Q2.4 Deviations affects outcome? NI Q2.5 Deviations balanced across groups? NI The authors did not report dropout or deviations. There is no sufficient information. Q2.6 Appropriate analysis to estimate the effects of assignment? NI Q2.6 Substantial impact of analyse in wrong groups? NI Unable to tell whether the study used ITT or PP analysis, due to a lack of information regarding deviations. Information and clarification were needed. Algorithm judgement: High risk Reviewer judgement: Some concerns (There was insufficient information to show low risk. However, there was also no sufficient information to indicate that the authors excluded the patients improperly or that the participants declined because of the true effect.)
Domain 3: Bias due to missing outcome data	Some concerns	Q3.1 Were outcome data available for all or nearly all participants randomised? NI Q3.2 Evidence that result is not biased? NI Q3.3 Missingness could depend on true value? NI Q3.4 Likely that missingness depended on true value? NI The authors did not report any missing data or describe any dropout. Thus, the information is lacking. Algorithm judgement: High risk Reviewer judgement: Some concerns (There was insufficient information to show low risk. However, there was also no sufficient information to indicate that the authors excluded the patients improperly or that the participants declined because of the true effect.)

Domain 4: Bias in measurement of the outcome	Some concerns	Q4.1 Methods of measuring the outcome inappropriate? NI Q4.2 Measurement or ascertainment of outcome differ between groups? NI Q4.3 Outcome assessors aware of intervention received? NI Q4.4 Could assessments have been influenced by knowledge of intervention? NI Q4.5 Likely that assessments have been influenced by knowledge of intervention? NI The authors did not describe the scales used to measure the outcomes. Thus, it is unclear whether the measurement was inappropriate. The authors did not provide how measurement was done. Thus, there is no sufficient information to enable judgement. Algorithm judgement: High risk Reviewer judgement: Some concerns (There was insufficient information to show low risk. However, there was also no sufficient information to indicate that the authors excluded the patients improperly or that the participants declined because of the true effect.)
Domain 5: Bias in selection of the reported result	Some concerns	Q 5.2 Selected from multiple eligible outcome measurements? NI The authors did not describe how outcome measurement was done. Thus, there is no sufficient information to enable judgement. Q 5.3 Selected from multiple analyses data? NI Q 5.1 Consistent with pre-specified analysis plan? NI We didn't find any pre-specified protocol or analysis plan for the study. Analysis intentions were not described in this study. Thus, there is not enough information to enable judgement. Algorithm judgement: Some concerns Reviewer judgement: Some concerns (Insufficient information.)
Overall	High risk	Algorithm judgement: High risk Reviewer judgement: High risk (Insufficient information for all domains, suggesting potential poor quality and standard of research reporting)

Appendix 5.4 Hinton 2020

Domain	Judgement	Rationale
Domain 1: Bias arising from the randomization process	Low risk	Q1.2 Was the allocation sequence concealed? Y Q1.1 Randomisation? Y The authors described that “randomization was conducted by one of the study investigators residing in the United States through the flip of a coin.” Thus, the allocation sequence was concealed and random. Also, the published study protocol mentioned that “Randomization will be conducted by one of the principal investigators (PIs) who is not familiar with the characteristics of the communes being randomized.”⁹⁹ Q1.3 Baseline differences? Y “There were no statistically significant differences in the characteristics at baseline except. for the number of hours providing care to the person with dementia each week, which was significantly higher in the enhanced control condition compared with the intervention group (P = .03).” Algorithm judgement: Low risk Reviewer judgement: Low risk
Domain 2: Bias due to deviations from intended intervention	Low risk	Q2.1 Participant aware of intervention? Y Q2.2 Personnel aware of intervention? Y “At the time of consent, participants were informed about their allocation to either the enhanced control or the intervention condition. Research assistants who conducted the 3-month outcome assessment were masked to allocation. Baseline assessments were conducted by research staff who were not masked to allocation.” However, the interventionists were not masked, as the planned intervention was only delivered to the intervention group. Q2.3 Deviations from intervention due to trial context? PN The intervention group showed deviation from protocol. “Based on a review of standardized treatment delivery forms completed by the interventionists at the end of each session for the REACH VN groups, all required elements were completed for 98% of session 1, 96% for session 2, 92% for session 3, 87% for session 4, and 100% of final (closure) sessions.” However, the level of non-attendance may also occur outside of the trial context. The study also attempted to minimise deviation from protocols in the control group through its design. “A cluster design was chosen because of concerns about possible contamination effects if participants in the control and intervention condition resided in the same community.” Q2.4 Deviations affects outcome? PN There is no evidence to indicate so. As the attendance rate was at least 87%, with the non-attendance even less than attribution. Thus, the outcome may not be significantly affected. Q2.6 Appropriate analysis to estimate the effects of assignment? Y The study mentioned that “primary analysis of the intervention effect used an intent-to-treat approach, in which groups were compared based on randomization assignment.” Of 60 participants, only 51 participants who completed the study were analysed, with those with missing data excluded. The sample was the proper sample included in this ITT analysis. Thus, this is the appropriate analysis. Algorithm judgement: Low risk

		Reviewer judgement: Low risk
Domain 3: Bias due to missing outcome data	Low risk	Q3.1 Were outcome data available for all or nearly all participants randomised? N The authors described dropouts in the study. “Of the 60 participants randomized (across nine clusters), 51 completed the study (26 in the control group and 25 in REACH VN intervention).” According to the RoB2 guideline, “appropriate study population for an analysis of the intention to treat effect is all randomized participants...For continuous outcomes, availability of data from 95% of the participants will often be sufficient. “The retention rate in the study was only 85%. Thus, we think the study has significant proportion of missing outcome data. Q3.2 Evidence that result is not biased? NI The authors did not mention such evidence. Q3.3 Missingness could depend on true value? N Q3.4 Likely that missingness depended on true value? N According to the authors, “reasons for caregiver attrition included care recipient passing away (n = 5, two in intervention and three in control), caregiver decided not to participate (n = 3, due to lack of time and other reasons), and one caregiver was found to be ineligible due to significant cognitive impairment.” These reasons were not related to the outcomes or adverse effects of the effects. Also, missingness was largely balanced between intervention groups and there was no direct evidence for dependency on true value. Algorithm judgement: Low risk Reviewer judgement: Low risk
Domain 4: Bias in measurement of the outcome	Low risk	Q4.1 Methods of measuring the outcome inappropriate? N “For preliminary efficacy, the primary outcome was caregiver burden measured by the 4-item Zarit Burden Inventory (ZBI).¹⁸ Secondary outcomes were depression/anxiety symptoms measured by the Patient Health Questionnaire 4 (PHQ4)²⁵ and Alzheimer’s knowledge measured by the 30-item Alzheimer’s Disease Knowledge Scale.²⁶ Primary and secondary outcomes were assessed at baseline and after completion of the intervention (≈3 months after enrollment).” ZBI, GHQ-28, AD knowledge questionnaire were all established outcome measurement methods. Q4.2 Measurement or ascertainment of outcome differ between groups? Y The authors mentioned in the published protocol⁹⁹ that “The feasibility of the study method will be assessed by using accepted guidelines with respect to recruitment, retention, treatment adherence, treatment fidelity, and assessment processes.” Thus, we can infer that the outcome measurement should be the same between groups. Q4.3 Outcome assessors aware of intervention received? PN “Baseline assessments were conducted by research staff who were not masked to allocation,” yet “assessments at 3 mo were conducted in the participants’ homes by bachelor’s level research staff masked to allocation”.

		The baseline data were expected to be similar due to randomisation, which minimised the impact of non-blinding of assessors. Post-intervention measurement was blinded. Thus, the risk of bias in data collection was low. Although ZBI, GHQ-28, AD knowledge questionnaire was all self-reported scales, the cluster randomisation design blinded the participants from other interventions. Algorithm judgement: Low risk Reviewer judgement: Low risk
Domain 5: Bias in selection of the reported result	Low risk	Q 5.2 Selected from multiple eligible outcome measurements? N In the published protocol:⁹⁹ “While the study will not be powered to examine efficacy, we will conduct analyses to examine caregiver outcomes. For the exploratory analyses of intervention effectiveness, we will use a 4-item version of the Zarit Burden Scale as the primary outcome for caregivers.[18] Secondary caregiver outcomes for this analysis will include change in depressive symptoms as measured by the Patient Health Questionnaire 4 (PHQ-4)[21] and change in Alzheimer disease knowledge.” In the published trial: “There were no significant baseline differences between the two groups in primary or secondary outcomes, including the ZBI (P = .9), PHQ-4 (P = .5), and Alzheimer’s knowledge (P = .7) (see Table 2)...” Thus, the authors reported all eligible outcomes. Q 5.3 Selected from multiple analyses data? N In the published protocol:⁹⁹ “Data will be analyzed using Stata 15 (StataCorp, College Station, TX)[22] or a similar statistical software to generate all statistical reports. Data will be analyzed on an intention to treat basis. For continuous data, the normality of data distribution will be tested using the Shapiro-Wilk test. Mean and standard deviation (SD) will be presented for normally distributed data while median and interquartile range (IQR) will be used otherwise. For categorical data, frequencies and percentages will be presented. To evaluate efficacy, a general linear model, analogous to an ANCOVA model, which includes the baseline assessment as a covariate, will be used to compare the groups; this model will further incorporate a cluster random effect to account for the clustered design of the trial.” In the published trial: “Primary analysis of the intervention effect used an intent-to-treat approach, in which groups were compared based on randomization assignment. To account for the clustering of participants by commune, mixed-effects models that included a cluster-specific random effect were used to estimate differences between groups. Models utilized the postintervention assessment as the outcome and included the baseline assessment as a covariate. The post-intervention assessment was divided by the standard deviation (SD) in the entire sample at baseline, so that interpretation of model coefficients was in terms of standard deviation units. Model assumptions were checked and met by the data. All analyses were conducted in SAS, version 9.4, and a P-value < .05 was considered statistically significant.” Q 5.1 Consistent with pre-specified analysis plan? Y See above. The study had published a study protocol⁹⁹ and an analysis plan.⁴¹ The plan was updated on 1 March 2020, which was later than the study completion date 31 October 2018. All outcomes were reported as stated in the methods section and the information on clinicaltrials.org. Algorithm judgement: Low risk Reviewer judgement: Low risk
Overall	Low risk	Algorithm judgement: Low risk Reviewer judgement: Low risk (low risk in all domains)

Appendix 5.5 Pan 2019

Domain	Judgement	Rationale
Domain 1: Bias arising from the randomization process	Low risk	Q1.2 Was the allocation sequence concealed? N “The researcher was initially blinded to the study allocation but supervised the intervention; data collectors were blinded to the treatment assignments; and caregivers received a sealed letter with an allocated number and were informed of their treatment when the intervention began.” Q1.1 Randomisation? N “After the initial assessment, caregivers were randomly allocated by a data processor into either the intervention or control group via a computerized random number, with 56 caregivers in each group.” Q1.3 Baseline differences? N Table 2 show the baseline data in the two groups, which doesn’t show any significant differences. Algorithm judgement: Low risk Reviewer judgement: Low risk
Domain 2: Bias due to deviations from intended intervention	High risk	Q2.1 Participant aware of intervention? Y “Caregivers received a sealed letter with an allocated number and were informed of their treatment when the intervention began” Q2.2 Personnel aware of intervention? Y “The researcher was initially blinded to the study allocation but supervised the intervention.” Q2.3 Deviations from intervention due to trial context? NI Q2.4 Deviations affects outcome? NI Q2.5 Deviations balanced across groups? N The authors mentioned that “although nurses were instructed to follow the protocol, minimal flexibility was allowed to adapt to individual caregivers” However, the authors didn’t further describe the deviations. “Family caregivers of PWD (N = 112; intervention group n = 56, control group n = 56) were recruited from the 276 caregivers referred and were evaluated at Time 1. At Time 2, 99 caregivers (intervention group n = 54, control group n = 45) completed the survey; and at Time 3, 82 caregivers (intervention group n = 47, control group n = 35) completed the full three evaluations. Among dropouts in the intervention group, two caregivers’ PWD died, six caregivers discontinued caregiving, and one caregiver relocated; in the control group, seven caregivers’ PWD died, three caregivers discontinued caregiving, eight caregivers relocated, and three caregivers refused the repeated surveys.” Here the authors still fail to explain the reasons for discontinued caregiving. The deviations were not balanced, as the control group had much more dropouts. Q2.6 Appropriate analysis to estimate the effects of assignment? N The guideline of RoB2 tool suggests that “both intention-to-treat (ITT) analyses and modified intention-to-treat (mITT) analyses excluding participants with missing outcome data should be considered appropriate.” The study used ITT analysis, where missing data were imputed using the missForest method. Thus, the analysis was different from the guideline. Moreover, it should be

		noted that there were more dropouts in the control group than in the intervention group. Thus, the algorithm didn't have a balanced impact on the data of two groups. Algorithm judgement: High risk Reviewer judgement: High risk (imbalanced deviations between groups + imbalanced impact of the machine learning algorithm on missing data)
Domain 3: Bias due to missing outcome data	Some concerns	Q3.1 Were outcome data available for all or nearly all participants randomised? N "Family caregivers of PWD (N = 112; intervention group n = 56, control group n = 56) were recruited from the 276 caregivers referred and were evaluated at Time 1. At Time 2, 99 caregivers (intervention group n = 54, control group n = 45) completed the survey; and at Time 3, 82 caregivers (intervention group n = 47, control group n = 35) completed the full three evaluations." Q3.2 Evidence that result is not biased? NI No such evidence was mentioned. Q3.3 Missingness could depend on true value? Y Q3.4 Likely that missingness depended on true value? NI "Among dropouts in the intervention group, two caregivers' PWD died, six caregivers discontinued caregiving, and one caregiver relocated; in the control group, seven caregivers' PWD died, three caregivers discontinued caregiving, eight caregivers relocated, and three caregivers refused the repeated surveys." Here the authors still fail to explain the reasons for discontinued caregiving. The deviations were not balanced, as the control group had much more dropouts. Algorithm judgement: Some concerns Reviewer judgement: Some concerns (due to unclear reasons for discontinuing the intervention)
Domain 4: Bias in measurement of the outcome	High risk	Q4.1 Methods of measuring the outcome inappropriate? N "Depression. The 10-item CES-D was used to assess frequency of depressive symptoms among caregivers. One example of an item was "I felt depressed." Items were rated on a 4-point response format from 0 (rarely or none of the time) to 3 (most of the time or all of the time). The scale score is the total of the item scores, with ≥ 10 indicating the presence of clinical depression. The Chinese version of the CES-D had a Cronbach's alpha of 0.79 among Chinese caregivers (Pan et al., 2016). Mutuality. The 15-item Mutuality Scale was used to measure caregiver mutuality. One example of an item was "To what extent do you enjoy the time the two of you spend together?" Items were rated on a 5-point Likert scale ranging from 0 (not at all) to 4 (a great deal). The scale score is the mean of the item scores; higher scores indicate higher levels of mutuality. The Chinese version of the scale had a Cronbach's alpha of 0.91 among caregivers in China (Pan et al., 2016). The Simplified Coping Scale. The 20-item Simplified Coping Scale (Xie, 1998) was developed to measure coping strategies of Chinese adults, with a Cronbach's alpha of 0.90 for the full scale, 0.89 for the 12-item subscale of active coping, and 0.78 for the eight-item subscale of passive coping. Examples of items included "Seeking suggestions from friends and relatives" and "Accepting the reality as there are no other choices." Items were rated on a 4-point scale ranging from 0 (never used) to 3 (often used). The subscale scores are the respective averages of their items. A higher score in active coping indicates better coping, whereas a higher score in passive coping indicates poorer coping. Care Recipients' Activities of Daily Living. The 14-item Activities of Daily Living (ADL) scale measures the basic and more complicated activities of patients' self-maintenance. One example of an item was "Are you having difficulty eating?" Items were rated on a 4-point scale from 1 (performed without difficulty) to 4

		(unable to perform).The scale score was the sum of the item scores. The Chinese version of the 14-item ADL scale (ADL-C) was used among Chinese family caregivers, with a Cronbach's alpha of 0.97 (Pan et al., 2016). Care Recipients' Mini Mental State Examination. The Chinese version of the MMSE was used to assess patients' cognitive function. This tool has been widely tested with satisfactory psychometrics and applied to older adults in China (Yu, Li, & Huang, 2012). The MMSE comprises 10 cognitive function domains, such as memory, orientation, and calculation, with a maximum of 30 points. Higher scores indicate better cognitive function. The cutoff for diagnosing cognitive impairment is 16 to 19 for individuals with primary education and below; thus, a 17-point cut-off was adopted in the current study (Yang et al., 2016). Demographic and Caregiving Characteristics. A questionnaire designed by the researchers was used to collect demographic data of caregivers and care recipients." Q4.2 Measurement or ascertainment of outcome differ between groups? NI The authors did not mention this. Q4.3 Outcome assessors aware of intervention received? N "Data collectors were blinded to the treatment assignments." However, self-report scales such as CES-D must be assessed by the participants themselves. Algorithm judgement: High risk Reviewer judgement: High risk
Domain 5: Bias in selection of the reported result	Some concerns	Q 5.2 Selected from multiple eligible outcome measurements? NI The analysis intentions were not reported. There is no study protocol or analysis plan available. Thus, we don't know whether the selection happened. Q 5.3 Selected from multiple analyses data? NI Q 5.1 Consistent with pre-specified analysis plan? NI The analysis intentions were not reported. There is no study protocol or analysis plan available. Thus, we don't know whether the selection happened. Algorithm judgement: Some concerns Reviewer judgement: Some concerns (There was insufficient information to show low risk. However, there was also no sufficient information to indicate obvious flaws or bias.)
Overall	High risk	Algorithm judgement: High risk Reviewer judgement: High risk (due to imbalanced missingness across groups)

Appendix 5.6 Uyar 2019

Domain	Judgement	Rationale
Domain 1: Bias arising from the randomization process	High risk	Q1.2 Was the allocation sequence concealed? NI No information provided in the article. Q1.1 Randomisation? Y “Randomisation criteria were dementia type, dementia stage and caregivers’ education level. We stratified the patients and caregivers based on the education status of caregivers, considering the randomisation criteria. Using simple random placement method, we determined groups of caregivers to be assigned to the groups. The patient–caregiver pairs were assigned to the randomised IG (n = 30) and CG (n = 31)” Q1.3 Baseline differences? Y Table 2 shows the baseline demographic data. Marital status showed significant difference between the control and intervention groups. Algorithm judgement: High risk Reviewer judgement: High risk (due to baseline differences and the lack of information for allocation sequence concealment)
Domain 2: Bias due to deviations from intended intervention	Some concerns	Q2.1 Participant aware of intervention? Y “We obtained written informed consent from all caregivers and some of the patients through the Volunteer Form.” Thus, the participants knew the intervention, which was different from the routine care they typically received. Q2.2 Personnel aware of intervention? Y The interventionists provided structured training and support, which was different from the routine care they typically provided. Thus, they should be aware of the intervention. Q2.3 Deviations from intervention due to trial context? NI Q2.4 Deviations affects outcome? NI Q2.5 Deviations balanced across groups? Y Figure 1 shows that the control group had two dropouts and two deaths, while the intervention group had 1 deaths and 2 who refused, which were balanced between groups. However, further detailed reasons about the dropouts were needed. Q2.6 Appropriate analysis to estimate the effects of assignment? Y Table 1-3 shows that the study used ITT analysis, excluding the participants with missing data. This was an appropriate method according to the guideline of RoB 2 tool, which says, “both intention-to-treat (ITT) analyses and modified intention-to-treat (mITT) analyses excluding participants with missing outcome data should be considered appropriate.” Algorithm judgement: Some concerns Reviewer judgement: Some concerns (lack of information for dropout reasons)
Domain 3: Bias due to missing outcome data	High risk	Q3.1 Were outcome data available for all or nearly all participants randomised? N Figure 1 shows that the control group had two dropouts and two deaths, while the intervention group had 1 death and 2 who refused. Table 2 shows that different number of PwD measuring QoL-AD and NPI-S, suggesting potential missing data.

		Q3.2 Evidence that result is not biased? NI No such evidence was mentioned. Q3.3 Missingness could depend on true value? Y Q3.4 Likely that missingness depended on true value? NI No such evidence was mentioned. Insufficient information to enable judgment. Algorithm judgement: High risk Reviewer judgement: High risk
Domain 4: Bias in measurement of the outcome	High risk	Q4.1 Methods of measuring the outcome inappropriate? N “Quality of Life Scale SF36 comprises eight subdimensions and defines two summary areas: mental health and physical health. The total score can be 0–100. Koçyiğit et al. (1999) assessed the Turkish reliability and validity of the scale. Cronbach’s α of the scale was 0.73–0.76 for sub-dimensions of the scale (14); in this study, it was 0.73–0.88. The Beck Depression Inventory (BDI) was developed by Beck et al. (1961) and enables the assessment of both depressive symptoms and cognitive content. The total score can be 0–63 points. Hisli (1989) assessed the Turkish validity and reliability. The total score can be 0–63 points. Cronbach’s α was 0.80 (15); in this study, it was 0.80. Beck Anxiety Inventory (BAI) is a 21-item Likerttype scale, developed by Beck et al. (1988), to determine the frequency of anxiety symptoms of individuals. The severity of anxiety increases as the total score of the scale increases. The total score can be 0–63 points. Ulusoy (1998) established the Turkish validity and reliability of the scale. Cronbach’s α of the scale was 0.93 (16); in this study, it was 0.87. Zarit Burden Interview (ZBI) the most widely used in caregiver burden. It encompasses the physical, emotional and financial burden as perceived by the caregiver. It asks 22 questions that are rated on a five-point scale (0 = not at all to 4 = nearly always). Cronbach’s α for the validity luck study of the scale was 0.95 (17); in this study, it was 0.91.” Q4.2 Measurement or ascertainment of outcome differ between groups? NI The authors did not mention this. Q4.3 Outcome assessors aware of intervention received? PY SF36, BDI, BAI, ZBI were all self-reported data. The assessors should be the participants themselves. As the study did not describe allocation sequence concealment and may likely be unblinded, the participants should know their intervention when they assess themselves. Q4.4 Assessment have been influenced by knowledge of intervention? PY Q4.5 Assessment likely influenced by knowledge of intervention? PY The knowledge of the assigned intervention may affect self-reported data, as those in the intervention arm tend to be believe that the intervention was effective and thus report better conditions, while those assigned to the control group may feel that the intervention was ineffective, thus reporting in a negative way. Algorithm judgement: High risk Reviewer judgement: High risk (due to self-reported scales)

Domain 5: Bias in selection of the reported result	Some concerns	Q 5.2 Selected from multiple eligible outcome measurements? NI The analysis intentions were not reported. There is no study protocol or analysis plan available. Thus, we don't know whether the selection happened. Q 5.3 Selected from multiple analyses data? NI Q 5.1 Consistent with pre-specified analysis plan? NI The analysis intentions were not reported. There is no study protocol or analysis plan available. Thus, we don't know whether the selection happened. Algorithm judgement: Some concerns Reviewer judgement: Some concerns (There was insufficient information to show low risk. However, there was also no sufficient information to indicate obvious flaws or bias.)
Overall	High risk	Algorithm judgement: High risk Reviewer judgement: High risk (due to baseline differences + self-reported scales)

Appendix 5.7 Zarepour 2020

Domain	Judgement	Rationale
Domain 1: Bias arising from the randomization process	Some concerns	Q1.2 Was the allocation sequence concealed? Y Q1.1 Randomisation? Y The study was randomised using “block randomization with a random sequence of 2 block sizes,” via a “block design software.” Thus, we can consider it a random, concealed. Q1.3 Baseline differences? NI The baselines for anxiety score were not significantly different. For other metrics, block randomisation was used to minimise the differences during randomisation. However, more data, such as age, would be helpful to validate the claim. Algorithm judgement: Some concerns Reviewer judgement: Some concerns
Domain 2: Bias due to deviations from intended intervention	Some concerns	Q2.1 Participant aware of intervention? Y The authors said that “The caregivers signed written informed consents. Besides, they received verbal and written information about the study objectives, design, and duration.” Thus, when the participants were aware when they received the intervention. Q2.2 Personnel aware of intervention? Y The control group did not receive any formal education, as compared with the intervention group. Thus, the interventionists, who were supposed to deliver the education, were aware of the intervention. Q2.3 Deviations from intervention due to trial context? NI The authors mentioned that “the stabilizing of information provided by the researcher to the family caregivers of the intervention group during the 4-week training sessions [..]” As correction for deviation part of the protocol, it is reasonable to believe that the deviations were minimised. Q2.6 Appropriate analysis to estimate the effects of assignment? Y The study used ITT analysis and reported no subjects with missing data. Thus, the authors didn’t need to exclude any participants in the analysis, and the method was appropriate. Algorithm judgement: Some concerns Reviewer judgement: Some concerns (lack of information for dropout reasons)
Domain 3: Bias due to missing outcome data	Low risk	Q3.1 Were outcome data available for all or nearly all participants randomised? N In Figure 1, the authors reported that the analysed sample size and the initial sample size were all 35 in each group. Thus, there were no subjects with missing data. Thus, Table 3 reported all the outcome data for all randomised subjects.
Domain 4: Bias in measurement of the outcome	High risk	Q4.1 Methods of measuring the outcome inappropriate? N “The psychometric properties of the inventory in Iranian population were confirmed by exploratory and confirmatory factor analysis. The correlation between the inventory and clinical specialists’ assessment also showed appropriate validity.¹⁹

		The reliability of the Iranian form of the inventory was Cronbach's alpha 0.88.2" Thus, the method was well-established in this population. Q4.2 Measurement or ascertainment of outcome differ between groups? NI The authors did not mention this. Q4.3 Outcome assessors aware of intervention received? PY The Spielberger anxiety questionnaire was completed by the caregivers themselves. Thus, the assessors knew the intervention. Q4.4 Assessment have been influenced by knowledge of intervention? PY Q4.5 Assessment likely influenced by knowledge of intervention? PY Figure 2 shows worsening anxiety scores for the control group, suggesting that the knowledge of intervention may affect the outcome. Algorithm judgement: High risk Reviewer judgement: High risk (due to self-reported scales)
Domain 5: Bias in selection of the reported result	Some concerns	Q 5.2 Selected from multiple eligible outcome measurements? NI The analysis intentions were not reported. There is no study protocol or analysis plan available. Thus, we don't know whether the selection happened. Q 5.3 Selected from multiple analyses data? NI Q 5.1 Consistent with pre-specified analysis plan? NI Although we didn't find the pre-specified analysis plan for this study, the clinical trial protocol of this study can be found at https://www.irct.ir/trial/14691 Algorithm judgement: Some concerns Reviewer judgement: Some concerns (There was insufficient information to show low risk. However, there was also no sufficient information to indicate obvious flaws or bias.)
Overall	High risk	Algorithm judgement: High risk Reviewer judgement: High risk

APPENDIX 6 META-REGRESSION

Year of publication

Mixed-Effects Model (k = 12; tau² estimator: REML)

tau² (estimated amount of residual heterogeneity): 0.9047 (SE = 0.5252)

tau (square root of estimated tau² value): 0.9511

I² (residual heterogeneity / unaccounted variability): 85.42%

H² (unaccounted variability / sampling variability): 6.86

R² (amount of heterogeneity accounted for): 0.00%

Test for Residual Heterogeneity:

QE(df = 10) = 60.0780, p-val < .0001

Test of Moderators (coefficient 2):

QM(df = 1) = 0.7939, p-val = 0.3729

Model results

	estimate	se	zval	pval	ci.lb	ci.ub
intrcpt	151.3695	171.1902	0.8842	0.3766	-184.1571	486.8960
Year	-0.0757	0.0850	-0.8910	0.3729	-0.2422	0.0908

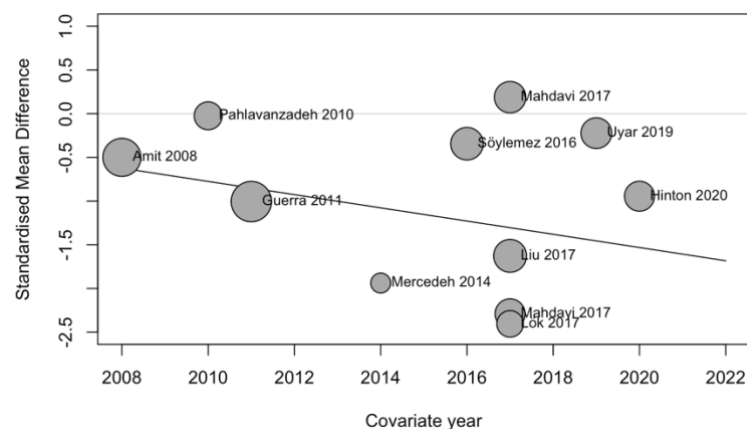


Figure 14 Bubble plot of meta-regression for effects on caregiver burden: Year of publication

Total attrition rate

Warning: Studies with NAs omitted from model fitting.

Mixed-Effects Model (k = 10; tau² estimator: REML)

tau² (estimated amount of residual heterogeneity): 0.9006 (SE = 0.5601)

tau (square root of estimated tau² value): 0.9490

I² (residual heterogeneity / unaccounted variability): 86.32%

H² (unaccounted variability / sampling variability): 7.31

R² (amount of heterogeneity accounted for): 1.73%

Test for Residual Heterogeneity:

QE(df = 8) = 64.9417, p-val < .0001

Test of Moderators (coefficient 2):

QM(df = 1) = 1.1377, p-val = 0.2861

Model results

	estimate	se	zval	pval	ci.lb	ci.ub	
intrcpt	-1.4587	0.5341	-2.7310	0.0063	-2.5057	-0.4118	**
Total_attrition.percent	0.0440	0.0413	1.0666	0.2861	-0.0369	0.1249	

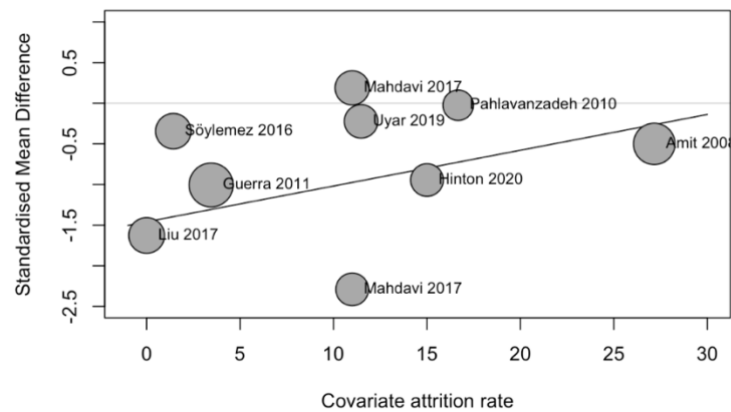


Figure 15 Bubble plot of meta-regression for effects on caregiver burden: Attrition rate

Measurement point

Mixed-Effects Model (k = 12; tau² estimator: REML)

tau² (estimated amount of residual heterogeneity): 0.9056 (SE = 0.5275)

tau (square root of estimated tau² value): 0.9517

I² (residual heterogeneity / unaccounted variability): 84.01%

H² (unaccounted variability / sampling variability): 6.25

R² (amount of heterogeneity accounted for): 0.00%

Test for Residual Heterogeneity:

QE(df = 10) = 60.5491, p-val < .0001

Test of Moderators (coefficient 2):

QM(df = 1) = 0.8540, p-val = 0.3554

Model Results:

	estimate	se	zval	pval	ci.lb	ci.ub	
intrcpt	-1.6407	0.6068	-2.7040	0.0069	-2.8299	-0.4515	**
Total_attrition.percent	0.0054	0.0059	0.9241	0.3554	-0.0061	0.0170	

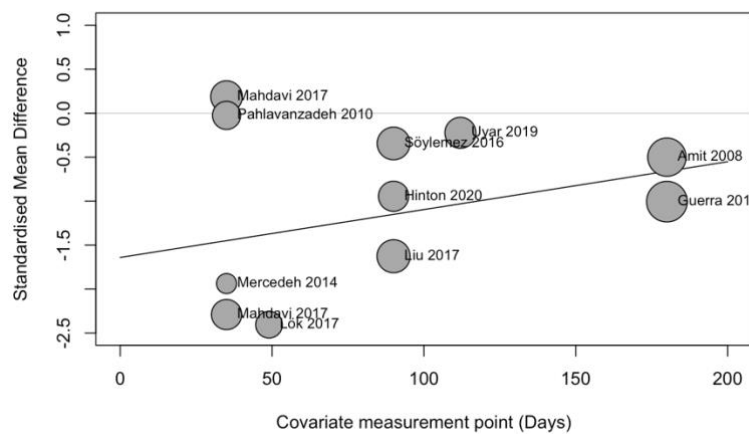


Figure 16 Bubble plot of meta-regression for effects on caregiver burden: Measurement point

Treatment duration

Mixed-Effects Model (k = 12; tau² estimator: REML)

tau² (estimated amount of residual heterogeneity): 0.9056 (SE = 0.5275)

tau (square root of estimated tau² value): 0.9517

I² (residual heterogeneity / unaccounted variability): 84.01%

H² (unaccounted variability / sampling variability): 6.25

R² (amount of heterogeneity accounted for): 0.00%

Test for Residual Heterogeneity:

QE(df = 10) = 60.5491, p-val < .0001

Test of Moderators (coefficient 2):

QM(df = 1) = 0.8540, p-val = 0.3554

Model Results:

	estimate	se	zval	pval	ci.lb	ci.ub	
intrcpt	-1.6407	0.6068	-2.7040	0.0069	-2.8299	-0.4515	**
Measurement.point	0.0054	0.0059	0.9241	0.3554	-0.0061	0.0170	

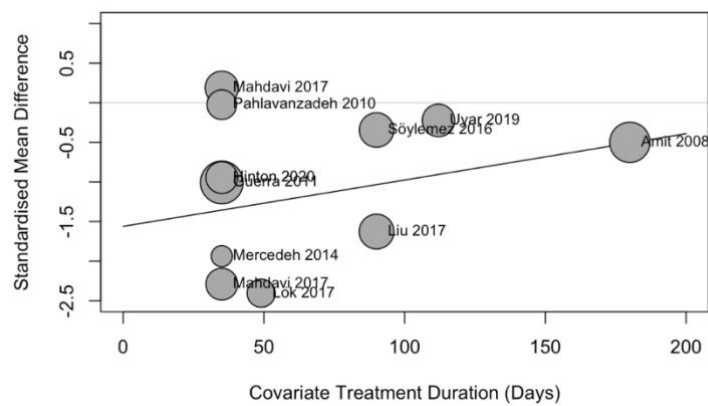


Figure 17 Bubble plot of meta-regression for effects on caregiver burden: Treatment duration

Caregivers' mean age

Warning: Studies with NAs omitted from model fitting.

Mixed-Effects Model (k = 11; tau² estimator: REML)

tau² (estimated amount of residual heterogeneity): 1.0457 (SE = 0.6005)

tau (square root of estimated tau² value): 1.0226

I² (residual heterogeneity / unaccounted variability): 88.87%

H² (unaccounted variability / sampling variability): 8.99

R² (amount of heterogeneity accounted for): 0.00%

Test for Residual Heterogeneity:

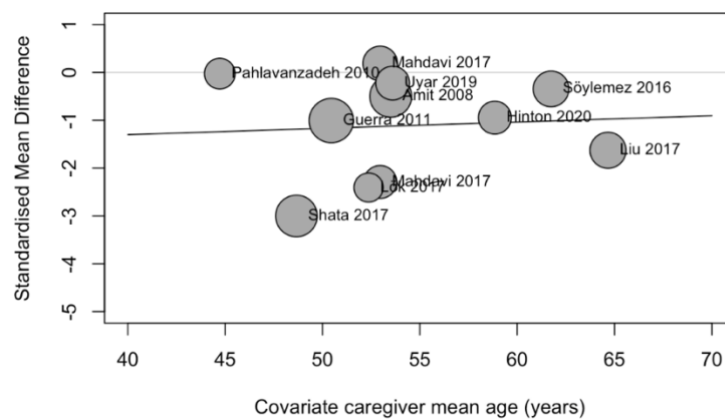
QE(df = 9) = 72.8532, p-val < .0001

Test of Moderators (coefficient 2):

QM(df = 1) = 0.0444, p-val = 0.8331

Model Results:

	estimate	se	zval	pval	ci.lb	ci.ub
intrcpt	-1.8301	3.4102	-0.5367	0.5915	-8.5139	4.8537
Caregiver.mean_age	0.0132	0.0628	0.2107	0.8331	-0.1099	0.1363



**Figure 18 Bubble plot of meta-regression for effects on caregiver burden:
Mean age of caregivers**

Patients' mean ages

Warning: Studies with NAs omitted from model fitting.

Mixed-Effects Model (k = 10; tau^2 estimator: REML)

tau^2 (estimated amount of residual heterogeneity): 0.9454 (SE = 0.5908)

tau (square root of estimated tau^2 value): 0.9723

I^2 (residual heterogeneity / unaccounted variability): 85.21%

H^2 (unaccounted variability / sampling variability): 6.76

R^2 (amount of heterogeneity accounted for): 8.26%

Test for Residual Heterogeneity:

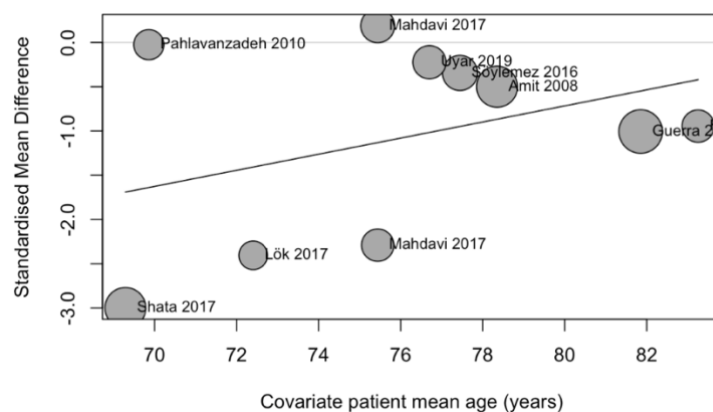
QE(df = 8) = 50.1654, p-val < .0001

Test of Moderators (coefficient 2):

QM(df = 1) = 1.3712, p-val = 0.2416

Model Results:

	estimate	se	zval	pval	ci.lb	ci.ub
intrcpt	-7.9971	5.9301	-1.3486	0.1775	-19.6198	3.6256
Patient.mean_age	0.0910	0.0777	1.1710	0.2416	-0.0613	0.2434



**Figure 19 Bubble plot of meta-regression for effects on caregiver burden:
Mean age of patients**

Female caregiver percentage

Warning: Studies with NAs omitted from model fitting.

Mixed-Effects Model (k = 10; tau^2 estimator: REML)

tau^2 (estimated amount of residual heterogeneity): 0.6762 (SE = 0.4607)

tau (square root of estimated tau^2 value): 0.8223

I^2 (residual heterogeneity / unaccounted variability): 81.63%

H^2 (unaccounted variability / sampling variability): 5.44

R^2 (amount of heterogeneity accounted for): 17.45%

Test for Residual Heterogeneity:

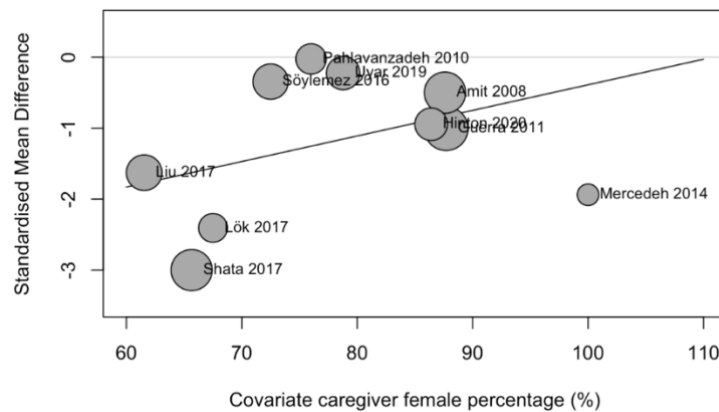
QE(df = 8) = 37.5723, p-val < .0001

Test of Moderators (coefficient 2):

QM(df = 1) = 1.6439, p-val = 0.1998

Model Results:

	estimate	se	zval	pval	ci.lb	ci.ub
intrcpt	-3.9937	2.2096	-1.8074	0.0707	-8.3244	0.3370
Caregiver.female_percent	0.0361	0.0281	1.2821	0.1998	-0.0191	0.0912



**Figure 20 Bubble plot of meta-regression for effects on caregiver burden:
Female% of caregivers**

Female patient percentage

Warning: Studies with NAs omitted from model fitting.

Mixed-Effects Model (k = 9; tau² estimator: REML)

tau² (estimated amount of residual heterogeneity): 1.0021 (SE = 0.6545)

tau (square root of estimated tau² value): 1.0010

I² (residual heterogeneity / unaccounted variability): 87.22%

H² (unaccounted variability / sampling variability): 7.82

R² (amount of heterogeneity accounted for): 0.00%

Test for Residual Heterogeneity:

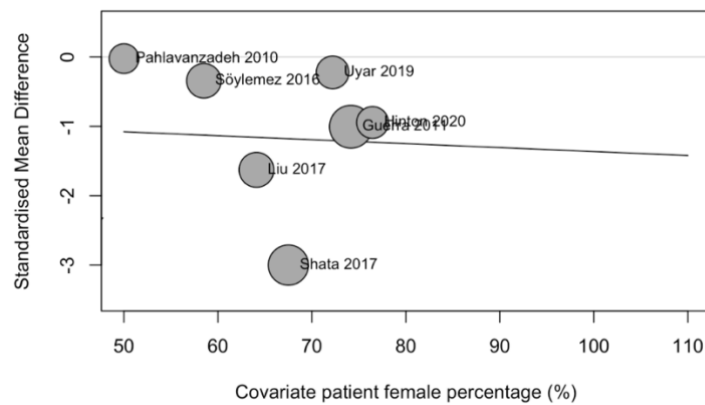
QE(df = 7) = 63.4283, p-val < .0001

Test of Moderators (coefficient 2):

QM(df = 1) = 0.0498, p-val = 0.8234

Model Results:

	estimate	se	zval	pval	ci.lb	ci.ub
intrcpt	-0.7915	1.5820	-0.5003	0.6168	-3.8921	2.3091
Patient.female_percent	-0.0057	0.0256	-0.2232	0.8234	-0.0559	0.0445



**Figure 21 Bubble plot of meta-regression for effects on caregiver burden:
Female% of patients**

APPENDIX 7 SUBGROUP ANALYSIS RESULTS

Table 6 Summary of subgroup analysis results

Variables	# of studies	Hedge's <i>g</i>	95% CI		PI		<i>p</i>	<i>I</i> ²	<i>p</i> _{subgroup}
			lower	upper	lower	upper			
Intervention type (pre-defined)									0.0277*
• PEI	4	-0.90	-1.72	-0.08	-4.35	2.55	0.03	0.63	
• GS	1	0.19	-0.83	1.21	NA	NA	0.72	NA	
• Misc.	1	-2.29	-3.41	-1.17	NA	NA	0.00	NA	
• CBT	1	-1.94	-3.84	-0.03	NA	NA	0.05	NA	
• MCI	4	-1.20	-2.49	0.08	-7.24	4.83	0.07	0.94	
• CCI	1	-1.63	-2.55	-0.70	NA	NA	0.00	NA	
Country (exploratory)									0.7466
• Turkey	3	-0.93	-2.32	0.46	-17.39	15.54	0.19	0.72	
• Iran	4	-0.95	-2.20	0.31	-6.36	4.47	0.14	0.77	
• Vietnam	1	-0.94	-3.35	1.46	NA	NA	0.44	NA	
• China	1	-1.63	-3.95	0.69	NA	NA	0.17	NA	
• Peru	1	-1.01	-3.15	1.14	NA	NA	0.36	NA	
• India	1	-0.50	-2.69	1.69	NA	NA	0.65	NA	
• Egypt	1	-3.00	-5.19	-0.81	NA	NA	0.01	NA	
Source of funding (exploratory)									0.3136
• Unknown	2	-1.22	-2.70	0.26	NA	NA	0.11	0.84	
• University	6	-0.95	-1.80	-0.10	-3.65	1.75	0.03	0.70	
• National government	1	-0.94	-2.99	1.10	NA	NA	0.36	NA	
• International organisation	2	-0.76	-2.00	0.48	NA	NA	0.23	0.66	
Intervention frequency (exploratory)									0.2333
• Weekly	8	-1.43	-2.26	-0.59	-4.24	1.39	0.00	0.89	
• Bi-weekly	1	-0.34	-1.29	0.60	NA	NA	0.48	NA	
• Multiple	2	-1.00	-2.09	0.10	NA	NA	0.08	0.77	
• Multiple	1	-0.22	-1.29	0.85	NA	NA	0.68	NA	

APPENDIX 8 ADDITIONAL META-ANALYSES

Caregiver QoL

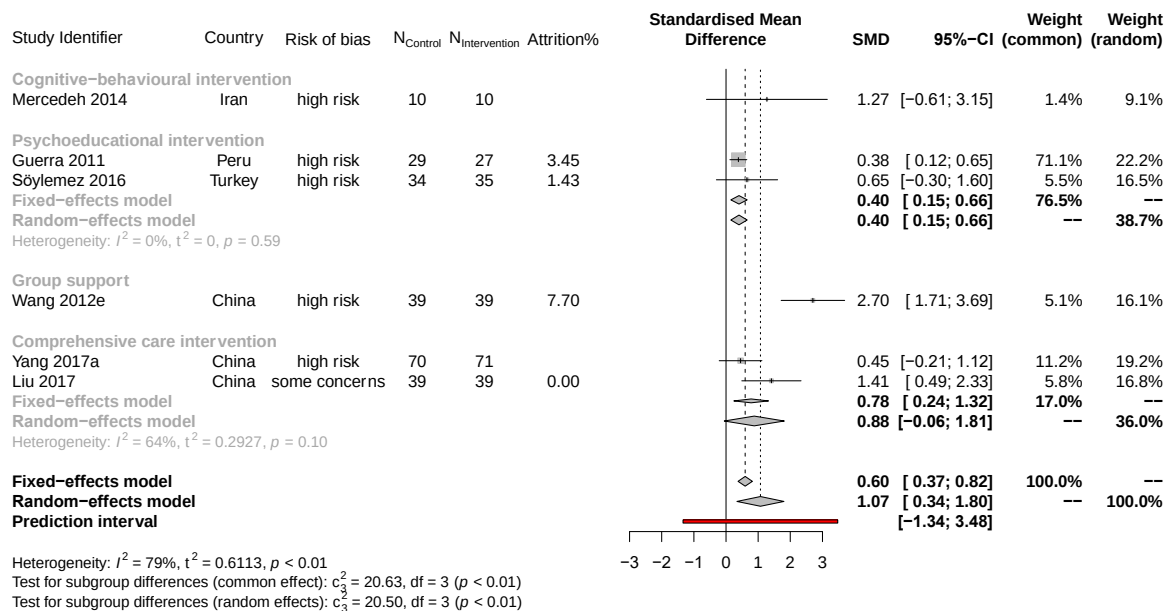


Figure 22 Forest plot of meta-analysis of effects on caregiver QoL

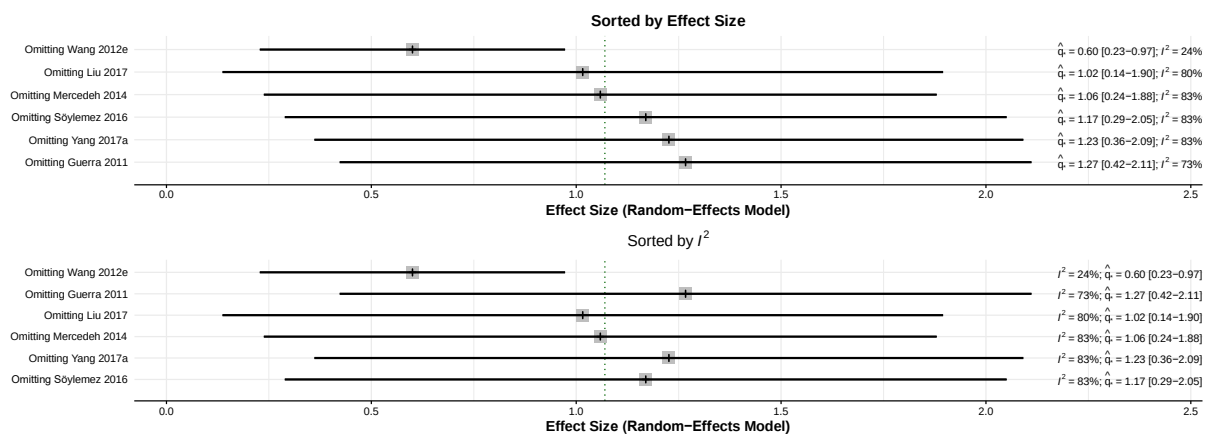


Figure 23 Leave-one-out analysis of effects on caregiver QoL

Caregiver distress

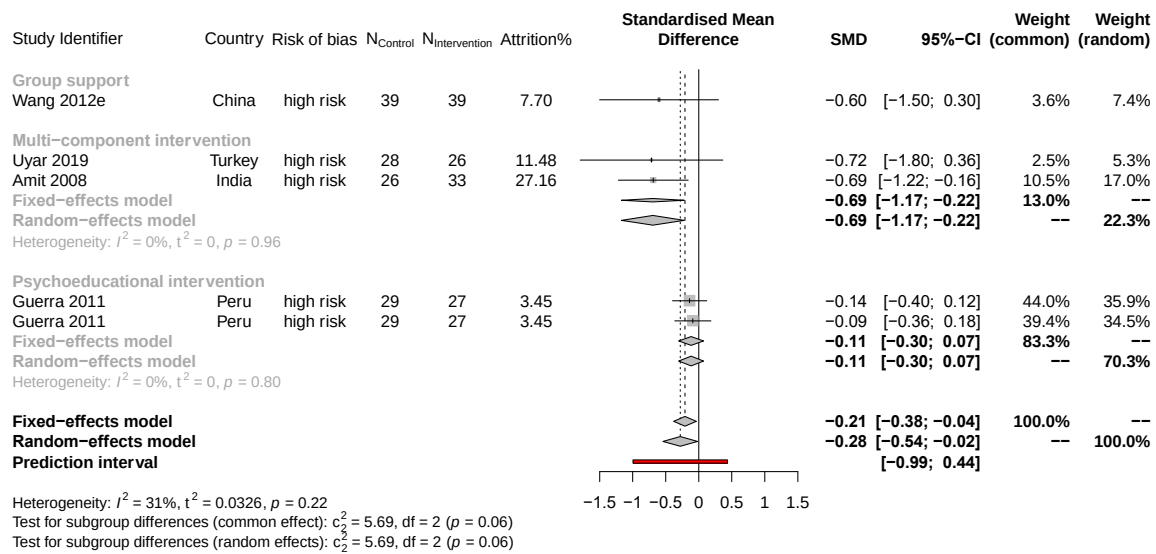


Figure 24 Forest plot of meta-analysis of effects on caregiver distress

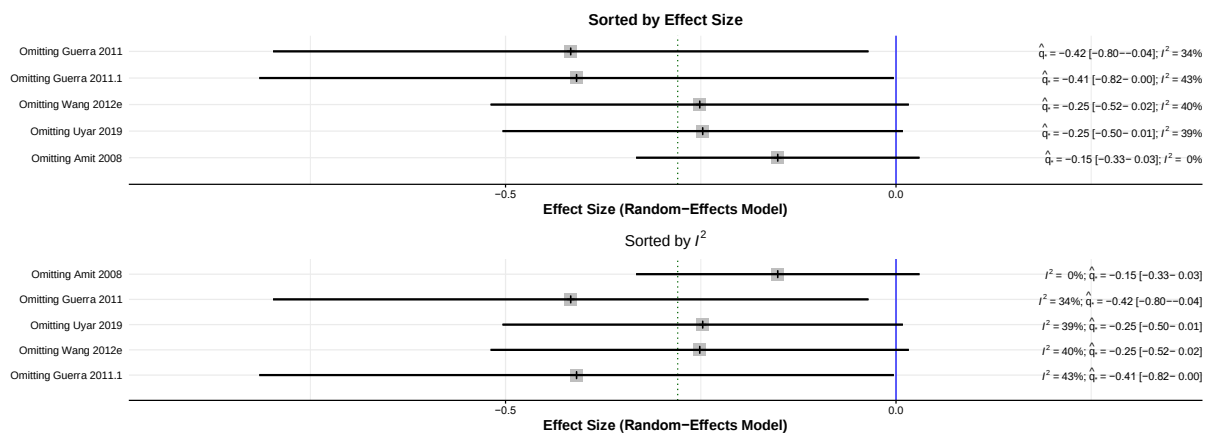


Figure 25 Leave-one-out analysis of effects on caregiver distress

Caregiver anxiety

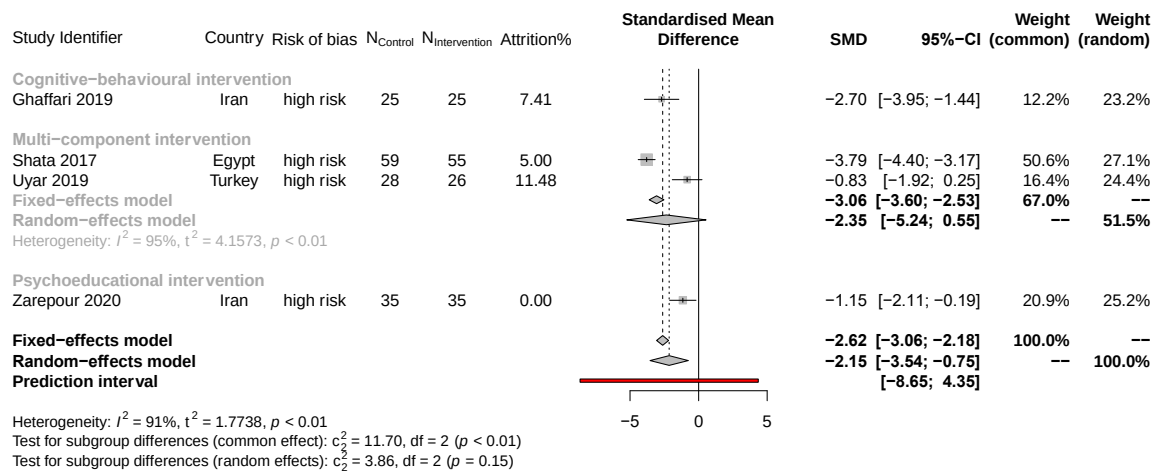


Figure 26 Forest plot of meta-analysis of effects on caregiver anxiety

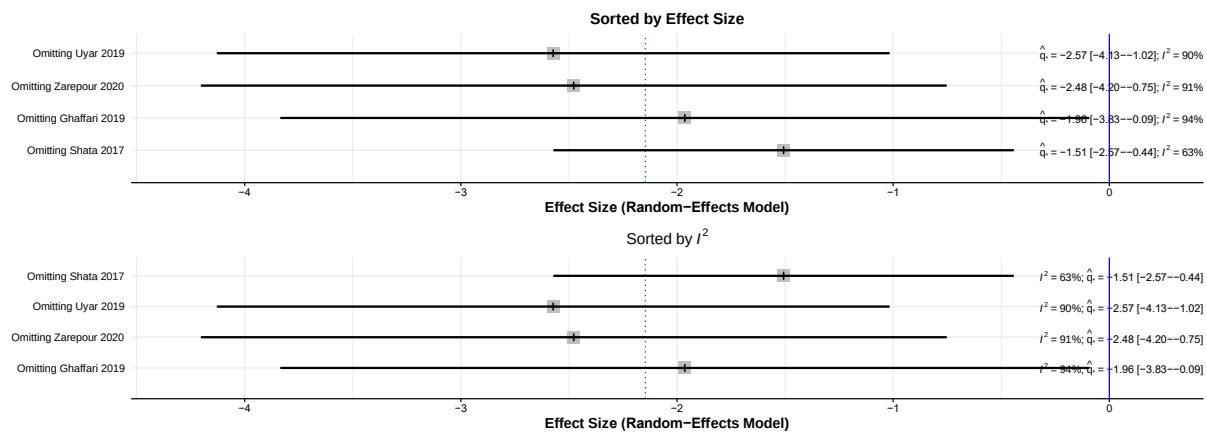


Figure 27 Leave-one-out analysis of effects on caregiver anxiety

Patient ADL

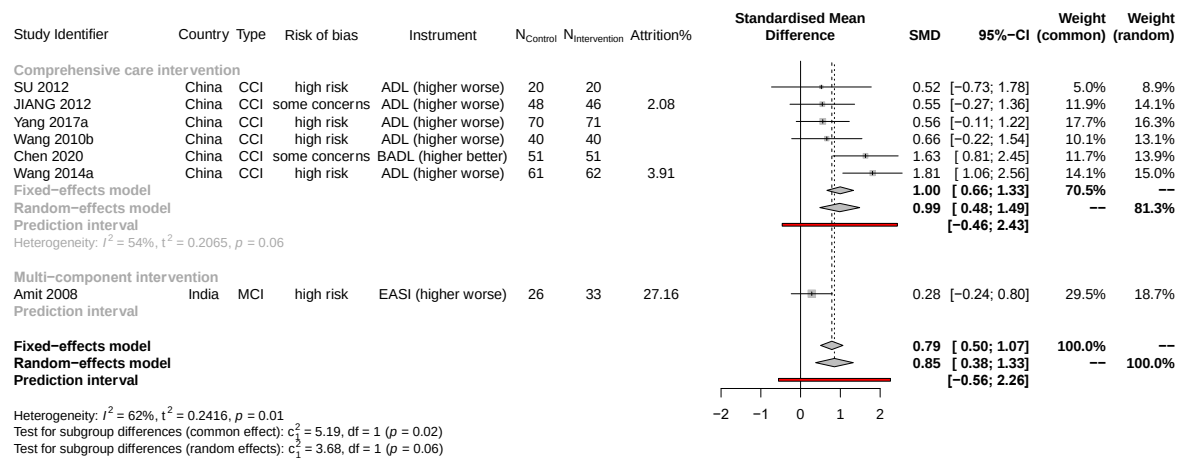


Figure 28 Forest plot of meta-analysis of effects on patient ADL

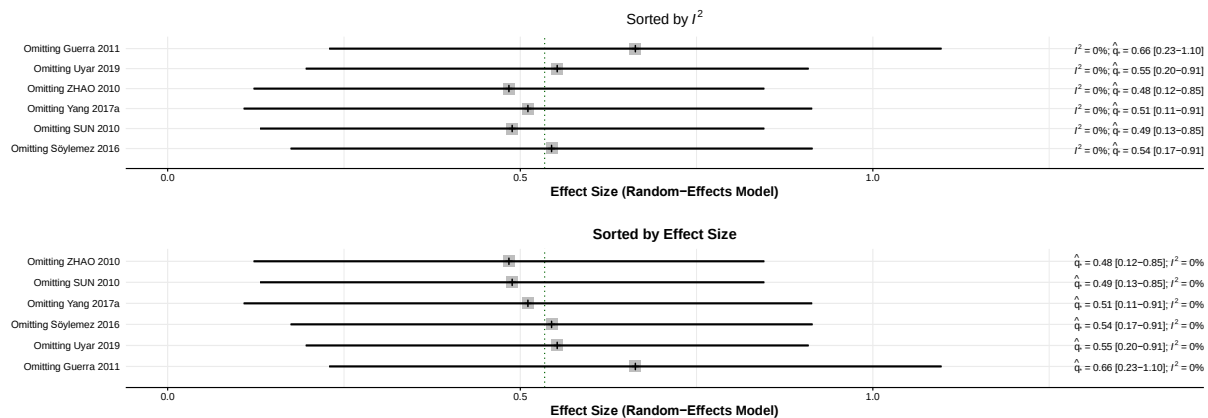


Figure 29 Leave-one-out analysis of effects on patient ADL

Patient cognitive function

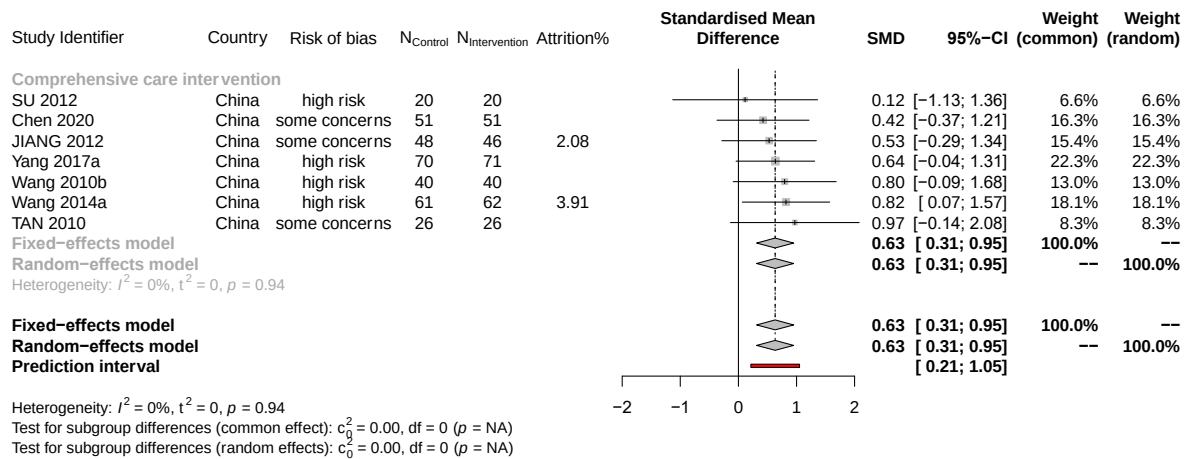


Figure 30 Forest plot of meta-analysis of effects on patient cognitive function

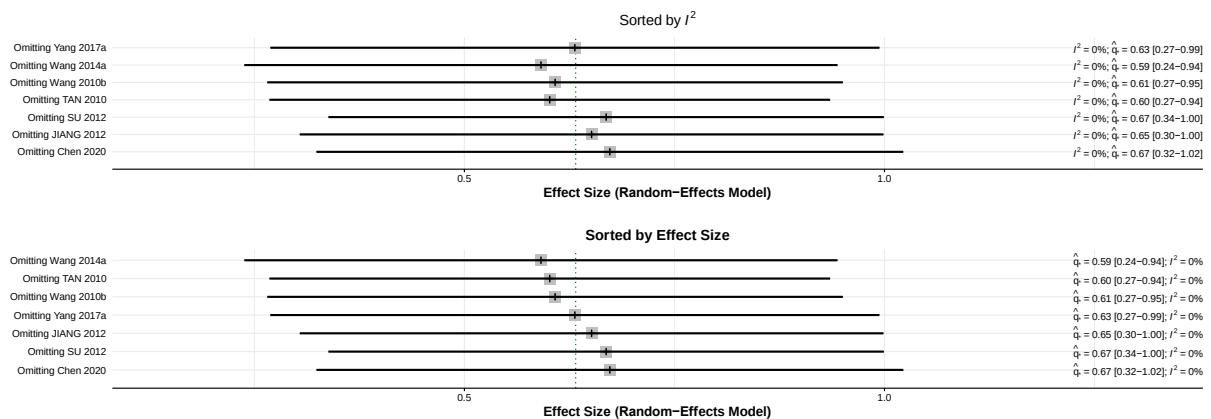


Figure 31 Leave-one-out analysis of effects on patient cognitive function

Patient QoL

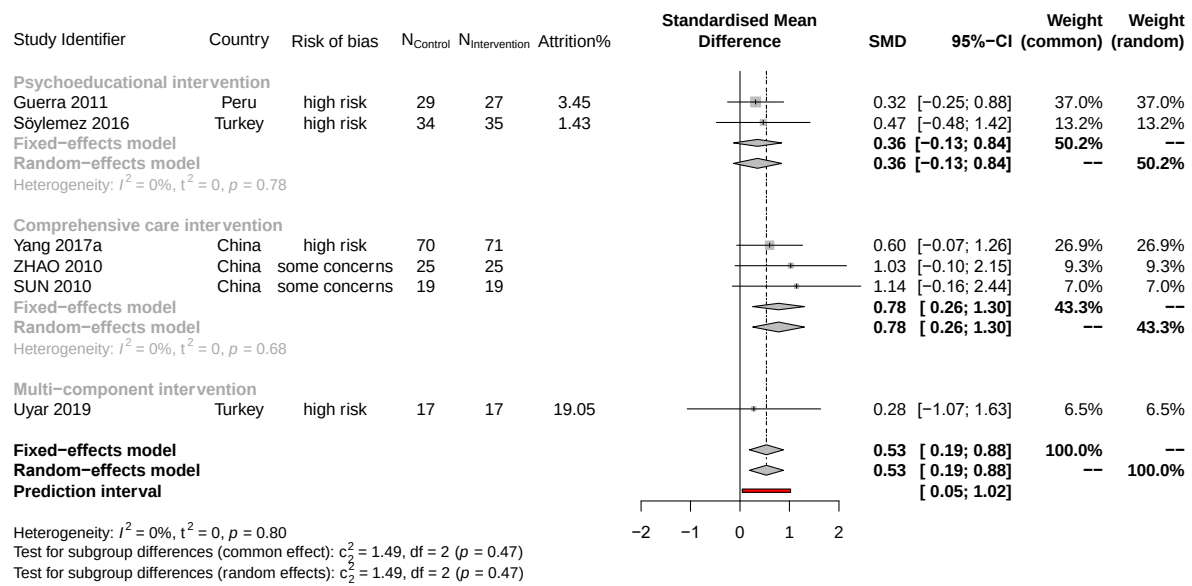


Figure 32 Forest plot of meta-analysis of effects on patient QoL

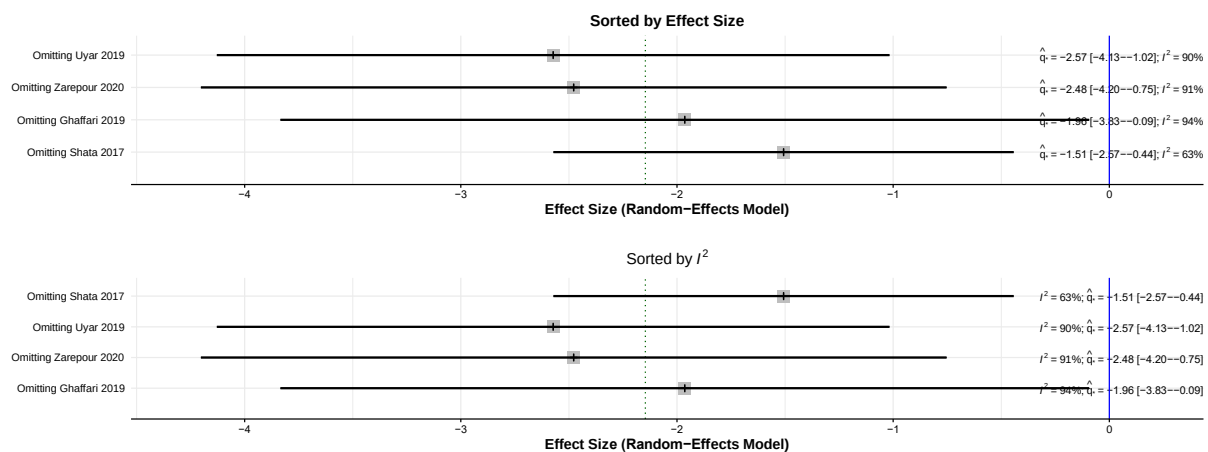


Figure 33 Leave-one-out analysis of effects on patient QoL

Patient neuropsychiatric symptoms

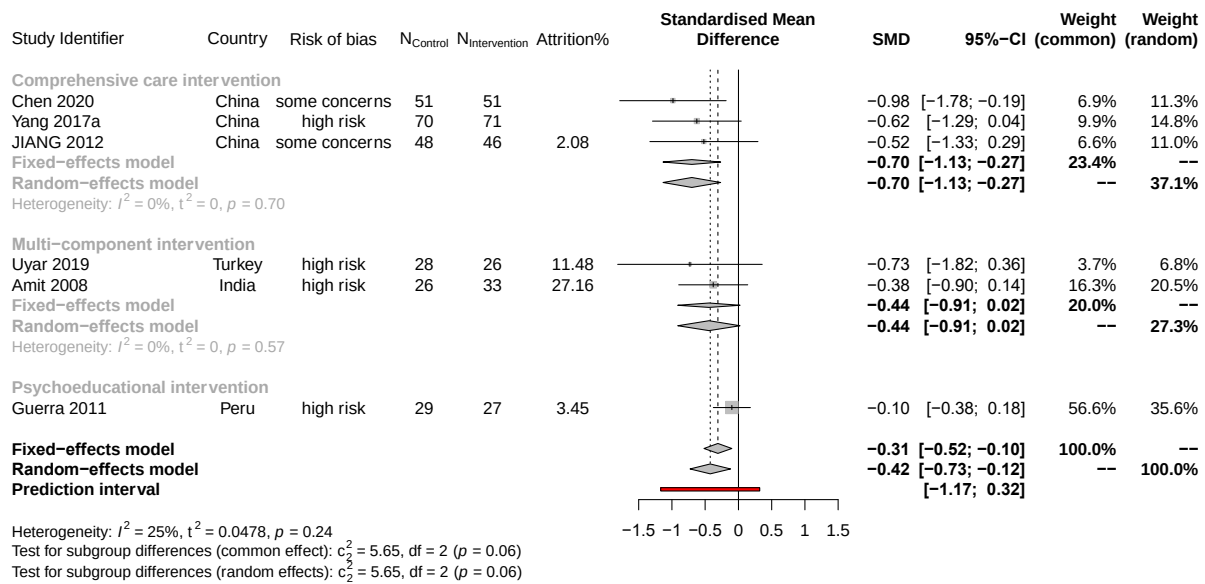


Figure 34 Forest plot of meta-analysis of effects on patient neuropsychiatric symptoms

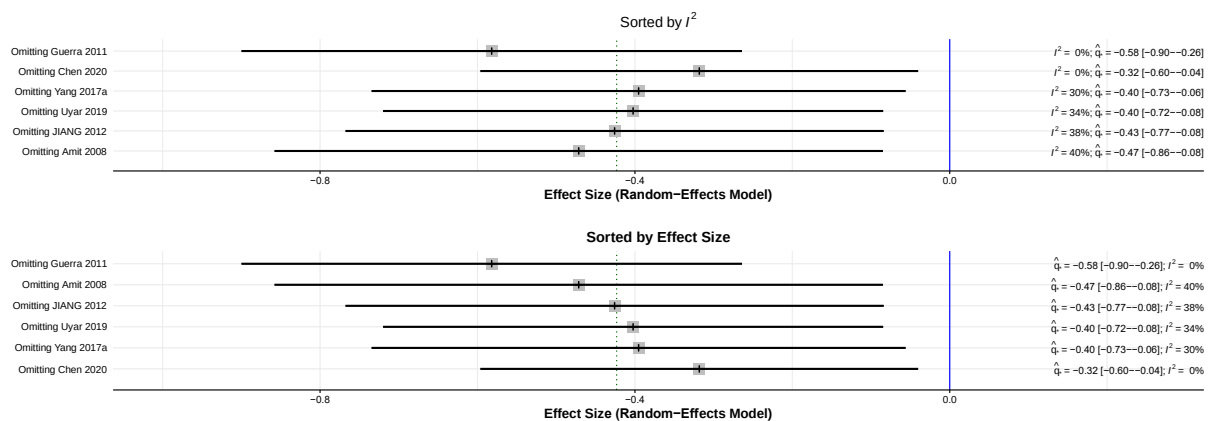


Figure 35 Leave-one-out analysis of effects on patient neuropsychiatric symptoms